Giesecke+Devrient

SECURITY TECH

ANNUAL REPORT 2023

Creating Confidence

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A major growth year, new positioning, and a move to the next level – welcome to our 2023 annual report

2023 was a special year for G+D. We launched a new, strong phase in the company's development, and it was a year of major growth with excellent results across all business areas that saw G+D achieving many new record highs.

Total sales rose significantly again last year, reaching the EUR 3 billion mark for the first time. Order intake also grew strongly by nearly 20 percent, which will translate into continued success going forward. In addition, G+D tightened its positioning as a SecurityTech company and is embarking on a new phase of development and transformation, which is aimed at helping to ensure the digitalization of our society happens in a secure manner.

To do this, G+D generates solutions in three segments – Digital Security, Financial Platforms, and Currency Technology. It makes the lives of billions of people more secure in these areas thanks to integrated digital and physical solutions. Innovation and investment go hand in hand at G+D. This is demonstrated, among other things, by recent successful acquisitions, such as that of Swiss software company Netcetera. Investment in our core business and in R&D that leads to healthy organic growth rounds out G+D's strategy. Never in the history of G+D has more investment been made in innovation and the customer proposition than in 2023.

In his annual interview, Ralf Wintergerst, CEO of G+D, talks about the financial figures, his forward-looking strategy against a challenging global backdrop, and the company's transformation. G+D has evolved into a comprehensive connectivity provider for the IoT, a full-service provider in the payment ecosystem, and a reliable innovator in the field of digital security technology for a wide range of industries. The company also provides answers to the major technology issues of our time, with development priorities including disruptive technologies, such as artificial intelligence, and sustainable solutions, e.g. in the SIM technology field, payment cards, and the cash cycle.

Comprehensive insights into current technology trends and innovation at G+D can be found in this year's lead article in the annual report; a further article explains why ESG is a core element of the company's strategy. The Group management report and consolidated financial statements provide detailed information on the company's business performance.

The fields of Digital Security, Financial Platforms, and Currency Technology cover fundamental aspects of life for companies, citizens, and consumers that need to function smoothly and efficiently, especially in turbulent times. With its modern security technology, G+D makes a crucial contribution here, enabling people to go about their daily lives with confidence.

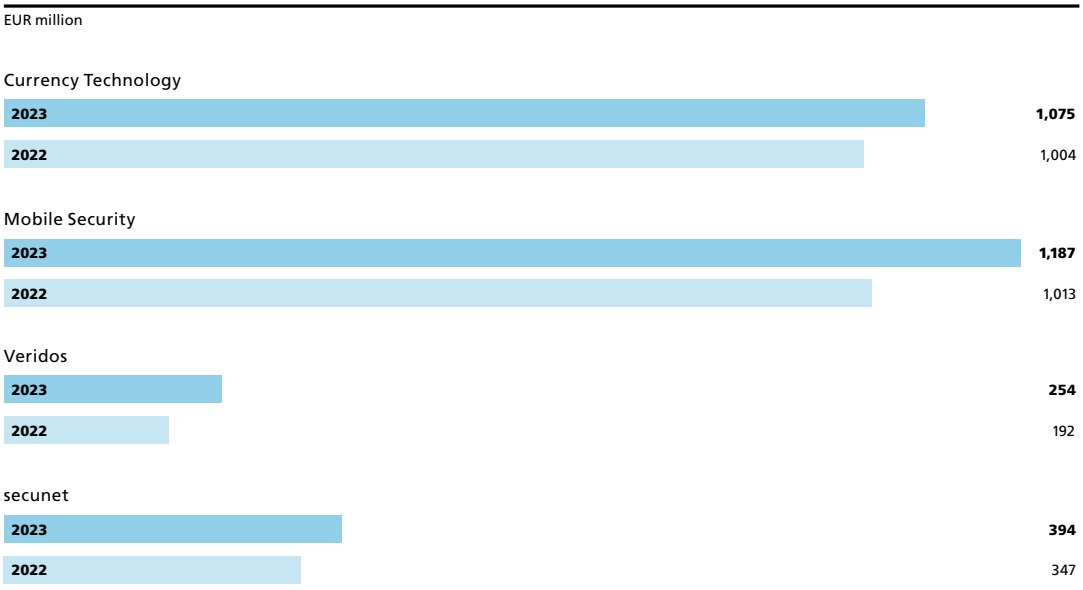
Information on the company's economic situation and its future is available in our audited management report and the notes to the financial statements.

Giesecke+Devrient at a Glance

Giesecke+Devrient Group

EUR million	2023	2022	Change in %
Sales	2,973.3	2,527.1	17.7
Total investments	489.0	340.3	43.7
Thereof research and development	200.2	164.5	21.7
Thereof investments in technology and other	111.2	82.1	35.3
Thereof equity interests	177.6	93.7	89.7
EBITDA	347.3	296.9	17.0
EBIT	176.4	168.2	4.9
Net income	92.1	80.6	14.3
Employees as of December 31	14,203	12,594	12.8

Sales by Business Sectors



Company Structure

Giesecke+Devrient GmbH

Head office	Number of employees	Sales in EUR millions
Munich	14,203	2,973.3

Group

Giesecke+Devrient GmbH

Business Sectors

Giesecke+Devrient Currency Technology GmbH	Giesecke+Devrient Mobile Security GmbH	Veridos GmbH 60 %	secunet AG 75.5 %
Munich	Munich	Berlin	Essen

Services		Other Group Companies	
Giesecke+Devrient Group Services GmbH & Co. KG	Giesecke+Devrient Immobilien Management GmbH	Giesecke+Devrient Ventures GmbH	Giesecke+Devrient advance52 GmbH
Munich	Munich	Munich	Munich

Interview with the Group CEO

» We have further tightened our positioning as a SecurityTech company. With our solid figures providing additional momentum, we are now entering a new phase of the company's development and moving forward with our successful transformation process. «

Mr. Wintergerst, if you were writing a report card for Giesecke+Devrient for 2023, what grade would you give the company?
2023 was an exceptional growth year for G+D, which saw us achieve excellent results across all business areas and set many new benchmarks. To give one example, we significantly exceeded our sales target and reached the EUR 3 billion mark. Our performance was thus very positive, as is my personal assessment of how we did. But what is much more important to me than a grade is the way we handled the year. Globally, 2023 was a difficult year but G+D's employees put their heart and soul into tackling the challenges we faced. That combined effort is the reason for our strong results.

Part of our success is also about looking to the future: we have further tightened our positioning as a SecurityTech company to ensure that we are ready for the coming years. With our solid figures providing additional momentum, we are now entering a new phase of the company's development and moving forward with our successful transformation process. We're excited about the developments that lie ahead of us and intend to actively shape the future.

How are the successes of 2023 reflected in G+D's business figures?
We increased sales by 18 percent to EUR 3 billion, evenly distributed across three areas. These three newly defined segments – Digital Security, Financial Platforms, and Currency Technology – provide a strong foundation for G+D as a security technology company. Successful acquisitions contributed to our robust growth – particularly the acquisition of a majority stake in software specialist Netcetera, with our holding recently being increased again to 95 percent. We are also growing organically thanks to our high-quality solutions, our competitiveness in the various markets, and our ability to innovate.

Ralf Wintergerst
Group CEO Giesecke+Devrient

Our order intake, which represents future sales, is proof of just how much our customers trust our products and services. Here the figure is up 19 percent, at EUR 3.2 billion, so we are well above revenues. Our EBITDA also increased by 17 percent to reach a new high – strong evidence that we are achieving profitable growth.

What impact is the difficult global situation having?
Economically and especially politically, the global situation has not eased, unfortunately. Quite the opposite, in fact, although in view of the many challenges around the world, close and committed collaboration across national boundaries would be highly desirable. As an international company with sites in 40 countries worldwide, G+D attaches great importance to sharing information and experience, and everyone pulling together. That works very well in our teams.

G+D is currently making major investments. What is your strategy here?
Last year, we invested more than ever before, particularly in expanding our portfolio. Acquisitions are a crucial part of this strategy, alongside investment in our core business, in research and development, and in evolving our IT infrastructure. We invest in acquisitions of international companies that are a good fit with us, complement our portfolio and services, and contribute to our digitalization strategy. Recent successful examples of such investments include German IoT company Mecomo, the payment and identity solutions business of a US competitor, and cloud expert SysEleven. In addition, we acquired Spanish company Pod Group, an international enterprise network operator, in 2021 and fully integrated it into G+D last year. I've already mentioned our latest acquisition, Swiss company Netcetera. Our corporate venturing activities are also part of this investment strategy.

New year, new strategy cycle: G+D has tightened its positioning. Can you tell us a bit more about that?
As a SecurityTech company, G+D's activities are organized in three segments, which are aligned with customer requirements and markets. Each area has different needs and opportunities but they all offer attractive prospects for growth: the Internet of Things, cloud security, digital identities, secure and customized payment cards, and solutions for banking and the cash cycle. Security technology is the common factor here. Our aim is to use our solutions in these areas to help people go about their daily lives with complete confidence and peace of mind.

What solutions are you thinking of?
G+D's transformation journey can be mapped very clearly by looking at examples from our portfolio. We have evolved from a SIM manufacturer (and inventor) into a comprehensive connectivity provider. We are currently continuing the success story of the SIM card with the eSIM. And we are developing services – on the basis of SIM, eSIM, and iSIM applications – for IoT connectivity, such as through international roaming agreements. As a second example, we have gone from being a provider of electronic payment methods to being a fully integrated digital player, with innovative new payment cards and solutions for e-commerce and banking. As a third example, G+D is sometimes still described as a banknote printer. That's not wrong, but we've been more than that for quite some time. We are now a holistic provider covering the entire payment ecosystem – offering a range of physical, electronic, and digital solutions, including one for central bank digital currency.

Speaking of digital payment, weren't there some important developments around the digital euro in 2023?
Central bank digital currency, or CBDC, is gathering pace. The go-ahead for the digital euro was given in 2023, among other developments. The European Commission passed corresponding legislation at the end of June. In October, after concluding a two-year investigation phase, the European Central Bank announced that it would start the preparation phase for the project.

To ensure broad acceptance among banks and citizens, a CBDC must be well designed. G+D has developed a solution that allows citizens to pay digitally even without technical knowledge and expensive devices. It also supports a variety of other use cases for consumers and businesses.

» We also keep working on what makes us special: our unique customer proposition as a **SecurityTech group that improves the lives of billions of people.** «

Ralf Wintergerst
Group CEO Giesecke+Devrient

Artificial intelligence gained real traction last year. What is G+D's stance on that?
Artificial intelligence has been an important field for many years, but last year there was a quantum leap in some applications. Like many others, G+D is interested in the technical possibilities of AI. We use it to enhance the security of products, to accelerate production processes, and for quality control, e.g. in the production of ID documents. secunet, the G+D division for securing digital infrastructures, uses AI to train facial recognition systems in order to avoid statistical bias in the underlying data, with airports being typical deployment scenarios for such systems.

AI is the latest big trend, ESG is already here to stay. Which goals did G+G achieve with its sustainability strategy in 2023?
All the aspects of sustainability covered by the term ESG are a top priority at G+D. The relevant department reports directly to the CEO's office.

We have converted all our sites to green energy and thus achieved an important milestone on the way to becoming a climate-neutral company. Overall, we cut our carbon emissions by around half last year.

Another strong focus is developing sustainable products, such as payment cards made from recycled plastic and our Green Banknote, which compared to the rest of the market has an excellent carbon footprint thanks to its material mix and lifespan. With these endeavors, we are also helping our customers in different industries to meet their own sustainability objectives and provide environmentally friendly products and services.

If you take a sneak peek into the future, what does the current year hold for G+D?
G+D has made huge strides in recent years and that work is now bearing fruit. Sticking with the same metaphor, we will continue to harvest that fruit in 2024. We want to continue growing – organically through our own efforts, and also by way of further acquisitions. That means maintaining the momentum in our existing businesses while also tapping into new business fields. We intend to take the development and integration of digital assets that complement our physical products to a new level. And we also keep working on what makes us special: our unique customer proposition as a SecurityTech group that improves the lives of billions of people.

Dr. Ralf Wintergerst

Group CEO
Giesecke+Devrient

Dr. Ralf Wintergerst has been Chair of the Management Board of Giesecke+Devrient since 2016. He is responsible for overseeing various central services, comprising Information Systems, Corporate Security, Compliance Management and Auditing, Mergers & Acquisitions, Corporate Brand Communications, Corporate Strategy and Development, Legal, and Corporate Governance. In addition to his role at G+D, Ralf Wintergerst is Chair of the Supervisory Board of secunet, Chairman of the Board of Directors of Netcetera, and president of German trade body Bitkom.

Dr. Peter Zattler

Group CFO
Giesecke+Devrient

Dr. Peter Zattler has been a member of the Giesecke+Devrient Management Board since 2001. As Chief Financial Officer, he oversees Controlling, Treasury, Accounting, and Tax. He is also responsible for Human Resources and Data Protection. In addition, Peter Zattler has been a member of the Supervisory Board of secunet since 2004.

Gabrielle Bugat

CEO
Giesecke+Devrient ePayments GmbH

Gabrielle Bugat has been Chair of the Management Board of Giesecke+Devrient ePayments since 2023. She manages the activities related to G+D's smartcard and associated services business and oversees sales, operations, the portfolio, and development. Ms. Bugat is also responsible for driving further expansion of business relating to secure digital payment solutions. In addition to her role at G+D, Gabrielle Bugat is a member of the Board of Directors of Netcetera.

Axel Deininger

CEO
secunet AG

Axel Deininger joined the Management Board of secunet Security Networks AG in 2018 and became its Chair in 2019. His particular responsibilities include Strategy/Business Development, International Sales, Marketing, and Human Resources. As an IT security partner to the Federal Republic of Germany, secunet enables digital sovereignty for governments, businesses, and society.

Group Executive **Committee**

Dr. Philipp Schulte

CEO
Giesecke+Devrient Mobile Security GmbH

Dr. Philipp Schulte is Chair of the Management Board of Giesecke+Devrient Mobile Security. In this role, he has been responsible for activities relating to connectivity and the Internet of Things since 2023. Philipp Schulte was previously CFO in the Mobile Security business unit. In addition to his role at G+D, he is a member of the Board of Directors of Netcetera.

Dr. Wolfram Seidemann

CEO
Giesecke+Devrient Currency Technology GmbH

Dr. Wolfram Seidemann has been CEO of Giesecke+Devrient Currency Technology since 2016. G+D Currency Technology is the market leader for solutions and services relating to banknotes and banknote processing systems. As a partner to central banks and the currency industry, G+D Currency Technology delivers comprehensive expertise and innovative technologies that increase the efficiency of the cash cycle.

Marc-Julian Siewert

CEO
Veridos GmbH

Marc-Julian Siewert has been Chair of the Management Board of Veridos GmbH, a joint venture between Giesecke+Devrient and Bundesdruckerei, since 2022. Veridos provides governments and public authorities with customized end-to-end solutions for secure identities – both physical and digital. Governments and public authorities in more than 100 countries rely on Veridos GmbH's project expertise and product portfolio in the field of integrated identity solutions.

Supervisory Board **Report**

Ladies and Gentlemen:

During the 2023 fiscal year, the Supervisory Board of Giesecke+Devrient GmbH performed all its duties as stipulated by legal provisions and the Articles of Incorporation. The Supervisory Board duly monitored the Management Board and discussed issues of note with its members.

At meetings of the Supervisory Board, the Management Board provided regular, comprehensive information about the situation of the company and the Group as a whole. The Supervisory Board additionally received updates on G+D's performance and finances in the form of quarterly reports. Outside the scheduled meetings, the Chairman of the Supervisory Board was also in regular contact with the Management Board and was kept informed of current issues.

The Supervisory Board held three scheduled meetings (on April 3, July 20, and December 7, 2023) based on detailed reports from the Management Board to review the company's economic situation, including major investment decisions and Group programs. At an extraordinary meeting (September 27), the Supervisory Board discussed and approved a change in management as part of succession planning and dealt with other HR matters.

In accordance with its duties, the Supervisory Board considered the Group's corporate governance, including the compliance and human rights management system, the risk report, the ICS and risk management system, and the internal audit report.

At the April meeting, the Supervisory Board discussed general topics relating to the annual financial statements. It also decided on the appointment of a hiring committee for succession planning in relation to the outgoing CFO, Dr. Zattler, who is retiring in 2024.

At the July meeting, the Supervisory Board discussed the current business situation, including the latest risk report and forecast. The Supervisory Board also decided on the introduction of a trust for funding pension obligations (Contractual Trust Agreement, CTA) and dealt with other HR matters.

At the extraordinary meeting in September, the Supervisory Board decided on the appointment of Dr. Wintergerst as Director of Labor Relations, in addition to his post as Chairman of the Management Board and CEO, and the appointment of Jan Thyen as Chief Financial Officer (CFO). The Supervisory Board also decided on the expansion of the Management Board to three members and decided on the appointment of Gabriel von Mitschke-Collande as Group Chief Digital Officer (CDO).

The appointments are due to take effect as of May 1, 2024.

In addition, the Supervisory Board decided on the introduction of a new remuneration system for the Management Board.

Prof. Klaus Josef Lutz
Chairman of the Supervisory Board

At the December meeting, the Supervisory Board reviewed the operational plans for 2024, in addition to the current business situation and the risk report. The meeting also addressed HR matters.

The Supervisory Board duly received the annual financial statements and management report of Giesecke+Devrient GmbH for the period ending December 31, 2023, prepared in accordance with the German Commercial Code (HGB), and the consolidated financial statements and Group management report for the period ending December 31, 2023, prepared in accordance with IFRS, along with the auditor's reports.

The annual and consolidated financial statements were examined by the auditor, KPMG AG, which issued an unqualified audit opinion in each case.

The auditor attended the meeting of the Supervisory Board on April 9, 2024, at which the financial statements were discussed. In the course of this meeting, the auditor reported on the main findings of the audit, including on the internal control system in relation to the financial reporting process, and answered questions from the Supervisory Board. No circumstances that would indicate a lack of impartiality on the part of the auditor were reported. The fees for consulting services and other services provided by the auditor of the Group accounts in addition to the audit are disclosed in the notes to the financial statements. The Supervisory Board accepted KPMG AG's audit opinion on both sets of financial statements.

The Supervisory Board concluded its review with no objections raised. It approved the annual and consolidated financial statements, including the corresponding management reports, at its meeting on April 9, 2024.

Giesecke+Devrient GmbH achieved exceptionally good results in fiscal 2023 despite again facing extremely challenging conditions. This was a remarkable performance by the Management Board and staff across the Group. On behalf of the Supervisory Board, I would like to thank the members of the Management Board, all employees, and the Works Councils of the Group for their efforts and high degree of personal commitment during fiscal 2023.

Munich, April 2024

Prof. Klaus Josef Lutz
Chairman of the Supervisory Board

Security in the digital age

G+D is paving the way for the technologies of tomorrow with trusted solutions

A longer, more comprehensive version of this article is available at **www.gi-de.com/spotlight**

Every day, many people around the globe use digital applications that make their lives easier and more convenient. Payment cards allow them to conduct contactless payments in stores and online through apps, while their smartphones connect them securely to the Internet; ID documents with electronic chips facilitate travel. Billions of people worldwide trust the technologies behind these solutions and expect them to function securely and reliably. At G+D, we are committed to creating that trust and confidence.

So that we can continue playing a key role in shaping digital transformation, G+D is committed to ongoing strategic development. Our activities are organized in three segments that reflect our target markets and the portfolio: Digital Security, Financial Platforms, and Currency Technology. In these areas, physical and digital products and services go hand in hand and complement each other. Permanent innovation is the engine that drives transformation. G+D has a tried-and-tested formula for innovation, which involves conducting its own R&D, engaging in business development projects, and making acquisitions (including corporate venturing). Ideas are jointly developed by G+D employees and then translated into marketable innovations with the aid of efficient, cross-divisional innovation management.

Digital Security: Solutions for the interconnected world, identities, and secure data traffic

Autonomous driving, smart homes and smart cities, sensor-controlled communication between machines in factories, and much more besides: the bigger and more comprehensive the Internet of Things (IoT) becomes, the more opportunities it offers. Simultaneously, trust in the security of the digital systems and devices involved grows ever more crucial.

Connectivity and IoT – a bright digital future for SIM cards:
Protecting our hyperconnected world against unauthorized access, data manipulation, and data theft is a hugely complex challenge. The SIM card and its latest eSIM and iSIM iterations can make a pivotal contribution to security. eSIMs, which are embedded in devices as a chip during the manufacturing process, have been gaining in importance for some time. More and more manufacturers of smartphones, tablets, and smartwatches have been equipping their products with this technology in recent years. Thanks to excellent performance and service availability combined with the highest security standards in its GSMA-accredited data centers, G+D is now the market leader in eSIM management. More than a billion mobile devices use G+D's eSIM operating system. G+D has already taken the next developmental step with the iSIM, which is integrated into an IoT device in a system-on-a-chip (SoC) architecture.

Supported by acquisitions, G+D is currently developing additional services for IoT connectivity on the basis of SIM, eSIM and iSIM applications. Thanks to our international roaming agreements, companies receive secure, easy access to more than 600 networks in 185 countries. Demand for IoT applications for tracking is growing rapidly, particularly in the goods transportation segment of the logistics industry. Knowledge of where goods carriers are located, what routes they are taking, and what condition the transported goods are in enables transport companies to manage their fleets with maximum efficiency and provide customers with additional services. As a result, G+D has also offered tracking solutions for the transport industry since 2023.

Identity Technology – a successful balance of security, trust, and convenience:
In the digital age, protecting identities takes on a new dimension. Digitalized ID documents need to be just as secure as their physical counterparts – and people should also be able to use them easily and intuitively. Although most ID documents and passports are still physical documents, they are increasingly being equipped with digital features. They also benefit from digitalized processes and modern security technology in their production and use.

Electronic passports and ID cards store the holders' data, including biometric information, on electronic chips. These integrated chips enable a number of services, such as digital use of ID documents via smartphone apps. Using the data stored in the app, travel documents can be registered or visa applications submitted. Veridos, a joint subsidiary of G+D and Bundesdruckerei, provides a solution that can manage one or more passport data sets, e.g. those of family members. Users can transmit this data to trusted third parties such as border authorities or private providers – to apply for a visa online, for example, or book a hotel room.

Digital Infrastructures – digital sovereignty in the cloud:
For companies that store electronic information about their customers and employees, commercial success and the trust of their customers is dependent on this information being inaccessible to unauthorized parties. In sensitive areas such as healthcare, critical infrastructures, and public institutions, the stakes are even higher. Digital interconnection therefore calls for a robust security architecture and technologies that reliably protect data from misuse. Especially in the cloud, there is a need for comprehensive protection of data, applications, and other services. secunet, G+D's Digital Infrastructures division, is developing an ecosystem of cloud services based on its tried-and-tested open-source-based platform. The ecosystem is modular and interoperable, which means it can be adapted to a wide range of user requirements. Underpinning this is secunet's proven encryption technology, combined with the high-security solution SINA, which already sets the benchmark for secure networks and workplaces in the public sector.

Artificial intelligence (AI) is also impacting more and more areas of life. With some AI applications, it is particularly important to avoid undesirable effects. Examples here include AI systems that are used at airports for live analysis of facial images to avert threats. secunet tests and trains these systems to ensure there is no statistical bias in the underlying data.

Financial Platforms: Secure payment across all touchpoints

Whenever consumers get out their payment card in a store or pay online, they trust that their money will reach the recipient safely. Debit and credit cards play an important role here for payment in brick-and-mortar retail and e-commerce stores alike. In 2023 alone, G+D shipped more than 500 million payment cards with a contactless payment function. Physical use of these cards goes hand in hand with the digital shopping experience.

Payment Technology: Holistic portfolio of payment and banking services
The trend towards customized and environmentally friendly payment cards is gaining momentum. Accordingly, G+D has become the first major payment card manufacturer to pledge to replace the virgin plastic in its payment card products with recycled plastic by the end of the decade. The transition process is already under way.

G+D also supports the social need for accessibility. In this context, G+D is developing cards with tactile elements for people with visual impairments, which help them to recognize more quickly whether they are holding a debit card or credit card, for example. G+D takes inclusiveness very seriously; these special features mean that the issued payment cards will prevent any form of discrimination and also offer greater convenience for users. Similarly, cards with a biometric security architecture eliminate the need to use passwords, PINs, or mobile devices to authenticate transactions – a fingerprint or facial scan is sufficient.

E-commerce operators also need to remain mindful of security. Netcetera provides 3-D Secure products that make online payments safe for consumers and retailers alike. Tokenization of customer and account information enables unique identification when making payments, so no information about the cardholder or sensitive card data is exposed during transactions. G+D's award-winning tokenization solutions protect more than three billion digital transactions a year. More and more consumers are doing their banking online, which gives added importance to secure authentication methods and digital differentiation. Netcetera's modular web and mobile banking solutions enable banks to present their brand's value proposition to their customers in the digital world as well.

Trusted Software: The key to new digital business models
Netcetera and G+D develop groundbreaking IT products and customized digital solutions. This makes them reliable partners for business-critical applications in highly secure environments, such as financial technology, mobility, healthcare, and access security, where data privacy is crucial. Netcetera combines in-depth expertise and technology skills with creativity and passion to deliver dependable, forward-looking solutions.

Currency Technology: Resilient, efficient, sustainable – that is the future of cash

Cash is part of modern life, and its many properties are astonishing. Apart from being a generally accepted means of payment, cash is easy and straightforward for consumers to use and a driver and stabilizer of national economies. As a public good, cash must meet high requirements. It needs to be extensively available, accessible to everyone, and widely accepted.

Digital Currency Ecosystem – the path to digital cash:
Secure, extensively available, and accessible without a bank account. That sums up the desired attributes of a digital currency issued by public authorities, i.e. a central bank digital currency (CBDC). Like with cash, there are no charges for using well-designed CBDCs, and no data has to be disclosed. A successfully implemented CBDC would also need to enjoy the trust of consumers, companies, banks, and other stakeholders in equal measure. G+D has developed a central bank digital currency solution that supports more than 24 use cases, from QR code payment to user-to-commerce transactions. The vision is that both fintechs and banks will develop payment solutions and build them on a digital public infrastructure. Such an infrastructure offers huge potential for innovation and can unlock great economic value.

Currency Management – automating the cash cycle:
The resilience of the cash cycle is exceptionally important for ensuring that cash fulfills its economic and societal role. This calls for a responsive system where the supply of cash is guaranteed at all times, whether in normal everyday life or in times of crisis. Efficient and automated processes are thus becoming increasingly important for the provision of cash, in addition to adaptable infrastructure. Cash center workflows need to be fast, smooth, and scalable. G+D's modular solutions include integrated automation solutions that meet the requirements of cash centers of all types and sizes in terms of speed, flexibility, and productivity.

Banknote Solutions – a green product portfolio for a sustainable cash cycle:
Security in tandem with innovation is essential for reliable and durable banknotes. G+D has designed and manufactured 160 billion banknotes worldwide to date, with sustainability becoming an increasingly important criterion. The aim is to use fewer resources throughout the entire lifecycle of a banknote in order to reduce the environmental impact. A special Green Banknote has been developed as part of this endeavor, for example. Compared to the rest of the market, this banknote has the best carbon footprint when considering the entire lifecycle, thanks to its material mix and lifespan. With its green products, G+D is pointing the way forward for a sustainable, environmentally optimized cash cycle.

New horizons, forward-looking technologies

Digital development is fast-paced and generates a constant need for new security solutions and business models. For G+D, it is therefore crucial to recognize future developments today and address the associated challenges for our portfolio and for our customers. This includes artificial intelligence: the implemented AI applications serve purposes ranging from accelerating production processes and improving product quality to boosting product security. AI also plays a role in data security throughout the entire lifecycle. Achieving trustworthy data, i.e. ensuring the trustworthiness of data from collection, storage, and processing through to transfer, is another key task in today's interconnected world. G+D is developing cross-segment solutions in this field. Digital identities are an additional area of emphasis, and in particular the principles of decentralization and autonomy. The concept of self-sovereign identities (SSI) involves allowing consumers to selectively decide when and how much sensitive data is released.

For G+D as a global security company, innovation and ongoing development of solutions are key components of our corporate strategy. In the future, our motto will therefore remain the same: G+D safeguards essential assets around the world through security technology that creates confidence.

Sustainability is at the heart of our corporate strategy

As an international company, we at G+D are responsible not only for our customers and employees but also for the impact of our activities on the world in which we and future generations will live. We take this responsibility very seriously. Sustainability is therefore a key element of our corporate strategy. Across all business areas, we are taking concerted and structured action to protect the environment and have a positive impact on society – because we firmly believe that sustainable conduct is vital for safeguarding the future viability of our company and of the world around us.

The major challenges of our time can only be tackled through sustainable transformation of the economy and society and also of our current way of life. These challenges involve climate and environmental protection as well as issues of global equity through the fair distribution of resources, prosperity, and quality of life.

International companies like G+D have a particularly important role to play in this transformation. Through their economic activities and with their products and services, they impact the environmental and living conditions of billions of people. They therefore have a responsibility to ensure that these conditions are sustainable. Our position is clear: a commitment to sustainable conduct is vital for creating long-term value – for our company and for the world in which we live. "Sustainability is the key to responsible, forward-looking corporate governance. By protecting assets that are important to people and at the same time conserving the planet's resources for coming generations, we are investing in the future and also strengthening the trust of our customers in our solutions. Sustainable conduct is vital for achieving long-term success and making a positive impact on our society," says Ralf Wintergerst, CEO of Giesecke+Devrient.

G+D has made sustainability a key element of its corporate strategy and set itself specific targets to bring about lasting change within the organization. We want to make the right decisions today so that our company is able to help create a more sustainable future. Our targets are ambitious. Among other things, we will

- significantly reduce our ecological footprint
- promote diversity, equity, and inclusion in our company
- set high ethical standards for our business activities across all areas

When making the necessary changes to our business, we are guided by the Sustainable Development Goals adopted by the United Nations in 2015.

Sustainability is a top priority at G+D

Maintaining a strong strategic focus on sustainable transformation of the company and its portfolio is critical for G+D. Accordingly, sustainability management is headed by CEO Ralf Wintergerst, with the dedicated Corporate Sustainability function handling the operational aspects. A broad-based organizational structure ensures that sustainability has high priority in all business units and across all locations. G+D benefits here from managers and staff being fully on board with its sustainability strategy and associated measures.

Our aim is to be a leader on ecological and social issues within our markets and help to create a healthy environment and a fulfilling society by doing business responsibly. Our understanding of sustainability is based on three pillars: environmental, social, and governance (ESG).

Decarbonization for climate protection

Given the challenges of climate change, decarbonization of all our business activities is a key element of our commitment to sustainability. Our ambitious plan is to reduce our Greenhouse Gas (GHG) emissions to net zero by 2040. As an intermediate step, we aim to achieve a 42 percent reduction by 2030 compared with 2022. To achieve these targets, G+D has developed a decarbonization roadmap for the company based on the internationally recognized criteria of the Science Based Targets initiative (SBTi). "We have deliberately chosen science-based targets because we know that only an absolute reduction of GHG emissions will mitigate the climate crisis," says Andreas Lamina, Head of Corporate Sustainability.

The decarbonization roadmap is just one example that shows our sustainability targets are just as important to us as our financial and economic targets – and that we take the same systematic approach towards achieving them. We are investing in sustainable remodeling of our production processes and in developing sustainable solutions for our customers.

Our decarbonization efforts cover direct emissions, which are mainly generated at our production sites, and indirect emissions, which come from the generation of purchased energy and from our supply chain. It is crucial that we reduce our carbon footprint directly and genuinely, thereby making an effective contribution towards protecting the climate. That is why we consciously avoid carbon offset programs like carbon reduction certificates, preferring instead to focus on specific, tangible reductions and sustainable replacements.

What we achieved in 2023 for a sustainable future – three examples

Converting to 98 percent green electricity worldwide
In 2023, G+D again made good progress on many sustainability-related projects. One important milestone was the switch to 98 percent green electricity across all locations, with 100 percent set to be reached in 2024. This enables us to save more than 54,000 metric tons of CO_2 equivalents each year – an amount that roughly corresponds to the per capita emissions of 7,000 EU citizens. In addition to generating some of our own energy on-site, such as in Malaysia through solar modules, we rely on Guarantees of Origin (GO) to ensure that we use renewable electricity efficiently and transparently. Among other things, they document how and when a specific amount of electricity is generated. G+D's annual power consumption is fully covered by a corresponding number of GO certificates. By using green electricity, G+D is taking a major step closer to its goal of climate neutrality while at the same time supporting producers of renewable energy.

Talent development program for long-term business success
For a SecurityTech company like G+D, it is crucial not only to retain qualified employees but also to attract and develop new, talented individuals who are eager to help shape the future of G+D. In 2023, a key focus of our sustainability strategy was on improving employee potential and introducing specific measures to boost their skills and prepare them for new challenges. Our Group-wide talent

development program, for example, encourages employees to take on more responsibility within G+D. When selecting new talent, it is important for us to reflect the diversity in the Group. In 2023, the 29 talented employees in this program included 11 different nationalities from various G+D locations worldwide, with an equal proportion of women and men. We strongly believe that diversity (e.g. different sexes, cultures, life experiences, mindsets) is key to the success of our company.

Global application of the Code of Conduct
The Group-wide Code of Conduct provides the foundation for all our business activities. It was updated in 2023 and rolled out across all locations. The Code reflects G+D's corporate values – innovative, reliable, collaborative – and applies to all our dealings with business partners and each other. Alongside integrity, the protection of essential assets and respect for social standards and human rights, the Code defines G+D's responsibility to society. With the updated version, we have taken the Code of Conduct to a new level in order to create confidence in our company everywhere and at all times.

Green products as drivers of transformation

The transformation of our company towards greater sustainability is also evident in our product portfolio. Here too, G+D is a pioneer in its industry. Increasingly, sustainable products are securing the future viability of G+D. "When it comes to sustainability, focusing on products pays off in a number of ways. Products that are more environmentally friendly support our customers' sustainability agendas, reflect the wishes of consumers, and are good for our own company and for environmental protection," says Andreas Lamina.

G+D is set to replace all virgin plastic in its payment card products by 2030. Recycled, industrially compostable, or biodegradable materials will be used instead. For cash transactions, we are producing green banknotes made from natural fibers. In this context, G+D launched the Green Banknote Initiative (GBI) in 2022 with the aim of optimizing the sustainability of banknotes through the use of alternative fibers, the development of new printing inks, and the use of other recycled raw materials. We are also using digitalization as a driver of our green transformation. Digital twins of analog products (e.g. eSIM and iSIM) significantly reduce consumption of resources and energy. In the identity solutions field, we will start producing environmentally friendly ePassports in 2024, leveraging efficient manufacturing techniques, environmentally friendly materials, and low-carbon transport.

Transparent reporting

G+D's transformation into a more sustainable company is a challenging task, but one that we are highly committed to tackling. We ensure our targets, actions, and progress in this field are documented transparently and comprehensibly. To support this endeavor, G+D has expanded its sustainability reports accordingly over recent years. These reports are based on international standards and now meet a wide range of reporting requirements. Starting in 2024, we will comply with the reporting requirements of the EU-wide Corporate Sustainability Reporting Directive (CSRD) in our annual sustainability report. The directive helps companies to measure and compare progress more effectively.

"We will gradually continue our sustainability transformation process throughout 2024," says Andreas Lamina. Further decarbonization of our activities, increasing the number of female managers, and extensive training on compliance and data protection are all on the agenda for 2024. From this year onwards, management will also be regularly assessed on how successfully they are meeting the sustainability targets in their business areas. Progress made towards achieving sustainability targets will be a core criterion for variable remuneration. "We know that our sustainable transformation is a marathon, not a sprint. But we've already come a long way."

This story and further information are available at **www.gi-de.com/spotlight**

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Please note: Figures shown may differ from the mathematically exact values (monetary units, percentages, etc.) due to rounding.

1. Group Profile

Giesecke+Devrient (G+D) is a global SecurityTech company that makes the lives of billions of people more secure. We create confidence through integrated security technologies geared to the digital age.

As a reliable partner for currency technology, we provide trusted and secure solutions for public currencies in physical and digital ecosystems. G+D has a unique portfolio that covers the entire cash cycle. **Currency Technology** at G+D comprises two divisions: Banknote Solutions produces and distributes banknote paper, banknotes, and high-tech security features, while the Currency Management Solutions portfolio includes banknote processing systems of different sizes and complete cash center solutions.

The Secure Transactions + Services (STS) division, part of **Mobile Security**, and our subsidiary Netcetera provide holistic solutions for payment and banking services. Alongside an innovative portfolio for payment cards and digital payment, these solutions include trusted software for the financial sector and other highly regulated industries. We enable an end-to-end customer experience – from onboarding, authentication, and card issuance to banking apps and streamlined payment processes for online shopping. We also provide solutions for access and identity management as well as highly secure software for healthcare, mobility, and publishing.

As a technology leader with extensive expertise in secure connectivity, G+D is both partner and pioneer when it comes to tackling all the challenges around the Internet of Things (IoT). Our Trusted Connected Devices (TCD) division is part of **Mobile Security** and enables connections to mobile networks and the IoT. This encompasses SIM cards, SIM management services, eSIM management including secure operating systems, and connectivity for enterprise IoT.

G+D supports countries around the world with technologies that address secure and efficient identity management while at the same time delivering seamless and convenient identification and authentication processes for citizens. **Veridos**, a joint venture with state-owned Bundesdruckerei, creates, manages, and verifies legal identities for the physical and digital space. Its portfolio comprises highly secure physical and digital documents and border control and identification systems.

G+D's high-quality cyber security solutions, meanwhile, enable digital sovereignty for governments, businesses, and wider society. Our listed subsidiary **secunet** is an IT security partner to the Federal Republic of Germany and the leading provider of cyber security solutions. We can thus offer public authorities and industrial enterprises encryption technology up to and including the highest security level. An extensive portfolio of products and consulting services enables protection of digital infrastructures.

Parent company G+D GmbH acts as the **Corporate Center**. As such, it manages the overall direction of the G+D Group and actively supports strategic development of the individual business sectors. It handles topics that are of strategic importance for the whole Group. These include M&A activities, strategic initiatives for developing future and new digital business activities, investment company G+D Ventures, the Corporate Technology Office (C-TO), and the Corporate Development Fund. Generic services – such as IT, procurement, accounting, and HR – are brought together in a shared services company, Giesecke+Devrient Group Services GmbH & Co. KG. G+D advance52, which has been responsible for G+D's activities around central bank digital currencies since 2019, is also organizationally attached to the Corporate Center. A dedicated G+D real estate company holds and operates the buildings at the Munich site and leases them to the Group companies. G+D employs more than 14,200 people across 40 countries.

In the second half of 2023, G+D reorganized its activities into three new segments, to which the individual divisions are assigned: Digital Security, Financial Platforms, and Currency Technology. For further information, see section 5.2: "G+D in Evolution: Successful Corporate Transformation."

2. Business Development

G+D enjoyed a year of particularly strong growth in 2023. The company achieved excellent results and set many new benchmarks, including for sales and investment. This growth took place against a challenging backdrop, thus demonstrating the fundamental relevance of G+D's portfolio. No matter how demanding the global environment is, digital security, IoT connectivity, identity credentials, and payment processes need to function smoothly and efficiently.

Around the world, 2023 was marked by challenges, including fluctuations in global demand and disruption to supply chains, Russia's ongoing invasion of Ukraine, and the subsequent sharp increase in prices of raw materials – especially affecting energy supply in Europe. Despite this disruption and the financial discipline required to rein in the highest inflation in decades, global economic growth continued, albeit at a slower pace. Looking back, the global resilience of the economy has been remarkable, but repeated interest rate hikes after years of cheap credit spell new challenges for many industry sectors.

According to the latest figures from the International Monetary Fund (IMF), global economic growth softened to 3.1 % in 2023. At 1.6 %, the slowdown was more pronounced in industrialized countries than in emerging markets and developing countries, where economic growth hit 4.1 %. Among the major industrial nations, the US saw surprisingly robust consumer and investment activity, but growth in the EU was weaker than expected, with German GDP declining by 0.3 % in year-on-year comparison. While China is currently facing a real estate crisis and rising youth unemployment, many other emerging markets are proving to be relatively resilient.

According to the German Federal Statistical Office, the average inflation rate in the EU was 6.4% in 2023. Prices fell compared to the previous year (2022: 9.2 %), but the rate of increase remains significant. In Germany, the inflation rate was slightly lower in 2023, at 5.9 %.

2.1. Business Performance

The Group is managed on the basis of net sales, earnings before interest and taxes (EBIT), earnings before interest, taxes, depreciation, and amortization (EBITDA), capital expenditure, working capital, free cash flow, and return on capital employed (ROCE)[1].

In a challenging market environment, G+D set new benchmarks in fiscal 2023 and further expanded its strategic position as a SecurityTech company. Sales rose by 18 % in the reporting period, reaching the EUR 3 billion mark for the first time. With order intake hitting an all-time high of EUR 3,169 million, representing growth of 17 %, the order backlog increased to its highest level to date at EUR 2,021 million (previous year: EUR 1,677 million). The company thus has grounds to be optimistic about the coming fiscal year. These figures show that G+D's focused growth strategy is paying off. The company invested more in 2023 than ever before. Alongside investment in R&D and its core business, G+D made systematic investments in new technology fields and further expanded its digital business portfolio. Carefully targeted M&A activities supplemented this strategy, as highlighted by the overview below.

[1] Ratio of EBIT to average capital employed (year-end value in each case);
capital employed = intangible assets + property, plant and equipment + financial investments accounted for under the equity method and other equity interests + inventories + accounts receivable trade – accounts payable trade

There is a growing need among businesses and consumers for trusted digital solutions in online payment and banking that are also viable in practice. Accordingly, G+D increased its stake in Swiss software company **Netcetera**, a leading provider of innovative software and digitalization solutions for payment, banking, healthcare, publishing, and mobility. G+D and Netcetera jointly provide comprehensive digital solutions and user-centric software, in addition to having a strong track record of applying them in highly secure environments with strict data protection requirements. G+D first invested in Netcetera in 2020 as a strategic partner, subsequently increasing its stake in the company to 61.5 % in 2023 and then to 95 %.

Via Netcetera, G+D has acquired **Kamino**, a Slovenian company that specializes in developing mobile apps and digital identities. Kamino has a strong track record in digital identities, self-sovereign identities (SSI), and eID and will contribute to the ongoing development of Netcetera's identity solutions. The two companies have collaborated successfully in recent years and see this merger as the next strategic step in their partnership.

The Internet of Things (IoT) is connecting more and more devices and machines. This generates huge potential for applications in the logistics domain and mobile connectivity, both nationally and internationally. In the Mobile Security (TCD) business sector, G+D therefore acquired **MECOMO AG**, an IoT software company specializing in tracking and tracing solutions. Headquartered in Unterschleißheim near Munich, MECOMO provides end-to-end telematics solutions for industrial and logistics companies and is the market leader in the DACH region for digitalization of fleet-based logistics processes. By acquiring MECOMO, G+D can further expand its product portfolio around the Internet of Things and provide more IoT solutions from a single source.

Reliable global connectivity and powerful end-to-end solutions are essential in today's world, alongside data authenticity and security. With the acquisition of **Pod Group**, an Enterprise Network Operator (ENO) specializing in scalable IoT solutions connected via cellular networks, G+D moved to expand its range of corresponding solutions back in 2021. Pod Group was fully integrated into G+D in 2023. In the Mobile Security (TCD) business sector, G+D also took a minority stake in **NetLync**. This Dublin-based company specializes in the development of Entitlements-as-a-Service, including eSIM transfers. Entitlement servers enable an eSIM to be directly transferred from one phone to another. This function is one of a growing number of services now available to consumers as eSIM technology is rolled out more widely. G+D has been working with NetLync as a sales partner since 2022; this investment strengthens its engagement in a strategically important business field. Mobile Security (STS) also acquired the remaining 50 % of the shares in **E-Kart**, a Turkish card development and production company headquartered in Istanbul, thus giving it a 100 % stake.

Veridos acquired a majority stake in **NetSeT Global Solutions**, a developer of identity management systems. This acquisition sees Veridos bolstering its position as a full-service provider of integrated identity solutions. Serbia-based NetSeT specializes in developing complex information systems for citizen data management and information security. The acquisition expands Veridos' product range, which covers the entire value chain of identity solutions: from registration and creating and personalizing ID documents to managing data of citizens and verifying documents. Veridos increased its holding from a minority stake to 67 % of the shares.

2.1.1. Results of Operations

In fiscal 2023, G+D posted total sales of EUR 3 billion, exceeding the strong sales performance of the previous year by 18 %. The increase in sales is split equally between organic growth and acquisitions, reflecting robust market demand for G+D's high-quality security technology solutions and also the success of its M&A strategy.

Sales by Business Sector

EUR million	2023	2022	Change absolute	Change in %
Currency Technology	1,075.0	1,004.1	70.9	7.1%
Mobile Security	1,187.1	1,013.3	173.8	17.2%
Veridos	253.8	192.3	61.5	32.0%
secunet	393.7	347.2	46.5	13.4%
Other[1]	63.7	(29.9)	93.6	
Total	**2,973.3**	**2,527.1**	**446.3**	**17.7%**

[1] Consolidated, sales for Corporate Center (a52) and Netcetera (prorated since acquisition)

Currency Technology achieved a significant year-over-year increase in sales, at EUR 70.9 million. Banknote Solutions and Currency Management Solutions both recorded sales growth compared to 2022. A key driver of growth in Banknote Solutions was buoyant demand for banknote paper; G+D met this demand with innovative products. In Currency Management Solutions, high-performance systems for banknote processing were responsible for a large proportion of its growth.

Mobile Security was again able to significantly improve on the high sales figures achieved in the previous year. Particularly strong growth was generated by Secure Transactions + Services (STS) in the area of payment cards, e.g. with cards made from sustainable materials, and card issuance services. The Trusted Connected Devices (TCD) division also performed well and was able to grow its sales compared to the previous year, especially with its integrated SIM card solutions (eSIM and iSIM).

Veridos, the joint venture with Bundesdruckerei, posted a significant year-over-year improvement in sales. This growth was primarily driven by strong demand for ID documents, which Veridos provides together with comprehensive consultancy and services.

secunet continued its growth trajectory and ended 2023 on another record sales high, comfortably surpassing the previous year's peak. The company thus generated increased sales for the tenth year in a row with its products and solutions in the cyber security segment.

Due to acquiring shares in the former associated companies Netcetera and NetSeT, the consolidated income statement contains non-recurring income and expenses that are causally linked. These are disclosed in different line items of the income statement and almost zero each other out in EBIT. To aid comparison, G+D has reclassified the non-recurring income of EUR 27.9 million in financial income (fair value of the shares already held in Netcetera and NetSeT prior to 2023) as cost of sales. The adverse impacts of the resulting purchase price allocation (depreciation of disclosed hidden reserves to short-term depreciable assets) and the project costs of EUR 28.3 million arising in 2023 are therefore shown as a combined item. This results in a true and fair view of the effects of our corporate acquisitions on Group earnings at the level of operating profit.

Consolidated Income Statement (IFRS)

EUR million	2023	2022	Change absolute	Change in %
Net sales	**2,973.3**	**2,527.1**	**446.3**	**17.7 %**
Cost of sales[1]	(2,135.1)	(1,784.2)	– 350.9	19.7 %
Gross profit	**838.3**	**742.9**	**95.4**	**12.8 %**
Gross margin (% of sales)	28.2 %	29.4 %	– 1.2 p.p.	– 4.1 %
Selling, R&D, and general administrative expenses	(660.0)	(571.7)	– 88.3	– 15.4 %
Other operating income and expenses	0.5	4.8	– 4.3	– 89.2 %
Operating profit	**178.8**	**176.0**	**2.8**	**1.6 %**
Financial income/(expenses)[1]	(2.4)	(7.8)	5.4	– 69.2 %
EBIT	**176.4**	**168.2**	**8.2**	**4.9 %**
EBIT margin (% of sales)	5.9 %	6.7 %	– 0.7 p.p.	– 10.9 %
Interest income	5.8	2.6	3.2	>+100 %
Interest expense	(38.6)	(19.2)	– 19.4	>– 100 %
Earnings before income taxes (EBT)	**143.7**	**151.6**	**– 7.9**	**– 5.2 %**
Income taxes	(51.5)	(71.0)	19.4	27.4 %
Net income	**92.1**	**80.6**	**11.5**	**14.3 %**
Reconciliation to EBITDA				
EBIT	176.4	168.2	8.2	4.9 %
plus depreciation and amortization[2]	(170.9)	(128.5)	– 42.3	33.0 %
EBITDA	**347.3**	**296.9**	**50.4**	**17.0 %**

[1] To aid comparison, income of EUR 27.9 million from the revaluation of existing shares in Netcetera and NetSeT has been reclassified from financial income to cost of sales.
[2] Depreciation and amortization = depreciation and amortization of property, plant and equipment and intangible assets + write-downs on investments in associated companies.

Expenditure in fiscal 2023 included acquisitions made and higher costs for procurement, transport, energy, and staff. Despite macroeconomic challenges, Group sales and all significant earnings metrics saw a further improvement over the previous year. These excellent results reflect G+D's strong organic growth and successful M&A activities.

Although the gross margin declined slightly by 1.2 percentage points year over year, gross profit improved due to a rise of EUR 95.4 million in sales volumes compared to the previous year. The gross margin was adversely impacted by higher materials costs, not all of which could be passed on to customers, and exceptional costs relating to a partially disrupted supply chain. At secunet, a different product mix consisting of a larger proportion of merchandise and a smaller proportion of licenses and services also weighed on the gross margin.

Structural costs for selling, research and development, and general administration rose overall by 15.4 % compared to 2022 and thus at a lower rate than sales. Selling costs totaled EUR 283.4 million, corresponding to a rise of 5.6 % over the previous year. This increase is attributable to both organic and inorganic growth. These costs were partially offset by reductions in impairments of accounts receivable compared to the previous year.

As an innovation-driven company, G+D places great emphasis on R&D. Targeted research activities in new technology areas, digitalization, and sustainable product development allow G+D to strengthen its position as an international player in the SecurityTech sector. Consequently, research and development costs rose by 27.6 % to EUR 169.4 million. This was mainly due to the acquisition of software company Netcetera, whose cost structure includes a significant proportion of R&D.

General administrative expenses increased by 21.3 % to EUR 207.2 million in the reporting year. Almost half of the increase in administrative expenses resulted from acquisitions. Administrative expenses also included costs for post-merger integration projects, other M&A activities, and legal costs.

Other operating income and expenses decreased by EUR 4.3 million, primarily because of lower proceeds from the sale of property, plant and equipment compared to the previous year. Operating profit was slightly above the previous year's level, at EUR 178.8 million.

Financial income was almost neutral at EUR -2.4 million in fiscal 2023 and showed a slight improvement compared to the previous year with an increase of EUR 5.4 million. Heightened exchange rate volatility had an adverse impact on financial income in fiscal 2023, with foreign currency transactions and currency hedging incurring net costs of EUR – 11.5 million (previous year: EUR – 0.2 million). Income from securities was significantly positive, making a EUR 8.1 million contribution to financial income (previous year: EUR – 9.4 million). Investments in companies consolidated at equity came in at EUR 0.9 million (previous year: EUR +1.6 million).

EBITDA was 17.0 % up over the prior-year level, at EUR 347.3 million, marking a record high. The EBITDA contribution of our most recent acquisitions is already firmly positive and a significant factor in the EBITDA increase.

At EUR 176.4 million, EBIT exceeded the record set in the previous year to reach a new all-time high in G+D's history. The EBIT margin was 5.9 %, just 0.7 percentage points below the prior year despite substantial adverse impacts from post-merger integration, volatile financial markets, and an often tight supply chain situation.

Net interest income declined in 2023 by EUR -16.1 million to EUR -32.7 million. This figure comprises interest of EUR – 18.8 million on pension obligations (previous year: EUR -9.6 million) and interest expenses, which increased mainly due to generally upward interest rates on financial and other liabilities of EUR – 19.8 million (previous year: EUR -9.6 million). Interest income totaled EUR 5.8 million (previous year: EUR 2.6 million).

The tax rate was 35.9 %, 10.9 percentage points below the previous year's level. The cause was mainly the fact that the tax rate in the previous year was impacted by provisions for anticipated tax audits, particularly in Slovakia and China. Tax-exempt income from the revaluation of existing shares in Netcetera and NetSeT in 2023 also played a role.

Net income increased by 14.3 % compared to the previous year and stood at EUR 92.1 million in the reporting year.

2.1.2. Investment in Innovation

Ongoing technological advancement drives the company's development and is part of G+D's identity. Accordingly, G+D invested nearly half a billion euros in its innovation capabilities in fiscal 2023 – an unprecedented amount. This substantial investment in innovation underlines G+D's focus on continuous growth and competitiveness and represents a significant milestone in the company's history.

Investment in Innovation

EUR million	2023	2022	Change absolute	Change in %
Total capital expenditure	489.0	340.3	148.7	43.7 %
thereof R&E (not capitalized)	200.2	164.5	35.7	21.7 %
thereof in technology and other investment (incl. capitalized R&D expenditure)	111.2	82.1	29.0	35.3 %
thereof equity interests	177.6	93.7	84.0	89.7 %

2.1.2.1. Research and Development

At EUR 227.5 million, total spending on G+D's own research and development in 2023 was significantly above the already high level of the previous year. This spending was spread across customer-specific development costs (EUR 30.8 million), capitalized research and development costs (EUR 27.3 million), and pure R&D expenditures (EUR 169.4 million). More than half of the increase in R&D expenditures was attributable to our most recent corporate acquisitions. In addition, there was an increase in capitalizable R&D costs and depreciation of capitalized development costs due to our latest acquisitions.

Research and Development

	2023	2022	Change in %
Number of R&D employees (FTE)	1,615	1,304	23.8 %
Proportion of total employees (%)	11.4 %	10.4 %	9.8 %
Spending on R&D (EUR million)	227.5	176.9	28.6 %
thereof pure R&D expenditure (EUR million)	169.4	132.7	27.6 %
R&D ratio (% of sales)	5.7 %	5.3 %	8.5 %
thereof cost of goods sold (EUR million)	30.8	31.8	– 3.2 %
thereof capitalizable costs (EUR million)	27.3	12.4	120.5 %
Capitalization ratio (%)	16.1 %	9.3 %	72.7 %
Amortization of capitalized development costs (EUR million)	25.2	13.4	88.1 %
Number of active patents	7,020	7,104	– 1.2 %
New patent applications	161	123	30.9 %

G+D's tried-and-tested formula for innovation involves investing heavily in its own R&D and core business as well as making targeted acquisitions that enhance and expand the portfolio in business areas with strong potential. Partnerships and investment in start-ups (corporate venturing) are also part of the innovation strategy.

G+D is driving forward the development of cutting-edge integrated hardware, software, and solutions to play a key role in shaping digital transformation. This includes investing in relevant technologies in the digital domain. We are also continuously growing and nurturing our robust, future-proof core portfolio. Consequently, digital and physical components complement each other in both solutions and services. Developing sustainable, climate-friendly products across all business areas is another key priority for G+D. In combination with other measures at the various sites, this contributes towards reducing the company's carbon footprint and helps our customers to achieve their own climate protection objectives.

To further strengthen its innovation capabilities, G+D runs cross-departmental workshops on IP harvesting, together with the newly structured Corporate IP (C-IP) department. A significant increase in patent applications is proof that this approach is delivering results, building on the already excellent figures of previous years. In addition, ideas jointly developed by G+D employees are translated into marketable solutions thanks to efficient, cross-divisional innovation management by the Corporate Technology Office (C-TO). These examples show our strong and consistent commitment to ideation and continuous technological development across all areas of the organization.

In **Currency Technology**, G+D provides trusted, efficient, and highly secure solutions for each step of the cash cycle in both physical and digital ecosystems. As a market leader in end-to-end management of the cash cycle, G+D is further expanding its market presence. Our solutions give customers attractive options for optimizing their value chains. The portfolio includes products and solutions for the entire cash cycle, from the provision of substrates, banknotes, design-based security features, and banknote processing systems to digital solutions for public money. Driven by R&D and investment in market intelligence, the offering is continuously being adapted to address dynamic and emerging market needs. The growing global market trend towards central bank digital currency (CBDC) is a notable example here. Central bank digital currencies offer a wide range of advantages for consumers, central banks, and the entire payments industry. Just like with cash, there are no charges for using well-designed CBDCs, and no data has to be disclosed. Currently, 130 countries are making preparations to introduce such a currency. Around half are at an advanced phase of exploration, i.e. are running a pilot or have already launched a CBDC.

G+D provides a leading, multi-award-winning solution for issuing tokenized money. This includes money issued from central banks in digital form as well as a non-physical commercial bank money token (CBMT). The Bank of Ghana has already piloted a digital currency called the eCedi using the G+D solution. The Central Bank of Eswatini and the Bank of Thailand are also both collaborating with G+D on CBDC initiatives. In fiscal 2023, **advance52**, which coordinates all CBDC activities at G+D, made a significant contribution towards getting Filia® market ready, doing important work on the core system. Key steps here included cooperation with standardization bodies to define security standards for backend systems and wallets, plus further development and optimization of Filia®'s offline payment function. In order to meet current and future security requirements, advance52, which is organizationally attached to the Corporate Center, participates in official research projects like CONTAIN, which was launched by Germany's Federal Ministry of Education and Research in 2023 to improve the effectiveness and efficiency of IT security incident handling procedures.

The **Banknote Solutions division** remains strategically committed to expanding its market and technology leadership. The main areas of focus are optical security solutions for banknotes based on a combination of state-of-the-art micro- and nanotechnology and new technologies around high-security features that offer the ultimate in protection. In this context, a number of joint development projects with leading central banks were continued or newly acquired worldwide. Work on creating a sustainable product portfolio, in which more and more components of the substrate and in manufacturing processes are re-engineered for maximum sustainability, was another key priority. Following the highly successful launch of the Green Banknote in 2022, another product, Green LongLife™, was introduced in 2023 and was likewise very well received by customers.

G+D also provides its customers with comprehensive software and intelligent automation solutions to enable an efficient, resilient, and sustainable cash cycle while remaining committed to the highest security standards. The R&D activities of the **Currency Management Solutions division** focused on expanding the existing portfolio, alongside work on major customer projects. To take one example, cash center workflows need to be fast, smooth, and scalable in accordance with the requirements of the cash cycle. G+D's modular solutions therefore include integrated automation solutions that meet the demands of cash centers of all types and sizes in terms of speed, flexibility, and productivity. The BPS® M evo represents the next generation of high-performance systems for banknote processing and exemplifies the trend.

Research and development activities in the **Mobile Security business sector**, specifically in the **Secure Transactions + Services (STS) division**, enable phygital payments, which comprise a combination of physical and digital payment methods. The development of digital solutions for banks continued and more e-commerce solutions were added to the portfolio, e.g. for tokenization. In addition, G+D has become the first major payment card manufacturer to pledge to replace the virgin plastic in its payment card products with recycled plastic by the end of the decade. The transition process is already under way. G+D is developing sustainable card products and expanding the range of materials that are recyclable, industrially compostable, or biodegradable. We are also strengthening our collaboration and partnerships with customers, joining with them to drive forward customized products that are tailored to specific customer needs. G+D also supports the social need for accessibility. In this context, we are developing cards with tactile elements for people with visual impairments, which help them to recognize more quickly whether they are holding a debit card or credit card, for example. A powerful, secure, and scalable cloud platform was also created to provide customers with services and solutions.

The R&D activities at **Netcetera** are focused on highly scalable products in the financial sector and on cloud operation. This includes payment solutions for card-not-present applications in e-commerce, wallets on mobile devices, web and mobile banking solutions, financial advice portals for insurers and banks, authentication and ID products, and a holistic platform for pension fund management. The company's business-critical applications meet the most demanding requirements with regard to security, reliability, scalability, and ease of use. As well as investing in developing and operating its own products and services, Netcetera is also committed to wider innovation in strategically important technology fields such as artificial intelligence and cloud native technologies. The latter involve designing applications specifically for cloud environments.

In terms of R&D, the **Trusted Connected Devices (TCD) division within the Mobile Security business sector** is particularly successful at developing innovative technological infrastructure for SIM solutions and IoT connections. G+D completed a new, innovative software platform for SIM operating systems, which will serve as a powerful foundation for future development. In addition, existing SIM operating systems were migrated to new semiconductor platforms and the latest standards incorporated. In backend solutions, we focused on expanding system performance for an even faster and more agile response to emerging market requirements. TCD was able to double the volume of eSIMs in the fiscal year and is well prepared for further growth in the years ahead, with February 2023 seeing eSIM downloads hit the 100 million mark. G+D is also currently developing additional services for IoT connectivity based on SIM, eSIM, and iSIM applications. Examples include international roaming agreements and tracking solutions for the transport industry, supported by acquisitions in the respective areas.

At **Veridos**, the portfolio for verifying ID documents, people, and ID-based entitlements and permissions comprises products and solutions ranging from passport readers to complete border control solutions. Work on the VeriCHECK M600 document reader, for example, was completed in 2023 and the device is currently undergoing final certification. In addition to covering physical documents, the portfolio was expanded to include verification of mobile ID documents to meet the requirements of increasing digitalization in the ID field. The border control systems segment includes traditional tools for manual border control as well as products for border control automation. Here, VeriCHECK eGate and VeriCHECK SelfKiosk were further optimized and their functionality expanded. Border control software was also updated in line with evolving international border control requirements. The main area of focus for the Identity Management Solutions (IDMS) unit were the products that enable registration of citizens and support the complete lifecycle of digital and physical IDs. The Veridos PTM (Portfolio and Technology Management) innovation team is tasked with evaluating innovative technologies and concepts, validating them via pilot projects, and translating them into products. The team's current emphasis is on artificial intelligence, biometrics, and digital identities.

The research and development work performed by **secunet** focuses on hardware and software solutions, applications, and architectures in areas with demanding IT security requirements, such as the cloud, the Internet of Things, eGovernment, eHealth, and biometrics. Alongside evolving its existing product portfolio, secunet is also working on new solutions. One of its key projects is a sovereign cloud ecosystem specifically tailored to public authorities and businesses with ultra-high security requirements. secunet also created SINA Mobile, an application suite that can be used to process and transfer sensitive data on standard mobile devices in a secure manner. AI is also an important topic at secunet: the company tests and trains systems to ensure there is no statistical bias in the underlying data used by this technology. With many AI applications, it is particularly important to avoid such undesirable effects. Examples include AI systems that are used at airports for live analysis of facial images to avert threats.

These examples from the various areas show the value and importance of research and development at G+D. We remain committed to driving forward innovation in security technology in the future.

[1] 100 millionth activation since launch of the eSIM.

2.1.2.2. Capital Expenditure

Acquisitions are an important part of G+D's growth strategy. Investment[1] totaled EUR 288.8 million in 2023, significantly exceeding the record prior-year level due to a further increase in M&A activities, which accounted for around 60 % of total investment in 2023.

Investment in other companies (EUR 177.6 million) makes up an increasing proportion of total investment. Having invested more than half of its total spending in equity interests in fiscal 2022, G+D's primary focus in the past fiscal year continued to be expanding the portfolio through outright acquisitions and equity interests. Just under two thirds of the company's total investment was in equity interests. In fiscal 2023, G+D acquired German company Mecomo and Slovenian company Kamino. G+D also invested in a new company, NetLync, and increased its existing stakes in Netcetera, E-Kart, and NetSeT. The second tranche of shares acquired in Netcetera will be recorded as a liability in the current year. This will lead to an outflow of funds in 2024 and is therefore not included in the statement of cash flows for 2023.

Investments in intangible assets and property, plant and equipment (excluding the acquisition of equity interests) were significantly higher than in the previous year and amounted to EUR 111.2 million in the fiscal year. G+D not only made selected investments in new technologies and markets via corporate acquisitions but also increased its own technology investment in IT, hardware, software, and capitalized development work by around 40 % compared to the previous year.

Capital Expenditure and Depreciation/Amortization

EUR million	2023	2022	Change absolute	Change in %
Investment in property, plant and equipment and intangible assets	111.2	82.1	29.0	35.3%
Investment in equity interests	177.6	93.7	84.0	89.7%
Total capital expenditure	**288.8**	**175.8**	**113.0**	**64.3%**
Depreciation/amortization[2]	134.6	103.9	30.6	29.5%

[2] Capital expenditure and depreciation/amortization before IFRS 16.

A focus of investment for the Banknote Solutions division was automation of the banknote finishing process at the Leipzig printing site. In terms of substrates, we extended our competitive lead with the additional purchase of a new system for producing state-of-the-art watermarks. In the security solutions space, the performance of our embossing and laminating machines was improved; another priority was reducing gas consumption. Similarly, a machine for manufacturing optimized fluorescent materials for security applications was commissioned for use in connection with authentication systems. In 2023, the Currency Management Solutions division mainly invested in capitalization of automation development work and BPS® X9 sensors as well as in replacement investment and test machines.

In the STS division, we made investments in relation to Valid, a company we acquired in 2022. This investment was focused on optimizing buildings and bringing machinery up to G+D standards. In addition, there was investment in state-of-the-art production machines for electronic personalization and automation in card production. We also drove forward infrastructure projects relating to buildings, including optimizing floor plans (e.g. in India) and making buildings more environmentally efficient. In addition to capitalization of development work, a project to optimize internal processes was capitalized (digital backbone).

In the TCD division, investment was focused on eSIM/AirOn with the aim of boosting the eSIM solutions business. Furthermore, data center infrastructure was upgraded with regard to processing capacity, speed, and resilience. The aim of this expansion is to future-proof our growth in the digital sector. At Veridos, investment was mainly focused on expanding production capacity at the Greek site and on constructing a new building in the US.

[1] Investment in intangible assets, property, plant and equipment and associated advance payments, and investment in equity interests (including revaluation of venture investments).

2.1.3. Assets and Liabilities

Balance Sheet Summary (IFRS)

EUR million	2023	2022	Change absolute	2023 % of total assets
Assets	**2,931.6**	**2,991.7**	**–60.1**	
Current assets	**1,835.8**	**2,026.6**	**–190.8**	**62.6 %**
thereof inventories	520.8	477.1	43.8	17.8 %
thereof other financial assets	22.9	104.1	–81.2	0.8 %
thereof current receivables	605.3	563.3	42.0	20.6 %
thereof contract assets	266.1	200.3	65.8	9.1 %
thereof cash and cash equivalents	323.5	610.0	–286.5	11.0 %
Non-current assets	**1,095.8**	**965.1**	**130.6**	**37.4 %**
thereof property, plant and equipment	508.0	513.4	–5.4	17.3 %
thereof intangible assets	385.6	246.6	139.0	13.2 %
thereof other non-current assets	202.1	204.9	–2.8	6.9 %
Liabilities and equity	**2,931.6**	**2,991.7**	**–60.1**	
Current liabilities	**1,030.2**	**1,046.8**	**–16.7**	**35.1 %**
thereof current financial liabilities	101.2	113.5	–12.2	3.5 %
thereof current lease liabilities	26.3	22.8	3.4	0.9 %
thereof provisions	73.6	76.8	–3.2	2.5 %
thereof trade payables	314.2	383.0	–68.8	10.7 %
contract liabilities	258.9	244.0	14.9	8.8 %
Non-current liabilities	**1,071.3**	**1,070.8**	**0.5**	**36.5 %**
thereof non-current financial liabilities	483.4	436.2	47.1	16.5 %
thereof non-current lease liabilities	60.4	66.9	–6.6	2.1 %
thereof pensions and similar obligations	420.0	466.2	–46.2	14.3 %
Equity	**830.1**	**874.1**	**–43.9**	**28.3 %**

Current assets decreased by EUR –190.8 million compared to 2022. Due to a significantly higher volume of business, inflation-related price rises, and increased decentralized warehousing (required to ensure delivery ability in a tight logistics situation), inventories increased by EUR 43.8 million compared to the previous year.

Other financial assets fell by EUR –81.2 million. In fiscal 2023, G+D converted its EUR 115.8 million special investment fund, originally disclosed under other financial assets (trading securities), into a Contractual Trust Arrangement (CTA). A CTA is a legal model used in occupational pension programs to manage and fund pension obligations. This model enables G+D to reduce its financial obligations while at the same time ensuring that employee pensions are better protected. As a result, financial assets and pension obligations were each reduced by EUR 115.8 million. Contract assets rose by EUR +65.8 million due to higher stocks of banknotes. These banknotes are scheduled for delivery at the beginning of 2024. Cash and cash equivalents decreased by EUR 286.5 million in fiscal 2023 and now stand at EUR 323.5 million. This significant reduction in funds is mainly a result of the increase in working capital due to sales growth, major investment, particularly in M&A activities, and repayment of our financial obligations. A detailed analysis of the change in free cash flow is provided in section 2.1.4.

As of December 31, 2023, **non-current assets** were above the previous year's level, at EUR 130.6 million. While property, plant and equipment and other non-current assets remained broadly unchanged over the previous year, intangible assets increased by EUR 139.0 million. The main driver of this increase was greater goodwill related to acquisitions made in 2023.

Current and non-current financial liabilities increased by a total of EUR 34.9 million. This included the raising of a promissory note loan of EUR 30 million and repayment of a promissory note loan of EUR 65 million. Financial liabilities also include the increased stake in Netcetera, up from 61.5 % to 95 %, as well as a put option for the remaining 5 % minority shares.

Current and non-current lease liabilities, provisions, and contract liabilities were comparable to the previous year. Accounts payable trade fell by EUR 68.8 million compared to the previous year.

Provisions for pensions were EUR 46.2 million lower than in the previous year. This is partly attributable to the reduction of EUR 115.8 million in pension obligations caused by switching to a Contractual Trust Arrangement (CTA). However, this effect was partly offset by the adjustment of the actuarial interest rate, which fell from an average of 4.2 % to 3.6 %.

The equity ratio decreased to 28.3 %, which was due, among other things, to actuarial losses and the fact that the acquisition of Netcetera in 2023 realized only part of the goodwill (present access method). The equity ratio was thus 0.9 percentage points lower than in the previous year.

Working Capital

EUR million	2023	2022	Change absolute
Inventories	520.8	477.1	43.8
Contract assets	265.6	199.8	65.8
Prepayments on inventories	65.6	49.9	15.8
Accounts receivable trade	542.0	517.5	24.5
Trade payables	– 312.8	– 377.0	64.2
Deposits/contract liabilities	– 304.0	– 288.4	– 15.5
Total	**777.4**	**578.8**	**198.6**

Working capital grew by EUR 198.6 million and stood at EUR 777.4 million. Major drivers behind the increase in working capital included a rise in inventories and contract assets. This involved both a quantity component (higher safety stocks of raw materials and spare parts as well as finished banknotes) and a price component due to in some cases significantly increased purchase prices.

In addition, accounts payable have been significantly reduced in the current fiscal year due to the lower purchasing volume at the end of the year. Working capital was primarily built up in the Currency Technology and Mobile Security business sectors. This was due to two factors. Firstly, high stocks of finished banknotes and cut-off effects at Currency Technology, and secondly, in times of tight supply chains and longer international logistics lead times, Mobile Security in particular has increasingly relied on decentralized warehousing and elevated safety stocks of chips and other intermediate products to ensure it can supply customers at all times.

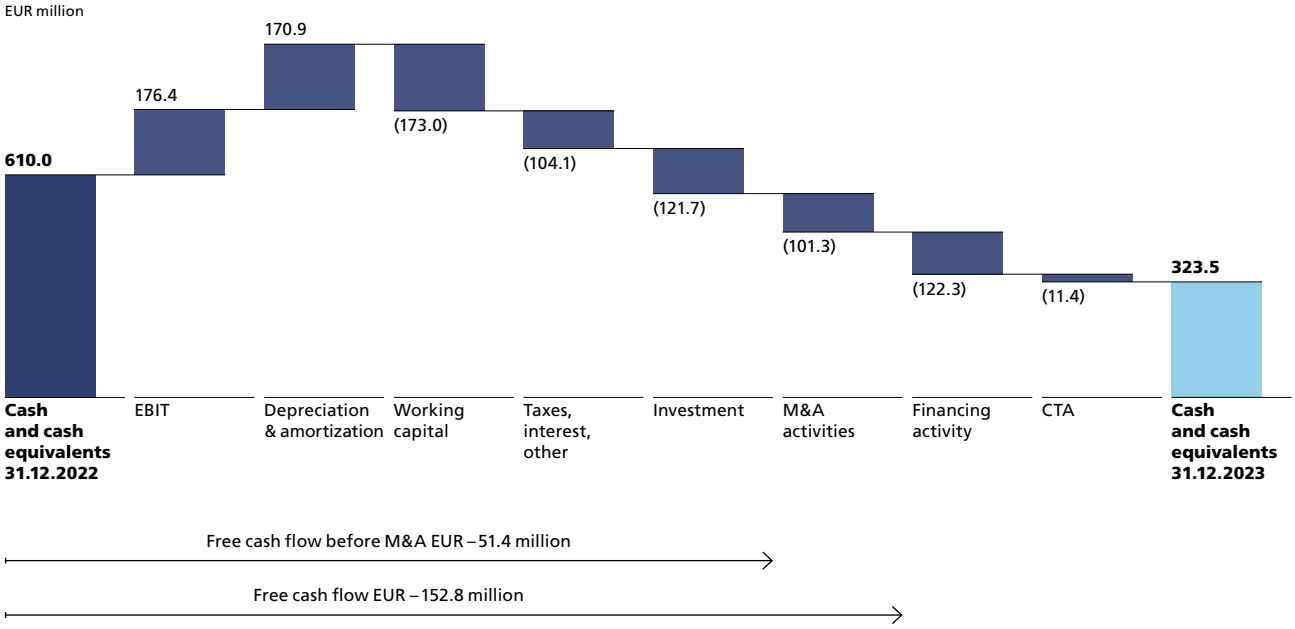
At 11.3 %, **ROCE** was 1.3 percentage points above the prior-year level due to higher capital commitment. Working capital and other capital employed increased significantly as a result of the acquisitions made in fiscal year 2023.

No significant effects are expected from off-balance-sheet commitments. Please see note 31 of the consolidated financial statements in this regard.

2.1.4. Financial Position

Free cash flow was strongly negative in 2023 at EUR –152.8 million. In particular, increased working capital and greater investment led to an outflow of funds compared to the previous year. Free cash flow before M&A activities was also negative at EUR –51.4 million due to high investment, exceptional sales growth, and the resulting build-up of working capital.

Change in Cash and Cash Equivalents



Other operating cash flow includes tax payments and interest paid. Cash flow from investing activities was EUR –223.0 million and included investment in property, plant and equipment and intangible assets (EUR –110.5 million) and acquisitions of EUR 101.3 million[1]. G+D also invested EUR 16.8 million net in securities.

Cash flow from financing activities amounted to EUR –122.3 million and included scheduled repayments on existing bank loans (EUR –91.5 million net). Part of this sum comprises the repayment of a promissory note loan of EUR –65 million from 2018 and a repayment of EUR –9 million to the European Investment Bank (EIB). G+D also replaced external financing of Netcetera and E-Kart with internal funding after its acquisition of majority stakes in these companies. A new promissory note loan of EUR 30 million was also taken out. In addition, lease liabilities were repaid (EUR –27.9 million) and a dividend payment of EUR 24.2 million was made to shareholders.

Please refer to note 13 of the consolidated financial statements for information on approved but unused credit lines in the amount of EUR 588.9 million and on the capital structure.

Cash and cash equivalents decreased by EUR 286.5 million to EUR 323.5 million in 2023. Including unused credit lines, G+D has total financial resources of EUR 912.4 million at its disposal.

[1] The second Netcetera tranche will be paid in early 2024.

2.1.5. Employees

As of December 31, 2023, G+D had a total of 14,203 employees worldwide. The headcount therefore reached a new record high, increasing by 1,609 employees or 12.8 % compared to the previous year. The significant rise in numbers can be attributed to the outstanding growth of the core business and to multiple acquisitions. Adjusted for acquisitions, headcount increased by 4.2 % compared to the previous year.

Number of Employees

FTE at reporting date	2023	2022	Change absolute	Change in %
Production	9,094	8,109	985	12.1 %
Sales	1,465	1,442	23	1.6 %
Research and development	1,615	1,304	311	23.8 %
Administration	2,029	1,738	291	16.7 %
Total	**14,203**	**12,594**	**1,609**	**12.8 %**

Personnel expenses increased to EUR 1,017.0 million (+23.7 %) in the reporting year. Two thirds of the rise in staffing costs is due to higher employee numbers (primarily as a result of M&A activity in Switzerland and the US). The remaining 8 % increase in salaries includes larger bonus payments to employees, the inflation compensation bonus in Germany, higher social security contributions, and increased vacation and overtime provisions.

Collaborative, reliable, and innovative: those are the values that drive us at G+D. Practicing authentic diversity is a key element here. Our promise of "Creating Confidence" applies not just to our products and services but to all employees, including with regard to diversity, equity, and inclusion (DEI). At G+D, we understand diversity as cooperation between people who are very different and unique as a result of their (professional and personal) experiences, values, and backgrounds. This variety of opinions and perspectives enriches us as a company and makes us successful. Diversity has many aspects: alongside origin, age, and personal orientation, to us it also means diversity of thought. Diversity in management and key positions is thus an important factor. G+D has set itself the goal of significantly increasing the proportion of women in management positions in the coming years. If there are more than three senior management positions in a G+D company or unit, at least one of these positions will be occupied by a woman. For global executive management, our aim is to gradually increase the proportion of women to at least 15 % by 2025 and to at least 20 % by 2030. The figure is currently 10 %. Worldwide, the gender balance in line management should, at the local level, correspond to the structure of the workforce locally by 2030.

When making appointments to our Supervisory Board, we ensure that we comply with legal requirements at the very least. The proportion of women on the Supervisory Board is currently 42 %.

3. Corporate Governance

As a global enterprise, G+D is exposed to significant uncertainty and change. The rules that determine how G+D operates in these conditions are defined by corporate governance. Intelligent integration of all elements of corporate governance allows G+D to

- consistently seize opportunities,
- develop appropriate strategies, including making decisions about its portfolio, and
- actively manage necessary risks and minimize the associated impact in a targeted manner.

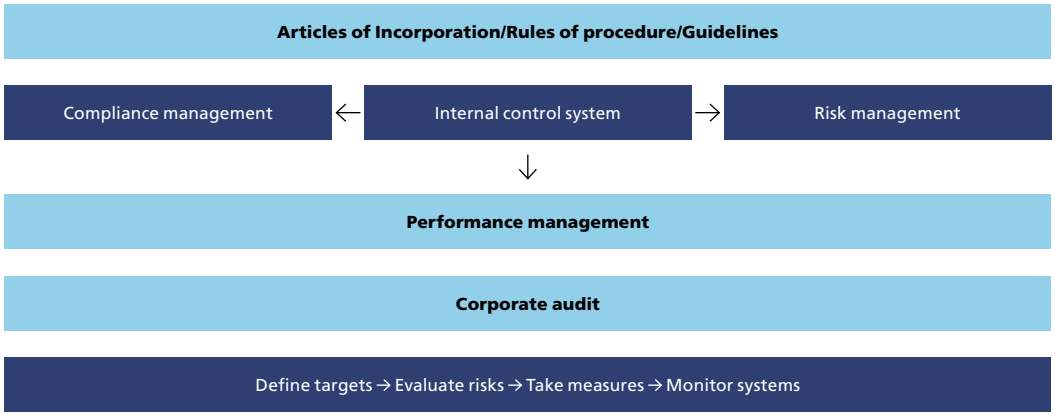
Balancing opportunities and risks safeguards the Group's business development, optimizes financial performance, and increases the company's total value.

The Articles of Incorporation, rules of procedure, and Group-wide guidelines constitute the highest level of corporate governance at G+D. The Advisory Board and Supervisory Board serve as controlling bodies.

Corporate governance at G+D is based on three pillars: compliance management, the internal control system, and risk management. These three pillars operate independently of each other. However, they are closely interrelated because they are directly linked to the performance management system and are coordinated centrally by the Group's Management Board. Individual responsibilities are clearly defined. In addition, as an independent function, Corporate Audit helps the senior management team to objectively monitor compliance with all values and rules. This internal check includes external (legal) requirements and internal rules.

By implementing all these elements, G+D has created a clear and effective organizational and management culture that reflects and also promotes G+D's values (collaborative, reliable, innovative).

Corporate Governance



The **compliance management system (CMS)** supports regulatory compliance across the Group. Among other objectives, the aim is to help prevent corruption and antitrust law violations in business transactions. The CMS thus underpins the confidence of customers and enables them to rely on G+D, while also protecting the company's overall reputation. The Group-wide compliance organization ensures that every employee and all members of the Management Board, Supervisory Board, and Advisory Board at G+D are familiar with compliance requirements and act accordingly. As part of an assessment process, risks to which the business units are exposed with regard to corruption, fraud, antitrust law, and conflicts of interest are reviewed and assessed across all Group entities. These global compliance-risk assessments are conducted regularly, and the results are used to continuously update G+D's CMS in order to fully meet requirements.

In the 2023 reporting year, the G+D Global Compliance Summit was held as an in-person event again for the first time following several pandemic-induced virtual events. Under the motto "Leadership & Integrity," the conference was attended by representatives from the global subsidiaries, all employees from the Compliance Offices, and representatives from the management teams in all business areas. In addition, awareness campaigns were run across the Group to refresh employees' understanding of compliance requirements, regular mandatory training sessions were held for all relevant employees, and compulsory training on the new code of conduct was rolled out globally.

Group-wide compliance is managed centrally by the Group Chief Compliance Officers and the Corporate Compliance department. The Compliance Offices of the various business units in G+D's business sectors and subsidiaries are functionally assigned to the Corporate Compliance department. Compliance managers report on a quarterly basis on activities in the core areas of prevention, detection, and response. G+D management is thus kept informed of the preventive measures that are in place to ensure the organization acts in a compliant manner. Any (potential) compliance violations and the countermeasures taken are also reported. This enables management to address any undesirable developments. A report is submitted to the Supervisory Board on an annual basis.

During the reporting year, the compliance organization was further expanded in terms of staff numbers, and the range of topics covered was extended. The aim is to achieve synergies in Group-wide compliance management, make processes more efficient, and meet the legal requirements of a compliance matrix organization. The CMS is also updated as required to reflect the latest legal developments. In 2023, the focus was on putting a dedicated human rights management system in place to meet the requirements of Germany's Supply Chain Due Diligence Act (Lieferkettensorgfaltspflichtengesetz).

Any issues arising are handled within the business sectors by the relevant Chief Compliance Officers or are dealt with at senior corporate compliance level, as required. Third parties are used where appropriate to investigate and advise on compliance matters. If necessary, issues are also reported on a case-by-case basis directly to the Chairman of the Management Board of the relevant business sector and to the Group CEO.

The **internal control system (ICS)** ensures that business is conducted properly. It defines responsibilities that serve to reduce operational and financial risks for G+D. In the core processes of HR, accounting, finance, order to cash (OTC), purchase to pay- (PTP), and IT security, control mechanisms are implemented, documented, regularly checked, and any gaps are rectified. In addition, care is taken to ensure that legal regulations and internal company policies are adhered to.

The internal control system for financial reporting includes key components such as:

- the Group accounting policy with stipulations on accounting practices and valuation methods,
- the definition of responsibilities for preparing the single-entity and consolidated financial statements and reporting packages,
- ensuring implementation of the dual-control principle for material matters, and
- plausibility checks.

The internal control system thus serves to mitigate risk. There are nonetheless risks inherent in any internal control system, including:

- Human error: despite putting controls and processes in place, there is always a risk of human error, be it as a result of carelessness, misinterpretation, or inadequate training.
- Fraud: an ICS cannot prevent all types of fraud, especially if several employees are involved in a coordinated fraud attempt.
- Technical failures: technology plays a crucial role in most internal control systems. Hardware or software failures, cyber attacks, and data loss can have a significant impact on the ICS.
- Lack of adaptability: an ICS that is not regularly updated and adapted to reflect the changing business environment or regulatory requirements may cease to be effective.

No control system, regardless of its rating, can be completely error-free or immune to all risks. However, a well-designed ICS should help to minimize these risks and enable a fast response to any problems that occur.

A **risk management system** is in place to identify, assess, and manage potential risks. Management of opportunities and risks is an ongoing process. Opportunities and risks associated with business operations are therefore subject to continuous monitoring. The impact of the identified risks on G+D's performance is also analyzed and quantified as part of risk management. Active risk management is thus an integral part of performance management, with its various strategy, planning, and controlling mechanisms.

On behalf of the Management Board, Corporate Audit regularly reviews the effectiveness of Group management and monitoring processes. The focus is on risk management, the internal control system, legal regulations, and internal corporate guidelines.

4. Opportunities and Risk Report

4.1. Risk Management System

The following points have been defined as key elements of an effective and efficient risk management system:

- The focus is on early detection of risks.
- Risks are systematically identified and assessed.
- Where possible, risks are taken on as the result of a conscious decision.
- Risks and measures are assigned to owners.
- Continuous monitoring and reporting of risks and actions is ensured.

Operational and financial risks are addressed on an ongoing basis in the course of day-to-day business management and assessed during quarterly performance reviews. Strategic risks are subject to an annual review as part of the strategy process. Compliance risks are managed by the compliance organization and covered by separate reporting, including notification of Corporate Controlling in the event of financial implications. The information below primarily relates to operational and financial risks.

G+D's risk management system is based on a comprehensive and interactive enterprise risk management approach that is integrated into the global organizational structure. Risk management is organized locally but managed centrally by Corporate Controlling.

All Group employees have a duty to report risks in their immediate environment. With support from the Local Risk Officer in each Group company and drawing on other departments (e.g. Legal, Tax, and Controlling), risks are assessed, and risk/measure owners defined. Risks are monitored by the business sectors and Corporate Controlling. The Management Board reports to the Supervisory Board on a quarterly basis.

4.2. General Risk Analysis and Assessment

The risks identified are evaluated using the gross and net methods. The gross impact is defined as the potential damage that might result if no measures were in place to mitigate the risk. The net impact is the risk remaining when mitigating measures are taken into account. Appropriate measures have a positive impact on the possible damage or on the probability of occurrence associated with the risk. The likelihood of occurrence is multiplied by the net impact to obtain the risk value.

Risk assessment	Net impact = [gross impact] – [measure]
	Risk value = [net impact] × [likelihood of occurrence]

The likelihood of occurrence indicates the estimated probability of the identified risk occurring, which is classified as follows:

Category	Description	Likelihood of occurrence
Very high	Expected to occur	x > 80 %
High	Probable	50 % < x ≤ 80 %
Medium	Not probable	10 % < x ≤ 50 %
Low	Possible, but largely theoretical	x ≤ 10 %

The individual risks are assigned to predefined risk categories (see section 4.4) and the risk values summarized at the level of the business sector and Group.

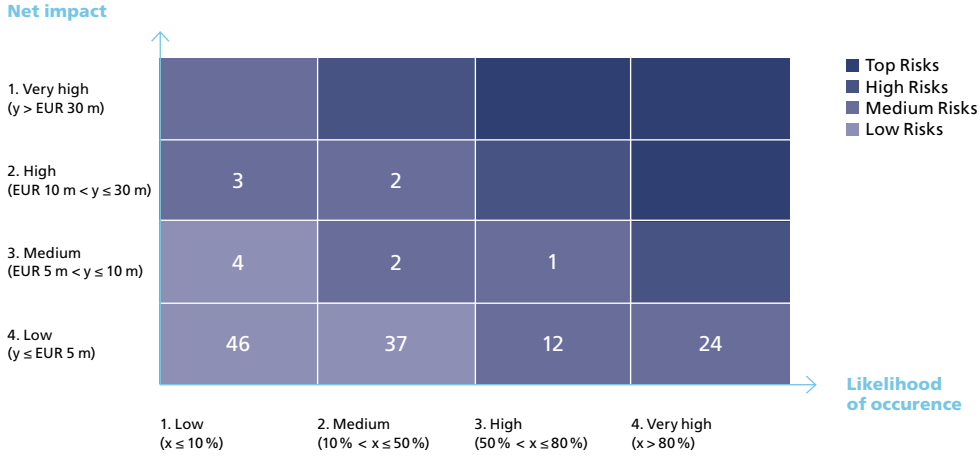
Overall risk is calculated by simple addition of the individual risk values. This is a largely theoretical exercise since simultaneous occurrence of all individual risks can be virtually ruled out due to the different correlations. Aggregate assessment offers the advantage that trends are easier to detect, and conclusions can be drawn regarding risk resilience.

4.3. Summarized Risk Report

As of December 31, 2023, 131 risks had been reported to the Management Board and the Supervisory Board via the risk report. There were thus 40 more risks than on December 31, 2022. Following the acquisition of a majority stake in Netcetera in 2023, risks relating to Netcetera were incorporated into G+D's risk management system for the first time.

The risks were evaluated according to net impact and likelihood of occurrence, as shown in the matrix below:

Risk Categories



In fiscal 2023, as in the previous year, there were no risks in the highest category of the G+D risk matrix, top risks.

In the previous year, there was one risk remaining in the high risks category, which was further reduced in 2023, allowing it to be reassigned to the material risks category as of December 31, 2023. The risk relates to a major project being handled by Veridos.

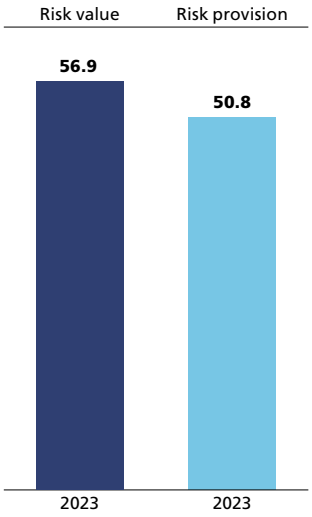
The overall risk value for the Group was EUR 56.9 million and thus virtually unchanged over the previous year.

To manage this risk in financial terms, the Group has made provisions and allowances on receivables or applied corresponding calculation surcharges to major projects totaling EUR 50.8 million. In 2023, financial risk provisioning was in place for 89 % of the risk value.

These risks have been taken into consideration in these financial statements and in the forecast, in accordance with the Group's accounting policy. If the risks for which no provisions have been made due to the low likelihood of occurrence (≤ 50 %) do occur or the risk premiums calculated for major projects turn out to be too low, this would have a negative impact on net assets, the financial position, and results of operations. If the risks covered by provisions occur, there would be a cash outflow.

Risk Value and Risk Provision

EUR million



in Mio. EUR

	Risk value	Risk provision	Δ PY Risk Value	Δ PY Risk Provision
Tax and customs risks	15.5	13.6		
Major project risks	14.0	13.6		
Value creation risks	12.9	14.3		
Product development risks	4.1	3.6		
Market risks	3.4	1.5		
Bad debt risks	2.3	2.0		
Political risks	2.1	1.6		
IT risks	1.2	0.4		
Security risks	0.7	0.0		
Legal risks	0.6	0.1		
Total risks	**56.9**	**50.8**		

Tax and customs risks are assessed as having the highest risk values, followed by risks associated with major projects and value creation risks.

For each major project the overall risk is recorded in the risk register by assessing and aggregating the individual risks accordingly. Overall, the risk assessment of these major projects has improved further as work has progressed. The potential net impact included in the assessment was reduced accordingly. Risk premiums were taken into account when calculating the project costs. Comprehensive risk provisioning is thus already in place.

After careful analysis, the Group-wide risks are not deemed existential in nature, either individually or overall. Due to G+D's strong market positioning, strategic development of the portfolio, capacity for technological innovation, globally standardized processes, and committed employees, the Management Board is confident that the Group is well equipped to meet the challenges posed by these risks in 2024 and to leverage the opportunities that arise.

4.4. Risks and Opportunities by Category

G+D divides risks into the following risk categories: risks associated with major projects, tax and customs risks, value creation risks, product development risks, market risks, bad debt risks, (geo-)political risks, IT risks, security risks, and legal risks.

Risks associated with major projects
G+D is responsible for a number of major projects with high sales potential and lengthy implementation periods spanning several years. These projects have an elevated risk structure, which is taken into account accordingly in risk calculations for the business models. As work on these ongoing major projects progressed, the overall risk assessment in the reporting period improved.

Professional project management and coordination and a dedicated project controlling team have been put in place to support and oversee these major projects. The risks associated with major projects can be successfully addressed and reduced as far as possible through continuous risk management.

Having major projects in the portfolio can help create a long-term order backlog and provide access to new markets. A positive impact on brand perception is another possible benefit. Acceptance and implementation of large projects always involves weighing opportunities and risks.

Tax and customs risks
As a globally active company, G+D's business activities are subject to a wide range of tax-related laws and regulations. These include country-specific tax regimes, possible tax obstacles that make business more difficult, and import/export regulations. Trade barriers in the form of customs duties can lead to negative volume and cost effects, as additional levies become due. Intercompany netting may be challenged by the tax authorities during a tax audit, which can entail lengthy negotiations and the need to produce extensive documentation.

G+D seeks to counter these risks by continually adapting its internal processes to changing requirements. The company also takes advice from tax and auditing companies and law firms in the countries concerned. Where possible, risk exposure is substantially reduced by obtaining binding information from relevant local tax authorities.

Value creation risks
At G+D, supply chain management is handled individually in each of the business sectors. Possible risks may arise as a result of changes to regulatory frameworks, disruption of infrastructure, delays or interruptions to supply chains, an increase in the price of raw materials (e.g. semiconductors or cotton), and higher energy costs. These factors could have adverse effects on the availability, quality, and cost of G+D products and therefore impact sales and earnings. To mitigate these risks, various risk-reducing measures are continuously considered and implemented, such as strategic stockpiling of critical raw materials and supplies in order to avoid bottlenecks and production stoppages.

Undetected quality problems could result in higher costs for G+D and have an adverse impact both on demand for the company's products and on its reputation. This is countered by defining precise quality requirements and ongoing and efficient development of the quality management system, with a corresponding focus on customer needs. Our underlying approach is that "quality is everybody's business." Making this attitude a reality requires processes, organizational interfaces, tasks, and responsibilities to be clearly defined and communicated in quality handbooks and work instructions. Each employee therefore needs to be fully aware of the contribution they can make within their role.

Production could also be compromised by technical failures. Additional production capacity may be maintained as one way of minimizing this risk. G+D seeks to ensure optimum machine utilization and back-up capacity by means of detailed production planning and management. Machines that are outdated or no longer meet the latest technical standards could lead to a loss of production capacity, resulting in partial or complete failure to produce the planned quantities. Problems of this kind can result in project delays or late delivery of products to end customers. If G+D is late in delivering products, it could face contractual penalties for failing to comply with delivery deadlines. Prompt investment in replacement machinery is intended to prevent such issues. Capital expenditure is managed by the business sectors at G+D and closely monitored by the project controlling team.

In a dynamic market environment, G+D strikes a balance between effectively meeting the current needs of customers and investing in promising new products and solutions.

Increased vertical and horizontal expansion of value-adding activities offers additional opportunities. Global procurement management (global sourcing) is an important strategy that contributes to reduced costs. Dependence on particular suppliers is reduced and access to special resources is secured. This could strengthen G+D's competitive position, with a corresponding positive impact on net assets, the financial position, and results of operations. In turn, this would allow an upward revision of existing forecasts. Continuous monitoring is essential in order to exploit potential opportunities arising from global sourcing. This also allows the resulting new risks to be identified at an early stage and countered by appropriate action.

Product development risks
G+D offers its customers high-quality products and solutions. Development of new products, or product features, and solutions is based on market analysis. Misinterpretation of this analysis or delays in launching products could result in higher costs and have an adverse effect on demand. In new product development in particular, deadline compliance is crucial in order to meet customer requirements and avoid additional costs during the subsequent commercialization phase. In addition to these risks, intellectual property rights must be protected, licensed, or acquired as part of R&D activities. The Group's patent department works with external law firms to register and monitor patents. Legal disputes regarding violations of intellectual property may result in payment of compensation for damages and a ban on using certain technologies.

Maintaining an extensive vertical and horizontal portfolio of products and solutions allows the Group to diversify risk and take advantage of market opportunities.

With its products and solutions, G+D protects essential assets for its customers and makes the daily lives of billions of people more secure. New and improved security technologies are required in order to safeguard these assets and protect customers against cyber threats. Accordingly, these business areas will continue to offer huge potential for G+D going forward. Alongside product innovation, increased efficiency made possible by process improvements can create competitive advantages.

Market risks
G+D is subject to various market risks. Changes in exchange rates and interest rates significantly affect operational business as well as investment and financing activities. If necessary, derivative financial instruments are used in relation to foreign currency and interest rates to hedge underlying transactions. The purpose of these financial instruments is to reduce risks stemming from fluctuations in exchange rates and interest rates and to increase planning security. In accordance with risk management standards applying to international banks, all trading activity is subject to monitoring that is independent of the Group's Finance department.

For further details on financial risks, including sensitivity analyses, please see note 22 of the consolidated financial statements.

Bad debt risks
Like all market players, G+D is subject to typical liquidity risk and counterparty credit risk. Liquidity risk is the risk of being unable to meet financial commitments as they fall due. This is minimized by means of a carefully considered financing policy. If contract partners do not meet commitments when due or collateral loses value, this is referred to as counterparty credit risk. The Group primarily manages these risks as part of its ongoing business and financing activities, based on written policies. Bad debt risks form part of the risk reports submitted to the Management Board and are included in regular reporting to the Supervisory Board and Advisory Board.

If, contrary to expectations, it becomes possible for business partners to settle the outstanding receivables, this could lead to a reversal of derecognition. This would positively impact net assets, the financial position, and results of operations by the corresponding amount.

Political risks
Any escalation of the trade-related dispute between China and the US and other (geo-)political tensions could lead to a significant deterioration in global growth prospects. Political decisions could result in the introduction of further trade barriers at the international level.

G+D counters these risks by continually monitoring economic and political developments in the key markets. Production and capital expenditure are managed centrally, enabling a rapid response in the event of any economic slowdown. At the same time, G+D benefits from having a local presence. The strong regional focus of the sales organization allows G+D to recognize and address changing customer requirements at an early stage. G+D's proximity to its customers gives it an advantage here, with fast response times allowing it to seize emerging business opportunities as they arise.

IT risks
The availability of internal IT systems is essential for G+D's ability to do business. In the age of digitalization, IT infrastructure and IT applications are becoming ever more important because all the company's business activities are based on IT systems. This dependence entails huge potential for damage. Workflows are becoming more complex. In addition, dependency on information and communication technologies and on systems designed to harmonize processes is growing. G+D counters these risks by continually adapting its IT systems to changing requirements and ensuring redundancy of critical IT systems. In addition to availability, legally compliant use of licenses is also a possible risk area. Active license management is in place to counter this risk.

As digitalization progresses, new digital business segments are being created in the IT area. Considerable sales potential could be opened up through the development and provision of digital solutions as well as in the field of connectivity. This could result in new sales opportunities for G+D.

Security risks
G+D is not completely immune to cyber security risks, whether in the form of economic or industrial espionage, cyber attacks, or common non-specific attacks, such as ransomware. Targeted attacks may be carried out by certain countries and potentially by competitors, or a mix of these (countries or criminal organizations acting on behalf of, or in support of, competitors). Non-specific (opportunistic) attacks are mainly carried out by actors associated with organized crime.

Harm could be caused by unintentional disclosure of confidential information or intellectual property, product damage (e.g. through loss of IT system integrity), supply difficulties as a result of loss of production, or compromised (personal) data. Individuals may also gain unauthorized access to buildings or systems and misuse or steal information, products, or materials. Damage to plant and equipment is also possible.

G+D has taken a range of preventive measures to minimize these risks. They include various technical IT security measures, organizational and procedural security measures (e.g. identity management, monitoring of IT processes, staff security measures), and physical security measures (e.g. entry and access control systems, camera and alarm systems, site security). Regular training sessions are also conducted to raise the awareness of all employees around the various security issues.

G+D has thus implemented a security and control system that makes it possible to identify risks promptly and respond in a timely fashion. Through stringent security management, G+D has systematically developed a targeted, robust culture of security and prevention. This competence enables G+D to set itself apart from other companies with lower security levels and allows it to operate as a trusted business partner.

Legal risks
When selecting external partners, alongside considering quality aspects, care must be taken to ensure that they abide by internal rules and applicable laws and regulations. During and after the end of a contractual relationship, legal disputes may arise, in which claims could be brought against G+D or the costs of a legal defense may be incurred. Risks arising in connection with legal disputes are continuously monitored. Risks may also arise from business relationships as a result of possible violation of export control law or through warranty periods and warranty conditions.

With regard to data protection, the requirements of the GDPR must be met. Breaches of data protection law or the loss of sensitive data can lead to fines or reputational damage. G+D's competitive position may also be negatively impacted. Employees receive regular awareness training on this subject.

New legal requirements can open up opportunities for G+D, with a corresponding improvement in sales and earnings. For example, if the use of a technology that G+D is involved in developing or that is contained in its products were to be made mandatory, internal forecasts could be revised upward.

Environmental issues are becoming increasingly important for businesses. As part of its sustainability activities, G+D is committed to developing sustainable products and minimizing the impact of its business operations on the environment and the resulting risks. A certified environmental management system provides the basis for these endeavors. Further information is available in the G+D sustainability report. Having a positive reputation for sustainability provides G+D with an opportunity to boost its attractiveness as a business partner and employer. This could have positive effects on financial metrics as well as on employer attractiveness.

5. Overall Assessment of Economic Situation and Forecast

5.1. Overall Assessment of Economic Situation

2023 was a year of very strong growth for G+D. The company posted new record figures for sales and earnings at the end of the fiscal year and also tightened its positioning as a security tech company. All business sectors contributed to the strong upturn in sales during the past year. The big increase in orders and healthy order backlog highlight the potential for future growth.

Currency Technology exceeded its predicted sales. The Banknote Solutions division and Currency Management Solutions both exceeded sales forecasts.

Mobile Security significantly outperformed the previous year's expectations. In particular, the Secure Transactions + Services (STS) division closed out the year with a new sales record, significantly exceeding the budgeted figures. This growth was largely attributable to payment cards and card issuance services. The Trusted Connected Devices (TCD) division beat its budget targets with integrated SIM cards.

Veridos significantly surpassed sales expectations, with growth being driven by high demand for ID documents.

secunet continued its impressive growth trajectory, with sales hitting another record high at the end of 2023. secunet thus substantially outstripped sales forecasts and boosted its sales for the tenth year in a row.

G+D significantly exceeded its target for operating profit, EBIT and EBITDA. Increased EBIT also meant that the predicted ROCE was exceeded.

Due to higher sales than anticipated, working capital was above target, primarily as the result of higher inventories and contract assets.

Investment activity significantly exceeded the target due to unbudgeted acquisitions. Non-M&A-related investments in our own activities were also slightly above budget and were not reprioritized in favor of acquisitions.

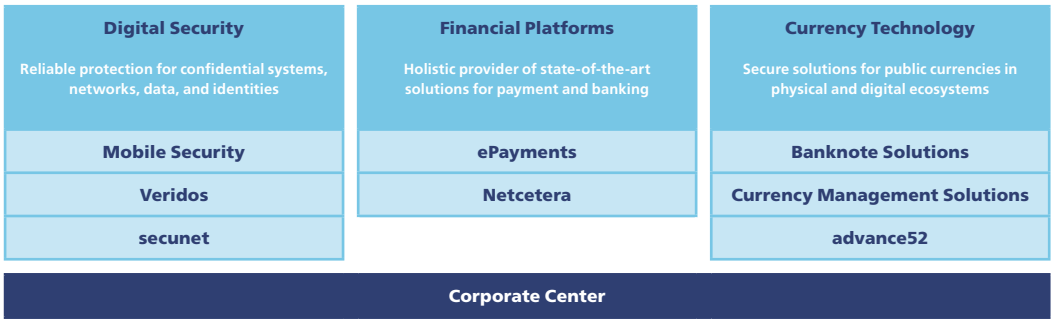
Due to high investment and the associated increase in working capital, free cash flow fell well short of the planned target.

5.2. G+D in Evolution: Successful Corporate Transformation

G+D ended fiscal 2023 with record sales. Reaching the EUR 3 billion mark means that the company has entered a new phase in its development. G+D has experienced significant and sustained growth in recent years. Our portfolio has become broader, more international, and more diversified through organic growth and M&A activities. This is also reflected in the company's strong digital transformation. We are also implementing an orderly generational change in the Management Board and Supervisory Board and continuing to strengthen our market position in the security tech field.

As a global security tech company, G+D will build on three equally strong pillars in the future: Digital Security, Financial Platforms, and Currency Technology. G+D is thus focusing on its core business. In all of our segments, across our physical and digital solutions, integrated security technology is a vital part of our value proposition for customers and the wider public. G+D Mobile Security (formerly TCD) and G+D ePayments (formerly STS) will operate as legally independent units from fiscal 2024.

New management structure in the segments



In the **Digital Security** segment, G+D protects and manages confidential systems, networks, data, and identities in physical, digital, and mobile environments. This segment includes Mobile Security, Veridos, and secunet. The **Financial Platforms** segment provides highly secure, state-of-the-art solutions for payment, banking, and other financial services, based on human-centric security technology. Financial Platforms includes ePayments and our new Netcetera division. The **Currency Technology** segment provides a comprehensive portfolio for the physical and digital ecosystems of official currencies. Currency Technology includes the Banknote Solutions division, the Currency Management Solutions division, and advance52.

Alongside the new positioning, G+D is also implementing a systematic and orderly generational change in the Management Board and Supervisory Board. After 23 years as Group Chief Financial Officer (CFO), Dr. Peter Zattler will retire after overseeing the annual financial statements for 2023. His successor, Jan Thyen, will take over on May 1, 2024. Mr. Thyen has been with the company since 2010 and is currently Head of Corporate Strategy. In addition to the change of CFO, the Management Board of the G+D Group is being expanded to include a third managing director. The Supervisory Board has appointed Gabriel von Mitschke-Collande as Group Chief Digital Officer (CDO) effective May 2024 in order to further drive forward digitalization of both the Group's business portfolio and its corporate structures and processes. Gabriel von Mitschke-Collande is a member of G+D's shareholder family. He joins G+D from BayWa AG, where he likewise held the position of CDO.

In the future, the Management Board of G+D will therefore comprise CEO Dr. Ralf Wintergerst, Jan Thyen, and Gabriel von Mitschke-Collande. With a forward-looking portfolio and tightened positioning as a SecurityTech company, G+D is in a strong position to meet the challenges of the future.

5.3. Forecast

After very strong sales growth of 18 % in fiscal 2023, consolidation and post-merger integration are on the agenda for the current year. We anticipate sales growth of 2.5 %, adjusted for exchange rate effects. The Group will top the EUR 3 billion mark in 2024.

After extremely strong growth by Digital Security in fiscal 2023, we are expecting sales that roughly match the high level of the previous year. Financial Platforms will likely fall slightly short of the exceptionally high sales volume seen in the previous year. Netcetera, which is part of the Financial Platforms segment, will focus on its portfolio of solutions with the aim of ensuring sustained positive earnings performance but will also seek to compensate for the decline in sales triggered by last year's bank merger in Switzerland. The Currency Technology segment expects to achieve sales growth in 2024.

The sales forecast is supported by the order backlog, which grew significantly by EUR 465 million to EUR 2,021 million in the 2023 reporting period.

However, we expect EBIT and EBITDA to come under pressure due to the ongoing rising cost of materials and logistics in the current geopolitical situation, and also as a result of the anticipated wage settlements and associated significant increases in personnel costs. In addition, function costs will be impacted by rising R&D expenditure and by the project launched at the start of the year to migrate the global ERP system to S4/HANA technology.

In the 2024 planning year, EBITDA will be below the prior-year figure, which was bolstered by the revaluation of shares in Netcetera, NetLync, and E-Kart. However, depreciation and amortization will be reduced by decreasing effects over time from the purchase price allocation related to acquisitions and increases in stakes in 2023, which will improve EBIT. Changes in the mix from physical products to the digital solutions business will also reduce depreciation and amortization. EBIT is expected to remain at roughly the prior-year level.

A slight decline in the Group's annual net income is expected due to increased interest expenses and a relative Group tax rate at the previous year's level. In fiscal 2024, we intend to exceed Giesecke+Devrient GmbH's net income of EUR 30 million, calculated in accordance with the German Commercial Code (HGB). Working capital will be significantly reduced compared to the 2023 reporting year; this will be supported by a cross-segment working capital program in 2024.

In 2023, G+D's investment activity, including M&A, reached a record level. Overall, we intend to invest less in the coming year because no new acquisitions are currently included in our planning, apart from venture investments. Despite this planned reduction, we will expand our investment activity before M&A. We will continue to focus on investing in technology. This includes IT investment and capitalized research and development expenditure.

Free cash flow before M&A will be firmly positive again in fiscal 2024. Both the contribution to operating income and the reduction in working capital, which was significantly increased in 2023, will have a positive impact on this KPI. All segments will make a positive contribution to free cash flow.

ROCE is expected to be slightly below the level achieved in 2023. Despite the improvements in working capital, capital employed will rise in 2024 due to acquisitions and increased stakes, only half of which were taken into account in the 2023 reporting year. Nevertheless, potential exists due to the planned reduction of working capital and this may counteract the increase in capital employed.

In addition, two non-financial KPIs will become important performance indicators as of fiscal 2024, i.e. CO_2e emissions and the proportion of women in management roles. CO_2e emissions amounted to 33,356 metric tons of CO_2e for scope 1 and 2 in fiscal 2023. We intend to come in below this figure in fiscal 2024. We plan to significantly increase the proportion of women in global executive management in fiscal 2024.

Actual results may, of course, vary from expected performance.

Consolidated Financial Statements

as of December 31, 2023

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Independent Auditor's Report

To Giesecke+Devrient Gesellschaft mit beschränkter Haftung, Munich

Opinions

We have audited the consolidated financial statements of Giesecke+Devrient Gesellschaft mit beschränkter Haftung, Munich, and its subsidiaries (the Group) – which comprise the consolidated statement of financial position as of December 31, 2023, the consolidated income statement, the consolidated statement of comprehensive income, the consolidated statement of changes in equity and the consolidated statement of cash flows for the financial year from January 1, 2023, to December 31, 2023, and notes to the consolidated financial statements, including the accounting policies presented therein. In addition, we have audited the group management report of Giesecke+Devrient Gesellschaft mit beschränkter Haftung for the financial year from January 1, 2023, December 31, 2023.

In our opinion, on the basis of the knowledge obtained in the audit,

- the accompanying consolidated financial statements comply, in all material respects, with the IFRSs as adopted by the EU, and the additional requirements of German commercial law pursuant to Section 315e (1) HGB [Handelsgesetzbuch: German Commercial Code] and, in compliance with these requirements, give a true and fair view of the assets, liabilities, and financial position of the Group as of December 31, 2023, and of its financial performance for the financial year from January 1, 2023 to December 31, 2023, and
- the accompanying group management report as a whole provides an appropriate view of the Group's position. In all material respects, this group management report is consistent with the consolidated financial statements, complies with German legal requirements and appropriately presents the opportunities and risks of future development.

Pursuant to Section 322 (3) sentence 1 HGB [Handelsgesetzbuch: German Commercial Code], we declare that our audit has not led to any reservations relating to the legal compliance of the consolidated financial statements and the group management report.

Basis for the opinions

We conducted our audit of the consolidated financial statements and of the group management report in accordance with Section 317 HGB and the German Generally Accepted Standards for Financial Statement Audits promulgated by the Institut der Wirtschaftsprüfer (IDW) [Institute of Public Auditors in Germany]. Our responsibilities under those requirements and principles are further described in the "Auditor's Responsibilities for the Audit of the Consolidated Financial Statements and of the Group Management Report" section of our auditor's report. We are independent of the group entities in accordance with the requirements of German commercial and professional law, and we have fulfilled our other German professional responsibilities in accordance with these requirements. We believe that the evidence we have obtained is sufficient and appropriate to provide a basis for our opinions on the consolidated financial statements and on the group management report.

Responsibilities of Management and the Supervisory Board for the Consolidated Financial Statements and the Group Management Report

Management is responsible for the preparation of consolidated financial statements that comply, in all material respects, with IFRSs as adopted by the EU and the additional requirements of German commercial law pursuant to Section 315e (1) HGB and that the consolidated financial statements, in compliance with these requirements, give a true and fair view of the assets, liabilities, financial position, and financial performance of the Group. In addition, management is responsible for such internal control as they, in accordance with German Legally Required Accounting Principles, have determined necessary to enable the preparation of consolidated financial statements that are free from material misstatement, whether due to fraud or error.

In preparing the consolidated financial statements, management is responsible for assessing the Group's ability to continue as a going concern. They also have the responsibility for disclosing, as applicable, matters related to going concern. In addition, they are responsible for financial reporting based on the going concern basis of accounting unless there is an intention to liquidate the Group or to cease operations, or there is no realistic alternative but to do so.

Furthermore, management is responsible for the preparation of the group management report that, as a whole, provides an appropriate view of the Group's position and is, in all material respects, consistent with the consolidated financial statements, complies with German legal requirements, and appropriately presents the opportunities and risks of future development. In addition, management is responsible for such arrangements and measures (systems) as they have considered necessary to enable the preparation of a group management report that is in accordance with the applicable German legal requirements, and to be able to provide sufficient appropriate evidence for the assertions in the group management report.

The Supervisory Board is responsible for overseeing the Group's financial reporting process for the preparation of the consolidated financial statements and of the group management report.

Auditor's Responsibilities for the Audit of the Consolidated Financial Statements and of the Group Management Report

Our objectives are to obtain reasonable assurance about whether the consolidated financial statements as a whole are free from material misstatement, whether due to fraud or error, and whether the group management report as a whole provides an appropriate view of the Group's position and, in all material respects, is consistent with the consolidated financial statements and the knowledge obtained in the audit, complies with the German legal requirements and appropriately presents the opportunities and risks of future development, as well as to issue an auditor's report that includes our opinions on the consolidated financial statements and on the group management report.

Reasonable assurance is a high level of assurance, but is not a guarantee that an audit conducted in accordance with Section 317 HGB and in compliance with the German Generally Accepted Standards for Financial Statement Audits promulgated by the Institut der Wirtschaftsprüfer (IDW) will always detect a material misstatement. Misstatements can arise from fraud or error and are considered material if, individually or in the aggregate, they could reasonably be expected to influence the economic decisions of users taken on the basis of these consolidated financial statements and this group management report.

We exercise professional judgment and maintain professional skepticism throughout the audit. We also:

- Identify and assess the risks of material misstatement of the consolidated financial statements and of the group management report, whether due to fraud or error, design and perform audit procedures responsive to those risks, and obtain audit evidence that is sufficient and appropriate to provide a basis for our opinions. The risk of not detecting a material misstatement resulting from fraud is higher than for one resulting from error, as fraud may involve collusion, forgery, intentional omissions, misrepresentations, or the override of internal controls.
- Obtain an understanding of internal control relevant to the audit of the consolidated financial statements and of arrangements and measures (systems) relevant to the audit of the group management report in order to design audit procedures that are appropriate in the circumstances, but not for the purpose of expressing an opinion on the effectiveness of these systems.
- Evaluate the appropriateness of accounting policies used by management and the reasonableness of estimates made by management and related disclosures.
- Conclude on the appropriateness of management's use of the going concern basis of accounting and, based on the audit evidence obtained, whether a material uncertainty exists related to events or conditions that may cast significant doubt on the Group's ability to continue as a going concern. If we conclude that a material uncertainty exists, we are required to draw attention in the auditor's report to the related disclosures in the consolidated financial statements and in the group management report or, if such disclosures are inadequate, to modify our respective opinions. Our conclusions are based on the audit evidence obtained up to the date of our auditor's report. However, future events or conditions may cause the Group to cease to be able to continue as a going concern.
- Evaluate the overall presentation, structure and content of the consolidated financial statements, including the disclosures, and whether the consolidated financial statements present the underlying transactions and events in a manner that the consolidated financial statements give a true and fair view of the assets, liabilities, financial position and financial performance of the Group in compliance with IFRSs as adopted by the EU and the additional requirements of German commercial law pursuant to Section 315e (1) HGB.
- Obtain sufficient appropriate audit evidence regarding the financial information of the entities or business activities within the Group to express opinions on the consolidated financial statements and on the group management report. We are responsible for the direction, supervision and performance of the group audit. We remain solely responsible for our opinions.
- Evaluate the consistency of the group management report with the consolidated financial statements, its conformity with [German] law, and the view of the Group's position it provides.
- Perform audit procedures on the prospective information presented by management in the group management report. On the basis of sufficient appropriate audit evidence we evaluate, in particular, the significant assumptions used by management as a basis for the prospective information, and evaluate the proper derivation of the prospective information from these assumptions. We do not express a separate opinion on the prospective information and on the assumptions used as a basis. There is a substantial unavoidable risk that future events will differ materially from the prospective information.

We communicate with those charged with governance regarding, among other matters, the planned scope and timing of the audit and significant audit findings, including any significant deficiencies in internal control that we identify during our audit.

Munich, March 22, 2024

KPMG AG
Wirtschaftsprüfungsgesellschaft

[Original German version signed by:]

Hachmann	**Höfelschweiger**
Wirtschaftsprüfer	Wirtschaftsprüferin
[German Public Auditor]	[German Public Auditor]

Consolidated income statement

for the years ended December 31, 2023 and 2022

EUR million	Note	2023	2022
Net sales	15	2,973.3	2,527.1
Cost of goods sold		(2,163.0)	(1,784.2)
Gross profit		**810.3**	**742.9**
Selling expenses		(278.5)	(246.6)
Research and development expenses		(169.4)	(132.7)
General and administrative expenses		(207.2)	(170.8)
Impairments on trade receivables and contract assets		(4.8)	(21.6)
Other operating income/(expenses), net		0.5	4.8
Operating profit		**150.9**	**176.0**
Share in earnings from equity investments	6	0.9	1.6
Other financial income/(expenses), net	17	24.6	(9.4)
Earnings before interest and income taxes		**176.4**	**168.2**
Interest income	18	5.9	2.6
Interest expense	18	(38.6)	(19.2)
Earnings before income taxes		**143.7**	**151.6**
Income taxes	19	(51.6)	(71.0)
Net income		**92.1**	**80.6**
thereof apportioned to non-controlling interests		4.8	4.0
thereof apportioned to the shareholders of Giesecke+Devrient GmbH		87.3	76.6
		92.1	**80.6**

The accompanying notes to the financial statements are an integral part of these statements.

Consolidated statement of comprehensive income

for the years ended December 31, 2023 and 2022

EUR million	Note	2023	2022
Net income		**92.1**	**80.6**
Other comprehensive income			
Items that will never be reclassified to the income statement			
Actuarial gains and losses	14	(49.1)	218.2
Deferred taxes on actuarial gains and losses	19	14.7	(67.6)
Fair value change in financial assets recorded in other comprehensive income		(3.6)	–
		(38.0)	**150.6**
Items that are or may be reclassified to the income statement			
Currency effects from the translation of foreign operations		(10.5)	5.4
Effective part of fair value changes in cash flow hedges		(1.6)	2.9
Deferred taxes fair value changes in cash flow hedges		0.5	(0.8)
Share of income and expenses recognized directly in equity resulting from the use of the equity method of accounting		(0.4)	0.7
		(12.0)	**8.2**
Other comprehensive income, net of tax		**(50.0)**	**158.8**
Total comprehensive income		**42.1**	**239.4**
thereof apportioned to non-controlling interests		2.0	6.6
thereof apportioned to the shareholders of Giesecke+Devrient GmbH		40.1	232.8
		42.1	**239.4**

The accompanying notes to the financial statements are an integral part of these statements.

Consolidated balance sheet

as of December 31, 2023 and 2022

EUR million	Note	Dec 31, 2023	Dec 31, 2022
ASSETS			
Current assets			
Cash and cash equivalents		323.5	610.0
Financial assets	2	22.9	104.1
Accounts receivable trade and other accounts receivable, net	3	605.3	563.3
Inventories, net	4	520.8	477.1
Income taxes receivable		45.4	23.1
Other current assets	5	51.8	48.8
Contract assets	23	266.1	200.3
Total current assets		**1,835.8**	**2,026.7**
Non-current assets			
Investments accounted for under the equity method	6	19.4	45.5
Investments in other related parties	6	18.7	12.6
Financial assets	2	20.7	25.3
Accounts receivable trade and other accounts receivable, net	3	14.5	13.6
Intangible assets	7	385.6	246.6
Property, plant and equipment	8	508.0	513.4
Deferred tax assets	19	106.6	88.6
Income taxes receivable		0.4	2.5
Other non-current assets		20.3	15.4
Contract assets	23	1.6	1.5
Total non-current assets		**1,095.8**	**965.0**
Total assets		**2,931.6**	**2,991.7**

The accompanying notes to the financial statements are an integral part of these statements.

EUR million	Note	Dec 31, 2023	Dec 31, 2022
LIABILITIES AND EQUITY			
Current liabilities			
Accounts payable trade and other accounts payable	10	314.2	383.0
Provisions	11	73.6	76.8
Financial liabilities	13	101.2	113.5
Lease obligations	9	26.3	22.8
Accrual for income taxes and income taxes payable		41.2	32.0
Other current liabilities	12	214.8	174.7
Contract liabilities	23	258.9	244.0
Total current liabilities		**1,030.2**	**1,046.8**
Non-current liabilities			
Accounts payable trade and other accounts payable	10	0.7	–
Provisions	11	18.3	16.6
Financial liabilities	13	483.4	436.2
Lease obligations	9	60.4	66.9
Pensions and related liabilities	14	420.0	466.2
Deferred tax liabilities	19	35.0	27.0
Other non-current liabilities		10.5	19.5
Contract liabilities	23	43.0	38.4
Total non-current liabilities		**1,071.3**	**1,070.8**
Equity			
Capital stock	20	25.8	25.8
Additional paid-in capital	20	29.5	29.5
Retained earnings	20	758.9	795.4
Accumulated income and expenses recognized directly in equity		5.8	19.1
Treasury stock	20	(60.1)	(60.1)
Non-controlling interests		70.2	64.4
Total equity		**830.1**	**874.1**
Total liabilities and equity		**2,931.6**	**2,991.7**

The accompanying notes to the financial statements are an integral part of these statements.

Consolidated statement of changes in equity

for the years ended December 31, 2023 and 2022

EUR million

	Capital stock	Additional paid-in capital	Retained earnings
Balance as of January 1, 2022	**25.8**	**29.5**	**594.3**
Net income	–	–	76.6
Other comprehensive income	–	–	148.3
Total comprehensive income	**–**	**–**	**224.9**
Total changes in equity	**–**	**–**	**224.9**
Contributions from non-controlling interests	–	–	–
Dividends paid	–	–	(23.8)
Balance as of December 31, 2022	**25.8**	**29.5**	**795.4**
Net income	–	–	87.3
Other comprehensive income	–	–	(33.9)
Total comprehensive income	**–**	**–**	**53.4**
Present access method for business acquisitions (see Note 24 “Business Combinations”)	–	–	(74.4)
Reduction of minority interests (see Note 24 “Business combinations”)	–	–	8.7
Total changes in equity	**–**	**–**	**(12.3)**
Dividends paid	–	–	(24.2)
Balance as of December 31, 2023	**25.8**	**29.5**	**758.9**

The accompanying notes to the financial statements are an integral part of these statements.

	Accumulated income and expenses recognized directly in equity resulting from currency translations	Reserves from changes in fair value	Accumulated income and expenses resulting from cash flow hedges	Treasury stock	**Subtotal**	Non-controlling interests	**Total**
	13.5	**(2.3)**	**–**	**(60.1)**	**600.7**	**50.3**	**651.0**
	–	–	–	–	76.6	4.0	80.6
	6.1	–	1.8	–	156.2	2.6	158.8
	6.1	**–**	**1.8**	**–**	**232.8**	**6.6**	**239.4**
	6.1	**–**	**1.8**	**–**	**232.8**	**6.5**	**239.4**
	–	–	–	–	–	18.0	18.0
	–	–	–	–	(23.8)	(10.5)	(34.3)
	19.6	**(2.3)**	**1.8**	**(60.1)**	**809.7**	**64.4**	**874.1**
	–	–	–	–	87.3	4.8	92.1
	(8.6)	(3.6)	(1.1)	–	(47.2)	(2.8)	(50.0)
	(8.6)	**(3.6)**	**(1.1)**	**–**	**40.1**	**2.0**	**42.1**
	–	–	–	–	(74.4)	18.6	(55.8)
	–	–	–	–	8.7	(8.7)	–
	(8.6)	**(3.6)**	**(1.1)**	**–**	**(25.6)**	**11.9**	**(13.7)**
	–	–	–	–	(24.2)	(6.1)	(30.3)
	11.0	**(5.9)**	**0.7**	**(60.1)**	**759.9**	**70.2**	**830.1**

Consolidated statement of cash flows

for the years ended December 31, 2023 and 2022

EUR million	2023	2022
Cash flows from operating activities		
Earnings before interest and income taxes	176.4	168.2
Adjustments to reconcile earnings before interest and income taxes to cash provided by operations		
(Increase)/decrease in inventories, net	(27.0)	(107.8)
Increase/(decrease) in accounts payable trade and other accounts payable	(72.1)	41.5
(Increase)/decrease in prepayments	(6.4)	(79.5)
(Increase)/decrease in accounts receivable trade and other accounts receivable, net	(10.8)	(1.6)
(Increase)/decrease in contract assets	(66.4)	68.7
Increase/(decrease) in contract liabilities	9.8	72.6
Change in working capital	**(172.9)**	**(6.1)**
Depreciation, amortization and impairment/recoveries	170.9	128.5
(Gain)/loss on sale of investments, net	(2.9)	–
(Gain)/loss on sale and disposal of intangible assets and property, plant and equipment	(0.3)	(4.6)
(Gain)/loss on the revaluation of old shares in associated companies	(27.9)	–
Undistributed earnings in associated companies and joint ventures excluding dividend payments	(0.9)	(1.6)
Dividends received from associated companies and joint ventures	3.6	0.8
(Increase)/decrease in investments in trading securities	–	8.0
(Increase)/decrease in prepaid expenses and other assets	(5.9)	(14.1)
Increase/(decrease) in provisions	(3.0)	(32.9)
Increase/(decrease) in pensions and related liabilities	(20.3)	(10.5)
Increase/(decrease) in other liabilities	38.1	1.1
Income taxes paid, net	(70.5)	(76.9)
Interest received	5.8	2.6
Interest paid	(19.9)	(9.2)
Changes in other operating activities	**66.8**	**(8.8)**
Net cash provided by operating activities	**70.3**	**153.3**

EUR million	2023	2022
Cash flows from investing activities		
(Increase)/decrease in short-term investments	(4.2)	51.4
Additions to and prepayments on intangible assets	(35.5)	(23.3)
Additions to and prepayments on property, plant and equipment	(75.0)	(60.3)
Acquisition of shares in associated companies and joint ventures	(5.9)	–
Capital increase in investments in related companies	(7.4)	(1.8)
Acquisitions, net of cash acquired	(88.0)	(90.4)
Proceeds from the sale of available-for-sale securities	5.3	21.3
Purchase of available-for-sale securities	(22.1)	(20.0)
Loans to shareholders	(3.0)	–
Loans to related parties	–	(5.2)
Payments received on loans to related parties	5.5	20.0
Proceeds from sale of investments	5.0	–
Proceeds from sale of intangible assets	0.6	0.4
Proceeds from sale of property, plant and equipment	1.7	7.7
Net cash used in investing activities	**(223.0)**	**(100.2)**
Free Cashflow[1]	**(152.7)**	**53.1**
Cash flows from financing activities		
Proceeds from issuance of long-term debt	63.6	223.2
Proceeds from issuance of long-term debt from the Giesecke+Devrient Foundation	15.0	–
Repayment of long-term debt	(118.4)	(17.9)
Repayment of long-term debt from the Giesecke+Devrient Foundation	(21.0)	–
Payments on lease obligations	(27.9)	(23.9)
Net (decrease)/increase in short-term debt and borrowings	(2.3)	(0.5)
Dividends paid to shareholders	(24.2)	(23.8)
Cash received from non-controlling interests	–	10.0
Dividends paid to non-controlling interests	(7.1)	(10.5)
Net cash provided by/used in financing activities	**(122.3)**	**156.6**
Effect of exchange rates on cash and cash equivalents	(11.4)	(0.8)
Net increase/(decrease) in cash and cash equivalents	(286.4)	208.9
Cash and cash equivalents at beginning of the year	610.0	401.1
Cash and cash equivalents at end of the year	**323.5**	**610.0**

[1] Free cash flow consists of net cash provided by operating activities less net cash used in investing activities.

The accompanying notes to the financial statements are an integral part of these statements.

Notes to the consolidated financial statements

for the years ended december 31, 2023 and 2022

(in EUR million, except where otherwise stated)

1. Summary of Significant Accounting Policies and Practices

A Description of Business

Giesecke+Devrient Gesellschaft mit beschränkter Haftung and subsidiaries ("G+D" or "Giesecke+Devrient") is in the business of printing banknotes and securities, as well as the development and production of security paper and currency automation equipment. Giesecke+Devrient also develops and manufactures magnetic stripe cards and smartcards mainly for the telecommunications, banking and health services industries. A further field of business includes security-related solutions for governments and public authorities, ranging from ID cards and travel documents to e-government solutions. New technologies comprise network solutions and secure mobile transaction solutions as well as a software system for mobile devices.

Giesecke+Devrient, headquartered in Prinzregentenstraße 161, 81677 Munich, Germany, is entered in the Commercial Register of the Munich District Court Dept. B under the number 4619. G+D has a strong international orientation with Germany being one of its major markets. Other key markets include the United States, Canada and China. As of December 31, 2023, G+D had subsidiaries in 39 countries and 14,203 employees (for simplification purposes the term "employee/employees" represents individuals of all genders) worldwide, including 9,329 outside Germany.

The consolidated financial statements were approved by the Management Board on March 22, 2024.

B Basis of Presentation

The consolidated financial statements as of December 31, 2023 have been prepared in accordance with International Financial Reporting Standards (IFRS) as adopted by the EU, and the applicable supplementary German statutory requirements in accordance with Section 315e no. 1 German Commercial Code (HGB).

MC Familiengesellschaft mbH was founded in 2012. MC Familiengesellschaft mbH based in Munich became the Group parent company and prepared statutory consolidated financial statements in accordance with IFRS as of December 31, 2023.

Some figures may not precisely add up in total due to rounding differences.

C Consolidated Group and Principles of Consolidation

Consolidated Group
All material G+D subsidiaries, joint ventures and associated companies are included in the consolidated financial statements.

Affiliated companies are companies that are controlled by the Group. The Group controls a company if it is exposed to or has rights to variable returns from its involvement in the company and has the ability to affect the amount of these returns by using its power. Financial statements of subsidiaries are included from the time the Group obtains control and ceases when the Group loses control. Non-controlling interests are valued at the respective share of the net assets of the company acquired that can be identified at the date of acquisition. Changes in the ownership interest in a subsidiary that do not result in loss of control are accounted for as equity transactions.

Group interests that are accounted for in accordance with the equity method comprise shares in associated companies and joint ventures. Associated companies are companies in which the Group has significant influence but does not control or jointly control with respect to financial and business policies. A joint venture is an arrangement whereby the Group has joint control of the arrangement and has rights to the net assets instead of rights to the assets and obligations for the liabilities of the arrangement.

The consolidated Group comprises 29 domestic and 82 foreign subsidiaries which are fully consolidated. Giesecke+Devrient has had a holding structure since January 2017, in which the divisions are fully consolidated as legally independent subgroups. As Giesecke+Devrient has more than half of the voting rights in Veridos Matsoukis S.A. Security Printing, Athens at each level, management has determined that G+D controls the company. This assessment is on the basis that G+D owns the majority of the voting rights in Veridos GmbH, Berlin, which in turn holds the majority of the voting rights in Veridos Matsoukis S.A. Security Printing, Athens. Additionally, eleven joint ventures and / or associated companies are accounted for under the equity method.

Principles of consolidation
The financial statements of the companies included in the consolidated financial statements are prepared using uniform accounting policies in accordance with IFRS.

Income and expenses, receivables, payables and provisions, as well as intragroup profits between companies included in the consolidated financial statements are eliminated.

A subsidiary is deconsolidated from the date it is no longer controlled by G+D.

Investments in joint ventures and associated companies accounted for using the equity method are initially recognized at cost and adjusted accordingly in subsequent periods. Intragroup profits from transactions with these companies are eliminated in proportion to the acquirer's interest.

Under IFRS, all business combinations are accounted for using the acquisition method. The acquirer allocates the cost of a business combination by recognizing the acquiree's identifiable assets, liabilities, and contingent liabilities that satisfy the recognition criteria at their fair value on the date control over the entity is obtained (acquisition date). The full amounts of identifiable assets and liabilities and contingent liabilities irrespective of the company's ownership interest are recognized at their fair values. Any excess of the purchase price over the fair value of the identifiable assets, liabilities, and contingent liabilities less any minority interests is recognized as goodwill. Where the fair value exceeds the purchase price, the resulting amount is recorded in the income statement.

Non-controlling interests are measured at the fair value of the proportionate identifiable net assets. In a business combination achieved in stages, interests held at the time of transfer of control are revalued and the resulting gain or loss is recognized in profit or loss. An adjustment of conditional purchase price components that were reported as a liability at the acquisition date is recognized in profit or loss for business combinations. Transaction costs are recognized as expenses at the time they are incurred.

After having gained control of a subsidiary, the difference between the purchase price and the proportionate share of equity for additional shares acquired is charged against retained earnings. Transactions which do not result in loss of control have no impact on the income statement and are recorded as equity transactions.

Remaining interests are measured at fair value at the time of loss of control. In the case of non-controlling interests, reporting negative balances are permitted, i.e. future losses are allocated in proportion to the participation without restriction.

D Use of Estimates

Preparation of the financial statements requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent amounts and liabilities as of the date of the financial statements and reported amounts of revenues and expenses during the reporting period.

Information about estimation uncertainties and where critical judgment in applying accounting policies that have the most significant effect on the amounts recognized in the consolidated financial statements and those through which a considerable risk can arise or a material adjustment will be required within the fiscal year ending on December 31, 2023 is included in the following notes:

- Note 1 (j) "Goodwill and Other Intangible Assets"
- Note 1 (n) "Provisions"
- Note 19 "Income Taxes"
- Note 24 "Business Combinations"

Interpretation 23 of the IFRS Interpretation Committee (IFRIC) clarifies the application of the recognition and measurement under International Accounting Standards (IAS) 12 if there is uncertainty regarding the income tax treatment. Estimates and assumptions must be made for recognition and measurement, e.g. whether an assessment is made independently or together with other uncertainties, whether a probable or expected value is used for the uncertainty and whether changes have occurred compared to the previous period. The risk of detection is irrelevant for the accounting of uncertain balance sheet items. The accounting is based on the assumption that the tax authorities are investigating the matter in question and that they have all the relevant information.

E Foreign Currency Translation

Transactions in foreign currency are translated into euros using the exchange rate on the date of the transaction. At the balance sheet date, monetary assets and liabilities are remeasured using the period-end exchange rate. Non-monetary assets and liabilities denominated in foreign currency are translated using the historical exchange rates as of the date of the transaction.

The individual functional currency for each of the Group companies is the currency in the primary economic environment in which the entity operates. The assets and liabilities of foreign subsidiaries with functional currencies other than the euro are translated using period-end exchange rates, while the revenues and expenses are translated using average exchange rates during the period. Differences arising from the translation of assets and liabilities in comparison with the translation of the prior periods are included in cumulative translation adjustment and reported as a separate component of equity.

The average and closing rates for significant currencies for the fiscal years ended December 31 are as follows:

1 euro equals X units of foreign currency	Rates – December 31, 2023		Rates – December 31, 2022	
	Average	Closing	Average	Closing
US dollar – USD	1.0816	1.1050	1.0538	1.0666
British pound – GBP	0.8700	0.8691	0.8528	0.8869
Chinese renminbi – RMB	7.6589	7.8509	7.0799	7.3582
Canadian dollar – CAD	1.4595	1.4642	1.3703	1.4440
Indian rupee – INR	89.3252	91.9090	82.7087	88.5950
South African rand – ZAR	19.9532	20.3477	17.2100	18.0986
Swiss francs – CHF	0.9717	0.9260	1.0052	0.9847

F Financial Instruments

A financial instrument is a contract that gives rise to a financial asset in one entity and a financial liability or equity instrument in another entity.

Financial assets include, in particular, cash and cash equivalents, accounts receivable trade, loans, other receivables, marketable securities, and derivative financial instruments.

For regular-way purchases and sales of all categories of financial assets, with the exception of derivative financial instruments, the date of initial recognition in the balance sheet or of derecognition is the settlement date, i.e. the date on which an asset is delivered to or by an entity. The trade date is determinant for derivative financial instruments.

Financial liabilities include accounts payable trade, liabilities to financial institutions, lease obligations, and derivative financial liabilities.

Financial assets and liabilities are generally measured at fair value at initial recognition. Accounts receivable trade that do not have a significant financing component are initially recognized at their transaction price. The fair value is the estimate of the price at which an orderly transaction to sell an asset or to transfer a liability would take place between market participants at the measurement date under current market conditions (i.e. to estimate an exit price). This is the price within a defined market which could be achieved in selling an asset or should be paid for a liability.

After initial recognition, financial assets are classified at either amortized cost, at fair value through other comprehensive income (FVOCI) or at fair value through profit or loss (FVTPL) under IFRS 9. A financial asset is derecognized when the contractual rights to the cash flows relating to the financial asset expire, that is, when the asset is realized, forfeited or is no longer under the control of the company. G+D did not record interest income on impaired financial assets.

The Group designates the spot element of forward foreign exchange contracts to hedge its currency risk and applies a hedge ratio of 1:1. The forward elements of forward exchange contracts are excluded from the designation of the hedging instrument. The Group's policy is to align the critical terms of the forward exchange contracts with the hedged item.

The Group assumes that the credit risk on a financial asset, unless specified otherwise, has increased significantly if it is more than 30 days past due. Lastly, the Group considers that its cash and cash equivalents based on the external ratings of the selected banks and financial institutions and the internal monitoring and limits per institutions ensures a low default risk ("low credit risk exemption").

The creditworthiness of a financial asset is impaired when one or more events occur that adversely affect expected future cash flows. Indicators for this are, for example, significant financial difficulties of the debtor or the probability that the debtor will go into insolvency or other restructuring proceedings.

The Group considers a financial asset to be in default if the debtor is unlikely to pay its credit obligations in full to the Group without recourse by the Group with actions such as realizing collaterals (if any are held) or the financial asset is more than 90 days past due.

The Group defines a financial asset based on trade receivables as default by 360 days or more past due.

Derivative financial instruments
Derivative financial instruments are used to manage the foreign currency exposure incurred in the normal course of business in the form of forward exchange contracts. In addition, interest rate swaps are used to hedge the interest rate risk on long-term loans with variable interest rates.

G+D has made use of the option to continue applying hedge accounting requirements of IAS 39.

Hedging of major projects by applying Cash Flow Hedge Accounting (CFH)
Currency risks from contracts with a nominal volume exceeding EUR 8.0 million are generally secured and verified in applying and presenting cash flow hedge accounting in the balance sheet.

In these individual transactions, fluctuations in earnings are avoided by the pro rata temporis accounting of the valuation effects of the derivatives in equity.

The Group designates the spot element of forward foreign exchange contracts to hedge its currency risk and applies a hedge ratio of 1:1. The forward elements of forward exchange contracts are excluded from the designation of the hedging instrument and are separately accounted for as hedging costs in financial result, shown analogously to free-standing derivatives. The Group's policy is to align the critical terms of the forward exchange contracts with the hedged item.

There are stringent requirements related to the underlying transaction in applying cash flow hedge accounting. They are constantly monitored at inception date of the project as well as during the applicable period. If, during the term of the project, the criteria are no longer met, the cash flow hedge account is dissolved and any effects are continuously reported in profit and loss. The commercial part remains in force.

In connection with the prospective effectiveness test, the Group specifies that there is a high balance between expected cash flows initiated by the hedging instrument and the hedged underlying transaction based on currency, amount and term of the corresponding cash flows (critical terms match). Retrospectively, the Group assesses whether the hedging instrument of the designated derivate was offset by the changes of cash flows of the secured transaction were effective using the hypothetical derivative approach.

In these hedging relationships, sources of ineffectiveness can arise from:
- the effect of the counterparties' and the Group's own credit risk on the fair value of the forward foreign exchange contracts, which is not reflected in the fair value change of the hedged cash flows attributable to the change in exchange rates; and
- mainly based on changes in the value of the hedged transactions.

Hedging of major projects by applying Fair Value Hedge Accounting (FVH)
G+D does not otherwise apply hedge accounting in managing foreign currency exposure. These derivative financial instruments therefore qualify as held-for-trading and are recorded at fair value at the balance sheet date as either an asset or a liability. Changes in fair value are recognized in the income statement as financial income or expense. The fair market values of forward exchange contracts are calculated on the basis of the applicable spot market rates as well as the forward contract premium or discount compared to the contracted forward contract rate.

Giesecke+Devrient identifies derivative instruments embedded in host contracts of financial liabilities and accounts for them separately in accordance with the provisions of IFRS 9 Financial Instruments: Recognition and Measurement. Financial derivatives that are embedded in financial contracts of financial assets are recognized at fair value. These derivatives consist solely of foreign currency derivatives embedded in certain firm sales and purchase contracts denominated in a currency that is neither the functional currency of G+D nor of the contractual counterparty and which is also not a currency in which transactions are commonly denominated in the jurisdiction in which the transaction is to occur.

The fair values of the external and embedded foreign currency derivatives are recorded as current financial assets and liabilities in the balance sheet.

Hedging of interest rate risk
Giesecke+Devrient has entered into interest rate hedging transactions for long-term bank loans. A hedging relationship was designated between the loan and the interest rate swaps and accounted for in the form of a cash flow hedge.

Cash and cash equivalents/Short-term investments
Giesecke+Devrient considers all highly liquid investments with an original maturity of three months or less to be cash equivalents. These are valued at amortized cost.

Highly liquid commercial paper with an original maturity up to three months is also classified as cash and cash equivalents and is measured at fair value.

Short-term investments with a duration between three months and one year are classified as current financial assets.

Cash and cash equivalents as of December 31, 2023 and December 31, 2022 include cash and bank balances in Russia with restrictions amounting to EUR 0.9 million and EUR 0.6 million, respectively.

Accounts receivable trade and other accounts receivable, net
Accounts receivable trade and other accounts receivable, net are allocated to the category "at amortized cost". They are measured at their transaction price at the time of initial recognition. The valuation at subsequent balance sheet dates at G+D is carried out on the basis of the business model for managing receivables and characteristics of contractual cash flows using amortized cost.

For doubtful accounts, credit risk impairments in the form of specific allowances are carried out. Specific defaults lead to derecognition of the receivables affected. In addition, according with IFRS 9, lifetime expected credit losses are calculated on a collective basis on the remaining balance on accounts receivable trade third party balances not subject to specific allowances. G+D first records impairments on an individual basis and then on a collective basis on the remaining balance of estimated credit losses in accordance with IFRS 9. Allowances on accounts receivable trade and other accounts receivable are recorded in separate allowance accounts.

The Group uses the simplified approach to calculate expected credit losses on accounts receivable trade using a provision matrix that includes an analysis of historical data over the past five years and current observable and future orientated data. Default risks within each default risk rating are segmented based on industry. Based upon the analysis of historical data as well as reasonable and supportable information regarding accounts receivable more than one-year past-due, G+D has assumed rebuttable rates equal to 90.0 % for all subgroup HUBs. The analysis therefore assumes a non-rebuttable rate of 10.0 % – being those receivables not expected to be recovered. The non-rebuttable rate is then reduced by a recovery rate of 25.0 % which represents the write-off portion expected to be collected in the event of insolvency proceedings. The calculation of collective allowances for the individual G+D companies incorporates the customer payment patterns which have been derived from the average ageing of accounts receivable trade third parties over the last five years as of December 31.

Income and expenses in connection with the recognition and reversal of specific allowances and allowances on a collective basis, as well as direct derecognitions of receivables are recorded in a separate line item in the income statement as a result of IFRS 9. Non- and low-interest-bearing non-current receivables are recorded at the present value of the expected future cash flows when the interest impact is material. For such amounts, the subsequent valuation is made applying the effective interest method. Assets are classified as non-current when the remaining duration at the balance sheet date exceeds 12 months.

Marketable securities and investments
G+D's marketable securities are classified as trading securities or as held-to-collect and sell securities. The classification under IFRS 9 is based on the business model in which a financial asset is managed and its contractual cash flow characteristics. IFRS 9 contains three general classification categories for financial assets: measured at amortized cost, at fair value through other comprehensive income (FVOCI) and at fair value through profit or loss (FVTPL).

G+D has designated the financial assets to the business model "Other" and therefore measures them at fair value through profit or loss. Investment funds classified as held-to-collect and sell do not meet the cash flow criteria and are therefore measured at fair value through profit or loss.

The equity instruments include investments in other related parties. These include the investments in Brighter AI Technologies GmbH with 7.8 % of the shares, FINANCIAL NETWORK ANALYTICS LTD with 6.5 % of the shares, Nano Corp. with 5.7 % of the shares, Yields NV with 5.6 % of the shares, Tenzir GmbH with 5.3 % of the shares, IDnow GmbH with 5.0 % of the shares, SALV TECHNOLOGIES OÜ with 4.3 % of the shares, Blindflug Studios AG with 4.1 % of the shares, Cognism Ltd with 1.2 % of the shares and Verimi with 0.5 % of the shares,. These investments are measured at fair value through other comprehensive income (FVOCI) and are shown in a separate line item in the balance sheet "Investments in other related parties". Trading securities include shares in a closed and fully consolidated special fund which invests in publicly traded equity and debt securities. These trading securities are measured at fair value through profit or loss as determined by the most recently traded price of each security at the balance sheet date. Highly liquid commercial paper with an original maturity of up to three months is classified as cash and cash equivalents and is measured at amortized cost.

Unrealized gains and losses on trading securities and held-to-collect and sell securities (investment securities) are included in net income on a current basis.

If, in a subsequent period, the fair value increases and the increase can be objectively related to an event occurring after the impairment loss was recognized, the impairment loss shall be reversed in the income statement.

Other financial assets
With the exception of derivative financial instruments, other financial assets recognized as assets are allocated to the measurement category "at amortized cost". The valuation is in accordance with the explanation provided for accounts receivable trade and other receivables, net. Impairments on financial assets are recognized using the impairment model for expected credit losses. An impairment is reversed when the reasons for the impairment recorded no longer prevail.

Financial liabilities
With the exception of derivative financial instruments, financial liabilities recorded as liabilities are allocated to the measurement category "financial liabilities measured at amortized cost". The initial valuation of these financial liabilities is at fair value and in subsequent periods at amortized cost using the effective interest method. Transaction costs are deducted from the acquisition costs to the extent to which they are directly attributable. Liabilities are classified as non-current when the remaining maturity as of the balance sheet date exceeds 12 months.

The valuation of accounts payable trade is in accordance with the procedures noted previously for financial liabilities.

A financial liability is derecognized when the underlying obligation relating to the liability is fulfilled, terminated or extinguished.

Giesecke+Devrient has not made use of the option to designate financial liabilities as financial liabilities measured at fair value through profit or loss at the time of initial recognition in the balance sheet.

Financial instruments that give rise to a right for the holder to repay the capital made available to the company must be classified as liabilities in accordance with IAS 32. In the case of companies in the legal form of a commercial partnership, the partner may demand repayment of the capital provided by exercising a statutory right of termination which cannot be excluded by the articles of association. This regulation also includes other comparable termination rights with severance agreements of minority shareholders. Liabilities from shares in commercial partnerships and minority interests with comparable termination rights are measured at amortized cost. Changes in liability due to valuation are recognized in net interest income in the consolidated income statement.

G Risk Management and Foreign Currency Exposure Policies

Risk management for the entire Group is coordinated centrally. Policies for risk management, foreign currency exposure, and documentation requirements are set forth in guidelines and procedures issued by the corporate treasury department. These guidelines are examined and updated on a regular basis. The approval of the guidelines is the responsibility of management.

Derivative financial instruments are used by G+D solely to reduce the risks inherent within its global business. As such, Giesecke+Devrient does not hold or issue derivative financial instruments for speculative purposes.

Refer to Note 22 “Financial Risk” and section 4.3 of the Group management report, “Summarized Risk Report” for additional related disclosures.

H Inventories

Inventories are carried at cost. Cost is determined using the weighted average, FIFO (first-in first-out) or standard cost method. Finished goods and work-in-progress inventories include direct material, labor, and manufacturing overhead costs which are based on the normal capacity of the production facilities. Items in inventory are written down at the balance sheet date if their net realizable value is lower than their carrying amount.

I Non-current Assets Held for Sale

Non-current assets are classified as held for sale if they are available for immediate sale in their present condition, subject only to terms that are usual and customary for sales of such assets and their sale is highly probable. They are measured at the lower of their carrying amount and fair value less costs to sell.

J Goodwill and Other Intangible Assets

Intangible assets consist of purchased intangible assets, such as standard software, licenses, patents, water rights, know-how, customer base, brands, goodwill, and internally developed intangible assets.

Intangible assets with definite useful lives are valued at cost and are amortized on a straight-line basis over their estimated economic useful lives.

Development costs are capitalized when the requirements of IAS 38 "Intangible Assets" are fulfilled. Capitalized development costs recognized include production costs less accumulated depreciation and impairments. Production costs comprise direct material and personnel costs as well as directly attributable material and manufacturing overhead costs and production-related depreciation. Such costs are amortized on a straight-line basis over the estimated economic useful lives. Research costs are expensed in the period in which they are incurred.

The useful lives of intangible assets with definite useful lives are generally as follows:

	Years
Capitalized development costs/Technology	3 – 10
Software, rights, customer base, brands etc.	2 – 15

Goodwill is not amortized but rather tested at least annually for impairment. Reversals of impairments on goodwill are not permitted.

At least once a year, Giesecke+Devrient evaluates the recoverability of goodwill at the cash-generating unit (CGU) level or group of CGUs applying a one-step impairment test. Where the recoverable amount (value in use equal to the present value of future cash flows) of the CGU or group of CGUs, to which the goodwill was allocated, is less than the carrying amount, an impairment loss is recognized. If the impairment loss exceeds the goodwill of the CGU, the excess is allocated to the other assets (generally property, plant and equipment and intangible assets) of the CGU or group of CGUs pro rata on the basis of the carrying amount of each asset.

The most critical assumptions in the calculation of the fair value less costs to sell and the calculation of the value in use are based upon include estimated growth rates, weighted average capital costs and tax rates. Such assumptions, as well as the underlying methodology, can materially influence the respective values and therefore impact the determination of a potential impairment of the goodwill. Where property, plant and equipment and intangible assets are included in the goodwill impairment tests, the determination of the recoverable amount is based on management estimates.

K Property, Plant and Equipment

Property, plant and equipment are valued at cost less accumulated depreciation. Depreciation of property, plant and equipment is calculated on a straight-line basis over the estimated economic useful lives of the assets. Depreciation of an asset commences once it has been placed in service.

The cost of self-constructed property, plant and equipment comprises the direct cost of materials and direct manufacturing expenses plus appropriate allocations of material and manufacturing overheads as well as production and output-related general and administrative costs.

Acquisition or manufacturing costs also include estimated dismantling and removal costs as well as costs relating to the restoration of the location to its original state.

Any investment allowances or grants received reduce the acquisition or manufacturing costs of the assets for which they were granted.

If an item of property, plant and equipment is comprised of several components with differing useful lives, the separate components are depreciated over their individual useful lives. Expenses for the day-to-day repair and maintenance of property, plant and equipment are normally charged against income.

Estimated economic useful lives of G+D's property, plant and equipment are as follows:

	Years
Buildings	up to 50
Technical equipment and machinery	2 – 25
Other plant and office equipment	2 – 23

L Impairment of Intangible Assets and Property, Plant and Equipment

Impairment of other intangible assets and items of property, plant and equipment is identified by comparing the carrying amount with the recoverable amount (the higher of fair value less costs to sell and value in use). If no future cash flows generated independently of other assets can be allocated to the individual assets, recoverability is tested on the basis of the cash-generating unit to which the assets can be allocated. With the exception of goodwill, impairment losses are reversed if the reasons for recognizing the original impairment loss no longer apply.

M Leasing

When concluding an agreement, the Group determines whether such an agreement is or contains a lease in accordance with the regulations of IFRS 16.

IFRS 16 defines a lease as a contract that conveys the right to control the use of an identified asset for a period of time in exchange for consideration. The contract conveys the right to control the use of an identified asset if the lessee has the right to obtain substantially all of the economic benefits from the use of the identified asset and has the right to direct the use of that asset.

Applying IFRS 16, as a lessee the Group records right-of-use assets and lease obligations in the amount of the present value of the payment obligations for all identified leases. According to the regulations of IFRS 16 the Group does not distinct between operating and finance leases anymore. No right-of-use assets or lease obligations are recorded in the balance sheet for short-term leases with a lease term of up to twelve months as well as leases for which the underlying asset is of low value (the original price of the underlying asset is below 5,000 EUR). The lease payments from these contracts are recorded as expenses in the amount of the monthly lease payments. The lease components of a contract are accounted for separately from the non-lease components on the basis of the stand-alone selling price.

Applying IFRS 16, the principal portion of lease payments on lease obligations is shown in the cash flows from financing activities. Corresponding interest payments as well as lease payments for short-term leases and leases for which the underlying asset is of low value are still recorded in the cash flows from operating activities.

Lease payments are discounted using the interest rate implicit in the lease. If that rate cannot be readily determined, the Group's incremental borrowing rate is used for discounting.

At initial recognition, right-of-use assets are measured at cost. The cost of the right-of-use assets comprise the lease liabilities as well as any lease payments made at or before the commencement date, initial direct costs and asset retirement obligations, less lease incentives received.

The subsequent measurement is carried out at amortized cost, deducting accumulated depreciation and accumulated impairment losses, in accordance with the cost model. Right-of-use assets are depreciated on a straight-line basis over the term of the lease.

If the lessee has a favorable extension option, so that it is reasonably certain that the lessee will exercise that option, the lease payments from the extension period are to be considered in the measurement of the lease obligation.

N Provisions

Pension provisions and similar obligations
Obligations for pensions and other postretirement benefits and related expenses and income are determined in accordance with actuarial measurements. These measurements are based on key assumptions, including discount rates, salary trends, pension trends, biometric probabilities and assumptions about the trend of health insurance benefits. The discount factors applied reflect the interest rates achieved at the balance sheet date for senior, fixed-interest bonds with commensurate maturities. As a result of a fluctuating market and economic situation, the underlying assumptions may deviate from the actual development, which can have a significant impact on the obligations for pensions and other postretirement benefits upon termination of employment.

Pension provisions under defined benefit plans are measured in accordance with the projected unit credit method. Thereby, not only the pensions and acquired expectancies known at the reporting date but also increase in compensation and pensions expected in the future are taken into account. Actuarial gains or losses and other remeasurements of the net obligation are determined at the reporting date and recorded through other comprehensive income (changes in equity not affecting profit or loss for the period).

In order to determine the discount rate for the measurement of the pension provisions and similar obligations, Giesecke+Devrient uses the Mercer Pension Discount Yield Curve Method. This is a method for determining the interest rate that conforms with IAS 19. The new method is based on a selection of AA-rated corporate bonds in accordance with Bloomberg analyses. Net interest expense, i.e. the interest portion of the allocation to the provision less the expected return on plan assets, is reported in interest expense. The amount payable in conjunction with defined contribution plans is reported as an expense.

When the benefits of a plan change or a plan is curtailed the resulting change in the relevant past service performance or the gain or loss from the curtailment is immediately recognized in the income statement. The Group recognizes gains and losses from the settlement of a defined benefit pension plan at the time of settlement.

Pre-retirement part-time work agreements
An obligation for pre-retirement part-time working arrangements is recognized from the point in time at which the employee is entitled under an individual agreement to the premature termination of employment. For pre-retirement part-time working arrangements in connection with the block model, the outstanding obligation for work performed by the employee during the work phase and the obligation to pay top-up amounts are measured separately. Both obligations are recorded in instalments applying actuarial principles from the start of the active phase until the end of the employment phase. In the passive phase, the present value of the future payments is provided. The net interest portion included in the pre-retirement part-time working arrangement expenses is reported as interest expense.

Product warranties
A provision for the expected warranty-related costs is established when the product is sold. Estimates of accrued warranty costs are primarily based on historical experience.

In 2022, there was a change in the methodology for estimating the amount and probability of possible warranty claims. In particular, thanks to new technologies for quality assurance already in the production process, the Group's management estimates the level and risk of warranty claims occurring in the production of banknote paper and the printing of banknotes as being significantly lower. In 2022, this assessment led to a reversal of warranty provisions in the amount of EUR 27.3 million. This income reduces the cost of sales for the 2022 financial year in the consolidated income statement.

Provision for restructuring costs
A provision for restructuring costs is recorded where a legal or constructive obligation exists. A constructive obligation for restructuring costs arises only when there is a detailed formal plan identifying key features of the plan and its implementation and a valid expectation on the part of those affected, either by starting to implement the plan or announcing its main features to those affected by it. A restructuring provision should include only the direct expenditures arising from the restructuring, which are those that are both necessarily entailed by the restructuring and not associated with the ongoing activities of G+D.

Provision for onerous contracts
The calculation of provisions for onerous contracts is to a great extent based on estimates. Such estimates are mainly related to the status of the projects, the fulfillment of the services requested, changes regarding the volume of the projects, the update of budgeted costs as well as applied customized and runtime-specific discount rates.

Giesecke+Devrient records provisions for onerous contracts for contracts in which the unavoidable costs of meeting the obligations exceed the expected benefits. The unavoidable costs under a contract reflect the minimum net costs of exiting from the contract, which is the lower of the cost of fulfilling it and any compensation or penalties arising from failure to fulfill it. Before a separate provision for an onerous contract is established, any impairment loss that has occurred on assets dedicated to that contract is recognized.

Other provisions
Other provisions are recognized where there are legal or constructive obligations to third parties on the basis of past transactions or events that will probably require an outflow of resources to settle, and this outflow can be reliably measured. They are recorded at their expected settlement amount, taking into account all identifiable risks, and may not be offset against potential reimbursements, for example, via insurance claims. The settlement amount is calculated on the basis of the best estimate. Non-current provisions are discounted where the effect of the time value of money is material.

Changes in the interest rate or the amount and timing of payments applied in measuring provisions for decommissioning, restoration, and similar obligations are recognized in the same amount as the related assets to be considered. Where the decrease in the amount of a provision is greater than the carrying amount of the related asset, the excess is recognized immediately in profit or loss.

O Recognition of Revenue, Interest and Dividends

In accordance with IFRS 15, revenue is recognized when the customer gains control of the asset. In several contracts for the sale of customer specific products, especially in the cards, passports and ID business, the Group transfers control during the production phase. Revenues relating to such contracts are realized based upon the percentage of completion of the products and therefore before the delivery of the assets to the customer.

In certain instances, G+D is the general contractor concerning the construction of paper mills, special facilities (e.g. production of security products), and personalization centres. The fulfillment of these types of contracts usually extends over a long period and can last up to several years until final completion. For construction contracts, the revenue is recorded over time provided that the revenue and expenses can be estimated reliably. The percentage of completion is generally determined using the output method (e.g. agreed milestones) or the cost-to-cost method. Profit recognized in the period is calculated by multiplying the contract revenues and costs by the percentage of completion less the results recognized in prior periods.

For long-term customer contracts in which the major components consist of the production, modification, or customizing of software, revenue is generally recognized upon customer acceptance as the percentage of completion cannot be reliable determined.

Giesecke+Devrient has contractual arrangements in which it performs multiple revenue-generating activities, for example, the delivery of card bodies and personalization services. Revenue allocation is based upon the relative stand-alone selling prices of the individual components of the total arrangement.

Across all business units, the increased scope of estimations referring to variable consideration affects the amount and the timing of revenue recognition.

Interest is recognized using the effective interest method. Dividends are recognized when the shareholder's right to receive payment is established.

P Grants

Where grants are received for certain assets, they are offset against the acquisition or manufacturing costs of the related assets and therefore reduce the acquisition costs. The grants/allowances are released to the income statement in instalments in the form of a reduction of depreciation expense.

Other types of grants are recorded in the income statement in the period in which the entitlement arises.

Q Deferred Taxes

Deferred tax assets and liabilities are recognized on temporary differences between the carrying amounts in the consolidated balance sheet and the tax base, as well as for tax loss carryforwards that are expected to reduce tax expense in future periods in accordance with the liability method.

R Statement of Cash Flows

The statement of cash flows is prepared in accordance with IAS 7 and indicates the cash inflows and outflows during the fiscal year classified by cash flows from operating activities, investing activities and financing activities. Cash flows from operating activities are presented using the indirect method, in which earnings are adjusted for non-cash transactions. Moreover, items attributable to cash flows from investing activities and financing activities are eliminated. Cash flows from interest received and interest paid, as well as dividends received, are allocated to cash flows from operating activities. Cash outflows for the acquisition of additional shares in affiliated companies under common control are classified as cash flows from financing activities.

The cash flow funds comprise the balance sheet line item "cash and cash equivalents". Cash and cash equivalents include cash on hand and cash at banks, as well as cash from funds and investments with an original maturity of up to three months.

S Change in Accounting and Valuation Policies in the 2023 financial year

The IASB has published a series of pronouncements that were applied for the first time in fiscal year 2023. These had no major impact on the consolidated financial statements of G+D.

T New and Changed Accounting Pronouncements

The IASB has published further standards, interpretations and amendments to existing standards whose application is not yet mandatory and which have not been early adopted in the consolidated financial statements. These new standards and amendments are expected to have no material impact on the net assets, financial position, and results of operations of the Group.

Amendments to IFRS 16 leases	The amendment relates to the accounting treatment of lease liabilities from sale-and-leaseback transactions.	Periods beginning on or after January 1, 2024
Amendments to IAS 7 Statements of Cash Flows and IFRS 7 Financial Instruments	The amendments relate to disclosure in connection with supplier finance agreements.	Periods beginning on or after January 1, 2024
Amendments to IAS 1 Presentation of Financial Statements	Improvement in consistency regarding the classification of liabilities as current or non-current. Specifically, the intention is to provide rules on the classification of liabilities with an uncertain settlement date.	Periods beginning on or after January 1, 2024
Amendments to IAS 21 The Effects of Changes in Foreign Exchange Rates	Addition of provisions and corresponding disclosure requirements to determine the exchange rate in the event of a long-term lack of exchangeability.	Periods beginning on or after January 1, 2025

2. Financial Assets

Financial assets are comprised of the following as of December 31, 2023 and 2022:

EUR million	Dec 31, 2023	Dec 31, 2022
Current		
Short-term investments (> 3 months and < 1 year)	4.4	0.2
Trading securities	–	86.8
Investment securities	9.8	9.1
Derivative financial instruments	5.7	2.8
Loans receivable from shareholders	3.0	5.2
	22.9	**104.1**
Non-current		
Cash surrender value of reinsurance	17.1	20.1
Loans receivable from related parties	–	0.2
Loans to third parties	0.8	0.5
Investment securities	1.8	1.9
Derivative financial instruments (interest swap)	1.0	2.6
	20.7	**25.3**

Investment securities have been recorded at fair value in the amount of EUR 11.6 million as of December 31, 2023 and EUR 11.0 million as of December 31, 2022. The fair value represents the fair market value.

3. Accounts Receivable Trade and Other Accounts Receivable, net

Accounts receivable trade and other accounts receivable, net comprise the following as of December 31, 2023 and 2022:

EUR million	Dec 31, 2023	Dec 31, 2022
Current		
Accounts receivable trade	581.3	554.3
Accounts receivable from joint ventures and associated companies	1.3	3.5
Accounts receivable from related parties	0.8	0.2
Accounts receivable from shareholders	0.1	0.2
Other	30.5	27.1
Prepayments	65.7	49.9
	679.7	**635.2**
Allowance for doubtful accounts	(74.4)	(71.9)
	605.3	**563.3**
Non-current		
Accounts receivable trade	2.4	4.1
Prepayments on property, plant and equipment	12.1	9.5
	14.5	**13.6**

Aging structure of accounts receivable trade and other accounts receivable (excluding prepayments) as of December 31, 2023:

EUR million	Not past due	past due 1–30 days	past due 31–90 days	past due 91–180 days	past due 181–360 days	past due more than 360 days	**Total**
Accounts receivable trade and other accounts receivable	384.3	92.4	33.9	15.9	19.8	70.1	616.4
Less allowance for doubtful accounts on accounts receivable trade and other accounts receivable	(5.1)	(0.2)	(0.1)	(0.1)	(7.8)	(61.1)	(74.4)
Accounts receivable trade and other accounts receivable, net	**379.2**	**92.2**	**33.8**	**15.8**	**12.0**	**9.0**	**542.0**
Average credit losses in %	1.3	0.2	0.3	0.6	39.4	87.2	12.1

Aging structure of accounts receivable trade and other accounts receivable (excluding prepayments) as of December 31, 2022:

EUR million	Not past due	past due 1–30 days	past due 31–90 days	past due 91–180 days	past due 181–360 days	past due more than 360 days	**Total**
Accounts receivable trade and other accounts receivable	382.8	59.4	49.0	14.6	14.9	68.7	589.4
Less allowance for doubtful accounts on accounts receivable trade and other accounts receivable	(9.0)	(0.2)	(0.3)	(0.2)	(0.5)	(61.7)	(71.9)
Accounts receivable trade and other accounts receivable, net	**373.8**	**59.2**	**48.7**	**14.4**	**14.4**	**7.0**	**517.5**
Average credit losses in %	2.4	0.3	0.6	1.4	3.4	89.8	12.2

The allowances for expected credit losses for accounts receivable trade and other accounts receivable are as follows:

EUR million			**2023**			**2022**
	Estimated credit losses (impaired)	Estimated credit losses (not impaired)	**Total**	Estimated credit losses (impaired)	Estimated credit losses (not impaired)	**Total**
Balance as of January 1	**68.1**	**3.8**	**71.9**	**50.9**	**3.0**	**53.9**
Additions (through profit or loss)	5.1	1.5	6.6	22.8	1.1	23.9
Recoveries (through profit or loss)	(0.4)	(0.9)	(1.3)	(4.1)	(0.3)	(4.4)
Write-offs	(2.3)	–	(2.3)	(1.4)	–	(1.4)
Transfers	(0.3)	–	(0.3)	(0.2)	–	(0.2)
Foreign currency effects	(0.2)	–	(0.2)	0.1	–	0.1
Balance as of December 31	**70.0**	**4.4**	**74.4**	**68.1**	**3.8**	**71.9**

The allowances for expected credit losses (impaired) relate to several customers that disclosed that they would not be able to settle the outstanding balances due to their economic circumstances (such as change in credit rating or breach of contract or financial problems or similar).

Accounts receivable trade and other accounts receivable which have neither been impaired nor are past due as of the balance sheet date amounted to EUR 379.2 million and EUR 373.8 million as of December 31, 2023 and 2022. G+D anticipates that the full amount of accounts receivable which have neither been impaired nor are past due are collectible. There is no indication that the debtors will not be able to meet their payment obligations. This estimate is based on the payment history as well as extensive analysis relating to the customer default risk.

Allowances for doubtful accounts on accounts receivable from joint ventures, associated companies, as well as related parties were not recorded.

In 2015, the Group entered into a service concession arrangement with a foreign authority for the construction and operation of a common factory for the production of identification cards and passports (BOT model = Build-Operate-Transfer). In addition to the one-year construction phase, it also provides for a ten-year operational phase. The recognized accounts receivable from the service concession agreement in the amount of EUR 12.8 million as of December 31, 2022 were fully written off. In fiscal year 2023, G+D did not record any net income or expense from ongoing operations.

4. Inventories, net

Inventories are comprised of the following as of December 31, 2023 and 2022:

EUR million	Dec 31, 2023	Dec 31, 2022
Raw materials	271.5	231.3
Work in process	93.8	83.7
Finished goods	28.2	17.9
Merchandise	55.3	76.0
Spare parts, modules, sensors	72.0	68.2
	520.8	**477.1**

In fiscal years 2023 and 2022, write-downs on inventory amounted to EUR 29.1 million and EUR 15.1 million, respectively.

There is no inventory which serves as collateral for financial liabilities as of December 31, 2023 and 2022, respectively.

5. Other Current Assets

EUR million	Dec 31, 2023	Dec 31, 2022
Taxes receivable (other than income taxes)	36.7	32.7
Restricted cash	9.3	10.6
Other	5.9	5.5
	51.9	**48.8**

6. Investments

Investments include the following:

EUR million	Dec 31, 2023	Dec 31, 2022
Investments accounted for under the equity method	19.4	45.5
Investments in other related parties	18.7	12.6
	38.1	**58.1**

The following investments (see Note 1c "Consolidated Group and Principles of Consolidation") are accounted for using the equity method of accounting:

	Interest in the company
Name of the joint venture	
Podsystem MX SAPI de CV, León	51.0 %
Shenzhen Giesecke & Devrient Currency Automation Systems Co. Ltd., Shenzen	50.0 %
UGANDA SECURITY PRINTING COMPANY (USPCL) LIMITED, Entebbe	29.4 %
Name of the associated company	
Netlync Mobile Communication Technologies Limited, Dublin	36.0 %
Blokverse d.o.o., Skopje	32.3 %
Braingroup AG, Zurich	31.4 %
Unitek Engineering AG, Zurich	28.5 %
D ONE Value Creation AG, Zurich	27.6 %
Securities Grid Ltd., London	23.8 %
Build38 GmbH, Munich	20.9 %
Relief Validation Ltd, Dhaka	15.0 %

Joint Ventures

Podsystem MX SAPI de CV serves as a sales office for the Pod Group.

Shenzen Giesecke & Devrient Currency Automation Systems Co. Ltd. sells and installs banknote processing systems.

On September 1, 2023, G+D acquired the remaining 50.0 % of the shares in E-Kart Elektronik Kart Sistemleri Sanayi ve Ticaret Anonim Sirketi, Gebze (see Note 24 "Business combinations").

UGANDA SECURITY PRINTING COMPANY (USPCL) LIMITED, Entebbe is manufacturing passports and identity documents for the local market.

Associated companies

Veridos acquired an additional 27.0 % of the shares in NetSeT Global Solutions d.o.o., Belgrade, effective January 16, 2023, and thus acquired controlin this company with 67.0% of the shares (see Note 24 "Business combinations").

With the acquisition of control of Netcetera Group AG (see Note 24 "Business combinations"), G+D acquired five additional associated companies: Blokverse d.o.o., Braingroup AG, Unitek Engineering AG, D ONE Value Creation AG and Securities Grid Ltd. Blokverse d.o.o., Braingroup AG, Unitek Engineering AG and Securities Grid Ltd. develop software solutions. D ONE Value Creation AG engages in software consulting.

With effect from June 26, 2023, G+D acquired 36 % of the shares in Netlync Mobile Communication Technologies Limited, Dublin for a purchase price of EUR 5 million. Netlync Mobile Communication Technologies Limited develops and distributes global software solutions. Due to new assessments of market expectations, an impairment in the amount of EUR 2.7 million was recorded in other financial results, net, in 2023.

The purpose of the company Build38 GmbH is the development and distribution of software solutions as service provider for third parties, including related consulting services.

Effective November 3, 2022 Veridos acquired 25.0 % of the shares in Relief Validation Ltd. As G+D holds 60 % of the shares of Veridos, 15.0 % in the shares of Relief Validation Ltd. are held by G+D. Relief Validation Ltd. provides digital certificates and electronic signatures to the local market in Bangladesh. Thus, the company offers modern and digital solutions for authorities (i.e. online applications for government documents and tax returns) as well as private companies (i.e. KYC for commercial banks) with the highest security standards.

Effective December 16, 2022 G+D liquidated the investment in Emirates German Security Printing L.L.C., Abu Dhabi, United Arab Emirates. The liquidation proceeds were recorded within share in earnings from equity investments using the equity method.

Investments in other related parties

G+D owns 7.8 % of the shares Brighter AI Technologies GmbH, Berlin, 6.5 % of the shares in FINANCIAL NETWORK ANALYTICS LTD, London, 5.7 % of the shares in Nano Corp S.A.S., Paris, 5.6 % of the shares in Yields NV, Brugge, 5.3 % of the shares in Tenzir GmbH, Hamburg, 5.0 % of the shares in IDnow GmbH, Munich, 4.3 % of the shares in SALV TECHNOLOGIES OÜ, Tallinn, 4.1 % of the shares in Blindflug Studios AG, Zurich, 1.2 % of the shares in Cognism Ltd., London and 0.5 % of the shares at Verimi GmbH, Frankfurt am Main. As G+D classifies its influence on these investments as not significant, these investments are classified as investments in other related parties. The investments are generally measured at fair value through other comprehensive income, as they represent strategic investments and are intended to be held for the longer term. The group focuses on companies whose innovations promote and protect trust in digital ecosystems.

Joint Ventures and associated companies

The following table summarizes the financial information for material joint ventures and associated companies based on their financial statements prepared in accordance with IFRS. This information is adjusted for fair value adjustments at acquisition and differences in the Group accounting policies:

EUR million	Joint Venture Shenzhen Giesecke & Devrient Currency Automation Systems Co. Ltd.	
	2023	**2022**
Revenues	14.2	12.8
Profit from continuing operations	1.7	4.6
thereof depreciation and amortization	(0.1)	(0.5)
thereof interest income/(expense)	0.1	0.1
thereof income taxes	(0.5)	(0.3)
Other comprehensive income	(0.6)	0.6
Total comprehensive income	**1.1**	**5.2**
Group's share of total comprehensive income	0.5	2.6
Elimination of intercompany profits	(0.1)	(0.2)
Group's share of total comprehensive income	**0.4**	**2.4**
Dividends received during the year	**(1.9)**	**(0.4)**
Current assets	11.4	16.7
thereof cash and cash equivalents	7.3	12.5
Non-current assets	2.4	2.5
Current liabilities	(3.9)	(6.4)
Net assets	**9.9**	**12.8**
Group's share of net assets	5.0	6.4
Elimination of intercompany profits	(0.4)	(0.3)
Carrying amount of interest in joint venture at year end	**4.6**	**6.1**

Non-material joint ventures

The following table summarizes the financial information for the Group's share in non-material joint ventures based on the amounts as reported in the Group's financial statements:

EUR million	Dec 31, 2023	Dec 31, 2022
Carrying amount of interest in non-material joint ventures	**4.8**	**7.9**
Share of		
Gain/(Loss) from continuing operations	(0.4)	1.9
Elimination of intercompany profits	(0.6)	–
Other comprehensive income	–	0.4
Total comprehensive income	**(1.0)**	**2.3**

Non-material associated companies

The following table summarizes the financial information for the Group's share in non-material associated companies based on the amounts as reported in the Group's financial statements:

EUR million	Dec 31, 2023	Dec 31, 2022
Carrying amount of interest in non-material associated companies	**10.1**	**31.5**
Share of		
Capital increase or changes in consolidation structure	13.2	0.6
Gain/(Loss) from continuing operations	(1.5)	(2.6)
Dividends received during the year	(0.8)	(0.4)
Impairments	(2.7)	–
Other comprehensive income	–	0.2
Total comprehensive income	**(5.0)**	**(2.8)**

7. Intangible Assets

A summary of the activity for goodwill and other intangible assets is as follows:

EUR million	Customer base/ Rights	Development costs/ Technology	Software	Goodwill	**Total**
Costs					
January 1, 2022	**41.7**	**156.1**	**196.5**	**87.3**	**481.6**
Additions	–	12.4	11.1	–	23.5
Changes in consolidation structure	25.6	–	7.4	37.3	70.3
Disposals	(0.3)	–	(8.9)	–	(9.2)
Foreign currency effects	–	(1.2)	0.1	(2.3)	(3.4)
December 31, 2022	**67.0**	**167.3**	**206.2**	**122.3**	**562.8**
January 1, 2023	**67.0**	**167.3**	**206.2**	**122.3**	**562.8**
Additions	1.2	27.3	5.0	–	33.5
Transfers	–	7.4	(5.5)	–	1.9
Changes in consolidation structure	49.0	15.9	0.8	98.0	163.7
Disposals	(1.2)	(5.4)	(2.5)	–	(9.1)
Foreign currency effects	0.7	0.5	(1.2)	5.5	5.5
December 31, 2023	**116.7**	**213.0**	**202.8**	**225.8**	**758.3**

The additions in 2023 and 2022 comprise self-constructed intangible assets in the amount of EUR 29.0 million and EUR 15.6 million, respectively.

EUR million	Customer base/ Rights	Development costs/ Technology	Software	Goodwill	**Total**
Accumulated amortization					
January 1, 2022	**30.5**	**109.2**	**151.7**	**1.7**	**293.1**
Additions	3.8	13.4	15.6	–	32.8
Disposals	(0.1)	–	(8.6)	–	(8.7)
Foreign currency effects	0.2	(1.1)	(0.1)	–	(1.0)
December 31, 2022	**34.4**	**121.5**	**158.6**	**1.7**	**316.2**
January 1, 2023	**34.4**	**121.5**	**158.6**	**1.7**	**316.2**
Additions	19.0	25.2	16.8	–	61.0
Transfers	–	0.4	(0.2)	–	0.2
Impairment losses	–	4.1	–	–	4.1
Disposals	(1.2)	(5.4)	(2.0)	–	(8.6)
Foreign currency effects	0.3	0.3	(0.8)	–	(0.2)
December 31, 2023	**52.5**	**146.1**	**172.4**	**1.7**	**372.7**
Carrying value					
January 1, 2022	**11.2**	**46.9**	**44.8**	**85.6**	**188.5**
December 31, 2022	32.6	45.8	47.6	120.6	246.6
December 31, 2023	**64.2**	**66.9**	**30.4**	**224.1**	**385.6**

The amounts of amortization of intangible assets recorded in the functional areas of the income statement are as follows:

EUR million	2023	2022
Cost of goods sold	55.9	26.0
Selling expenses	0.5	0.8
Research and development expenses	–	0.5
General and administrative expenses	4.6	5.5
	61.0	**32.8**

In fiscal year 2023, impairment losses of EUR 4.1 million were recognized on capitalized development costs, of which EUR 3.1 million was included in cost of sales and EUR 1.0 million in research and development expenses. In the previous year, no impairment losses were recognized on these assets. The impairment losses were recognized due to negative business cases and lower generation of substantial cash flows.

The goodwill from CI Tech Components AG in the amount of EUR 14.0 million (prior year EUR 14.0 million), as well as the goodwill from Giesecke+Devrient Currency Technology Netherlands B.V. in the amount of EUR 4.8 million (prior year EUR 4.8 million), were allocated to the CGU "Currency Management Solutions". As the CGU "Currency Management Solutions" business is mainly conducted in Euro, these goodwills are recorded in Euro. The goodwill from secunet AG in the amount of EUR 48.2 million (prior year EUR 48.2 million) was assigned to the "secunet" CGU. Sensitivity analyses are not required since the recoverability of these goodwills is not deemed to be critical.

In fiscal year 2023, there was an increase in goodwill in the amount of EUR 103.3 million. This goodwill includes currency effects. This increase results from acquired shares as part of share deals in Netcetera Group, NetSet Global Solutions D.O.O. and Mecomo Group.

Goodwill of EUR 87.3 million resulted from the shares in the Netcetera Group acquired as part of a share deal. This goodwill is allocated to the CGU "Netcetera". Goodwill of EUR 3.4 million from the additional shares in NetSeT Global Solutions D.O.O. acquired in a share deal is allocated to the CGU "Veridos". Goodwill of EUR 12.6 million related to the shares of the Mecomo Group acquired in a share deal is allocated to the CGU "Trusted Connected Devices". The goodwill from Veridos Matsoukis S.A. Security Printing in the amount of EUR 2.1 million (prior year EUR 2.1 million) and the goodwill from Veridos GmbH in the amount of EUR 4.4 million (prior year EUR 4.4 million), as well as the goodwill from E-SEEK Inc. in the amount of EUR 4.0 million (prior year 4.1 EUR million), recognized within the acquisition of the companies were allocated to the "Veridos" CGU. The goodwill from Giesecke+Devrient Mobile Security Sweden AB in the amount of EUR 26.4 million (prior year EUR 26.4 million) was assigned to the business sector "Mobile Security" and thereby to the divisional CGUs "Secure Transactions + Services" in the amount of EUR 7.5 million (prior year EUR 7.5 million), and "Trusted Connected Devices" in the amount of EUR 18.9 million (prior year EUR 18.9 million). The goodwill acquired in the connection with the purchase of C.P.S. Technologies S.A.S. in the amount of EUR 0.5 million (prior year EUR 0.5 million) was allocated to the CGU "Secure Transactions + Services" . The goodwill from the acquisition of the Crédit Agricole's personalization business in the amount of EUR 4.3 million (prior year EUR 4.3 million) was also allocated to the CGU "Secure Transactions + Services". The goodwill from the acquisition of the Pod Group in the amount of EUR 12.0 million (prior year EUR 11.8 million) was allocated to the CGU "Trusted Connected Devices".

In performing the impairment tests for goodwill, the recoverable amount of the CGU is based on the value in use. The value in use is the present value of the future cash flows expected to be derived from the CGU. Since 2014, the cash flow projections are based upon G+D's five-year plans. The cash flows for the CGU "Netcetera" were determined using the planning assumptions of an average EBITDA margin of 21.1 % during the planning period and perpetual growth rate of 1.0 %. The cash flows for the CGU "secunet" were determined using the planning assumptions of an average EBITDA margin of 16.5 % (prior year 21.5 %) during the planning period and perpetual growth rate of 1.0 % (prior year 1.0 %). The cash flows for the CGU "Veridos" were determined using the planning assumptions of an average EBITDA margin of 10.5 % (prior year 11.0 %) during the planning period and perpetual growth rate of 1.0 % (prior year 1.0 %). The cash flows for the CGUs "Secure Transactions + Services" und "Trusted Connected Devices" were determined using the planning premises of average EBITDA margins of 11.9 % and 12.4 % (prior year 11.9 % and 9.9 %) during the planning period and perpetual growth rate of 1.0 % (prior year 1.0 %). The cash flows for the CGU "Currency Management Solutions" were determined using the planning premises of average EBITDA margin of 13.1 % (prior year 12.3 %) during the planning period and perpetual growth rate of 1.0 % (prior year 1.0 %).

In discounting the cash flows of the "Netcetera" CGU, pre-tax interest rate of 11.6 % was used in 2023. In discounting the cash flows of the "secunet" CGU, pre-tax interest rate of 14.5 % was used in 2023 and 12.6 % in 2022. For the CGU "Currency Management Solutions", a pre-tax interest rate of 13.3 % was applied in 2023 and 12.2 % in 2022. In discounting the cash flows of the "Veridos" CGU, a pre-tax interest rate of 14.7 % was applied in 2023 and 14.1 % in 2022. In discounting the cash flows of the "Secure Transactions + Services", a pre-tax interest rate of 14.5 % was used in 2023 and 13.0 % in 2022. In discounting the cash flows of the "Trusted Connected Devices" CGUs, a pre-tax interest rate of 14.8 % was used in 2023 and 13.0 % in 2022. Impairments on goodwill were not recorded in fiscal years 2023 and 2022.

A sensitivity analysis was carried out for the goodwill in the CGU "Netcetera". An increase in the interest rate from 11.6% to 13.2 % ceteris paribus as of December 31, 2023 would result in a first time impairment loss. A reduction in the cash flows for the terminal value ceteris paribus from EUR 32.2 million to EUR 25.7 million as of December 31, 2023 would result in an impairment loss. The headroom amounted to EUR 36.7 million as of December 31, 2023.

A sensitivity analysis was carried out for the goodwill in the CGU "Veridos". An increase in the interest rate from 14.7 % to 18.1 % ceteris paribus as of December 31, 2023 would result in a first time impairment loss. A reduction in the cash flows for the terminal value ceteris paribus from EUR 28.8 million to EUR 19.5 million as of December 31, 2023 would result in an impairment loss. The headroom amounted to EUR 36.4 million as of December 31, 2023.

A sensitivity analysis was carried out for the goodwill in the CGU "Secure Transactions + Services". An increase in the interest rate ceteris paribus from 14.5 % to 16.9 % as of December 31, 2023 would result in a first time impairment loss. A sensitivity analysis was carried out for the goodwill in the CGU "Trusted Connected Devices". An increase in the interest rate ceteris paribus from 14.8 % to 20.4 % as of December 31, 2023 would result in a first time impairment loss. A reduction in the cash flows for the terminal value ceteris paribus from EUR 80.6 million to EUR 60.0 million for "Secure Transactions + Services" as of December 31, 2023 would result in an impairment loss. The headroom amounted to EUR 82.7 million as of December 31, 2023. A reduction in the cash flows for the terminal value ceteris paribus from EUR 39.7 million to EUR 18.8 million for "Trusted Connected Devices" as of December 31, 2023 would result in an impairment loss. The headroom amounted to EUR 81.2 million as of December 31, 2023.

No intangible assets serve as collateral for financial liabilities (see Note 13 "Financial Liabilities") as of December 31, 2023 and 2022, respectively.

8. Property, Plant and Equipment

A summary of the activity for property, plant and equipment[1] is as follows:

EUR million	Land and buildings[1]	Technical equipment and machinery[1]	Other plant and office equipment[1]	Construction in process	Total
Costs					
January 1, 2022	**592.0**	**845.4**	**243.3**	**16.2**	**1,696.9**
Additions	25.2	22.6	22.2	10.5	80.5
Transfers	0.1	7.3	2.4	(11.2)	(1.4)
Changes in consolidation structure	17.3	10.8	2.6	0.1	30.8
Disposals	(37.1)	(13.0)	(14.6)	–	(64.7)
Foreign currency effects	1.5	3.7	1.3	0.1	6.6
December 31, 2022	**599.0**	**876.8**	**257.2**	**15.7**	**1,748.7**
January 1, 2023	**599.0**	**876.8**	**257.2**	**15.7**	**1,748.7**
Additions	18.3	30.3	27.5	14.8	90.9
Transfers	2.1	11.7	(0.9)	(13.2)	(0.3)
Changes in consolidation structure	11.0	1.1	2.2	0.1	14.4
Disposals	(8.1)	(25.0)	(13.3)	–	(46.4)
Foreign currency effects	(4.3)	(7.4)	(4.0)	(0.1)	(15.8)
December 31, 2023	**618.0**	**887.5**	**268.7**	**17.3**	**1,791.5**

[1] Including assets under leases (see Note 9 "Leasing").

EUR million	Land and buildings[1]	Technical equipment and machinery[1]	Other plant and office equipment[1]	Construction in process	Total
Accumulated amortization					
January 1, 2022	**337.5**	**674.9**	**183.7**	**–**	**1,196.1**
Additions	30.6	41.3	24.1	–	96.0
Impairment losses	–	0.2	–	–	0.2
Disposals	(35.3)	(12.9)	(13.0)	–	(61.2)
Foreign currency effects	0.8	2.7	0.7	–	4.2
December 31, 2022	**333.6**	**706.2**	**195.5**	**–**	**1,235.3**
January 1, 2023	**333.6**	**706.2**	**195.5**	**–**	**1,235.3**
Additions	33.8	42.3	25.4	–	101.5
Transfers	0.3	1.0	–	–	1.3
Impairment losses	0.8	0.8	–	–	1.6
Reversal of impairment losses	–	(0.1)	–	–	(0.1)
Disposals	(6.7)	(25.0)	(13.1)	–	(44.8)
Foreign currency effects	(2.2)	(5.4)	(3.7)	–	(11.3)
December 31, 2023	**359.6**	**719.8**	**204.1**	**–**	**1,283.5**
Carrying value					
January 1, 2022	254.5	170.5	59.6	16.2	500.8
December 31, 2022	265.4	170.6	61.7	15.7	513.4
December 31, 2023	**258.4**	**167.7**	**64.6**	**17.3**	**508.0**

In fiscal years 2023 and 2022, Giesecke+Devrient recorded impairments amounting to EUR 1.6 million (previous year EUR 0.2 million) on property, plant and equipment in cost of goods sold. The impairment losses were recognized due to negative business cases.

The carrying value of property, plant and equipment which serves as collateral for financial liabilities (see Note 13 "Financial Liabilities") amounted to EUR 3.5 million and EUR 1.0 million as of December 31, 2023 and 2022, respectively.

Commitments for the purchase of property, plant and equipment amounted to EUR 12.7 million and EUR 15.4 million as of December 31, 2023 and 2022, respectively.

9. Leasing

Leases as lessee
As a lessee, Giesecke+Devrient rents various lease objects. This includes land and buildings, manufacturing facilities, electronic data processing equipment, motor vehicles and other office equipment. Lease contracts are negotiated for different terms and may include extension or cancellation options. The conditions are negotiated individually and include various conditions.

The Group includes right-of-use assets within property, plant and equipment in the consolidated balance sheet. A summary of the activity for right-of-use assets is as follows:

EUR million	Land and buildings	Technical equipment and machinery	Other plant and office equipment	**Total**
January 1, 2022	**62.8**	**2.2**	**4.7**	**69.7**
Additions	23.0	0.5	2.9	26.4
Disposals	(1.4)	–	(0.1)	(1.5)
Acquisitions	11.7	1.4	–	13.1
Depreciation	(20.1)	(1.5)	(3.0)	(24.6)
Foreign currency effects	0.1	0.1	–	0.2
December 31, 2022	**76.1**	**2.7**	**4.5**	**83.3**
January 1, 2023	**76.1**	**2.7**	**4.5**	**83.3**
Additions	13.6	0.2	5.1	18.9
Disposals	(0.4)	(0.2)	(0.1)	(0.7)
Acquisitions	9.0	–	0.3	9.3
Depreciation	(22.9)	(1.7)	(3.4)	(28.0)
Impairment losses	(0.8)	–	–	(0.8)
Foreign currency effects	(1.3)	–	(0.1)	(1.4)
December 31, 2023	**73.3**	**1.0**	**6.3**	**80.6**

The lease obligations are recorded as current lease obligations and non-current lease obligations in the consolidated balance sheet.

The following expenses were recorded in the income statement due to leases:

EUR million	**2023**	**2022**
Depreciation on right-of-use assets	(28.0)	(24.6)
Interest expense from lease obligations	(2.2)	(1.2)
Expenses for short-term leases	(0.9)	(0.6)
Expenses from leases for low value assets	(0.3)	(0.8)
	(31.4)	**(27.2)**

The total cash outflows for leases in 2023 amounted to EUR 31.3 million (prior year EUR 26.5 million).

The future cash outflows from lease payments amount to:

EUR million	2023	2022
Less than one year	28.2	24.1
Between one and five years	54.8	57.5
More than five years	9.2	12.1
Total minimum lease payments	92.2	93.7
Less amount representing interest	(5.5)	(4.0)
Present value of net minimum lease payments	**86.7**	**89.7**

The present value of net minimum lease payments is as follows:

EUR million	2023	2022
Less than one year	26.3	22.8
Between one and five years	51.6	55.0
More than five years	8.8	11.9
Present value of net minimum lease payments	**86.7**	**89.7**

Potential future cash outflows in the amount of EUR 12.8 million EUR (prior year EUR 17.9 million) were not included in the lease obligations as it is not reasonably certain that the leases will be extended or not cancelled.

Regarding the summary of the activity for lease obligations, refer to Note 13 “Financial Liabilities”.

Leases as lessor

G+D leases out devices in order to provide services as part of full solutions. All leases are classified as operating leases from a lessor perspective as they do not transfer substantially all of the risks and rewards incidental to the ownership of the assets.

In 2023 and 2022 rental income in the amount of EUR 1.2 million and EUR 0.0 million was recognized.

The following table sets out a maturity analysis of lease payments, showing the undiscounted lease payments to be received after the reporting date:

EUR million	
2024	1.9
2025	1.7
2026	1.4
2027	1.0
2028	0.5
thereafter	1.0
	7.5

10. Accounts Payable Trade and Other Accounts Payable

EUR million	Dec 31, 2023	Dec 31, 2022
Current		
Accounts payable trade due to third parties	311.0	375.7
Accounts payable due to related parties	0.1	–
Accounts payable due to associated companies and joint ventures	1.1	1.5
Deposits received/deferred income	2.0	5.8
	314.2	**383.0**

11. Provisions

EUR million	Warranties	Personnel related costs	Licenses and patent infringements	Onerous contracts	Restructuring	Other	Total
January 1, 2023	**33.1**	**13.8**	**1.1**	**16.3**	**1.9**	**27.2**	**93.4**
Additions	13.5	6.0	4.3	5.0	1.4	13.3	43.5
Transfers	–	–	–	–	(0.2)	0.1	(0.1)
Acquisitions	–	–	–	–	–	2.6	2.6
Interest component	–	0.3	–	–	–	–	0.3
Utilization	(5.2)	(6.0)	(0.3)	(10.5)	(0.6)	(6.4)	(29.0)
Release	(8.4)	(0.3)	(0.8)	(4.3)	(0.4)	(3.4)	(17.6)
Foreign currency effects	(0.7)	–	–	(0.1)	(0.1)	(0.3)	(1.2)
December 31, 2023	**32.3**	**13.8**	**4.3**	**6.4**	**2.0**	**33.1**	**91.9**
thereof current	32.3	8.3	4.3	5.6	2.0	21.1	73.6
thereof non-current	–	5.5	–	0.8	–	12.0	18.3

Personnel-related provisions include obligations for pre-retirement part-time working arrangements and long-service awards. The interest component on pre-retirement part time working arrangements and long-service awards in the amount of EUR 0.3 million is included in interest expense.

Provisions for restructuring mainly consist of provisions relating to personnel measures.

Other provisions include, in particular, provisions for penalties, withholding tax obligations, asset retirement obligations and litigation.

12. Other Current Liabilities

EUR million	Dec 31, 2023	Dec 31, 2022
Payroll and social security taxes	139.1	117.6
Sales and other taxes	52.5	31.4
Other liabilities	23.2	25.7
	214.8	**174.7**

13. Financial Liabilities

Financial liabilities consist of the following as of December 31, 2023 and 2022:

EUR million	Dec 31, 2023	Dec 31, 2022
Current		
Short-term borrowings due to financial institutions	6.1	–
Current portion of debt due to other third parties	1.7	3.2
Purchase price consideration	64.1	–
Liabilities from put-options	9.3	–
Current portion of debt due to financial institutions	17.5	86.6
Current portion of debt due to Giesecke+Devrient Foundation	–	21.0
Current portion of debt due to Giesecke+Devrient Foundation Geldscheinsammlung	–	0.4
Accrued interest on debt to financial institutions	1.8	1.8
Derivative financial instruments	0.7	0.5
Total current portion of financial liabilities	**101.2**	**113.5**
Non-current		
Unsecured notes payable to financial institutions, interest rates 1.20 % to 5.67 %, due through July 20, 2028	453.8	502.0
Unsecured notes payable to Giesecke+Devrient Foundation, interest rate 4.50 %, due through July 31, 2026	15.0	21.0
Unsecured notes payable to Giesecke+Devrient Foundation Geldscheinsammlung, interest rate 4.50 %, due through November 17, 2025	0.6	0.4
Unsecured notes payable to other third parties, interest rates 1.85 % to 5.15 %, due through September 30, 2026	25.8	23.0
Liabilities from put-options	2.8	–
Mortgage notes payable to financial institutions, interest rate 1.03 %, due through June 30, 2024	0.2	1.0
Secured notes payable to other third parties	4.4	–
Total	**502.6**	**547.4**
Less current portion of non-current financial liabilities	(19.2)	(111.2)
Total non-current portion of financial liabilities	483.4	436.2
Total financial liabilities	**584.6**	**549.7**

The aggregate maturities of financial liabilities for each of the following years are as follows:

EUR million	
2024	101.2
2025	161.4
2026	171.3
2027	77.1
2028	68.0
thereafter	5.6
	584.6

A summary of the activity for financial liabilities is as follows as of December 31, 2023 and 2022:

EUR million	Non-current borrowings (incl. short-term portion)	Current borrowings	Purchase price considerations[1]	Derivative financial instruments	Sum of financial liabilities	Lease obligations[2]	**Total**
Carrying value January 1, 2022	**349.2**	**1.9**	**–**	**0.7**	**351.8**	**72.4**	**424.2**
Repayments during the period	(17.9)	(0.5)	–	–	(18.4)	(23.9)	(42.3)
New borrowings	223.2	–	–	–	223.2	–	223.2
Total change in cash flow from financing activities	**205.3**	**(0.5)**	**–**	**–**	**204.8**	**(23.9)**	**180.9**
New lease contracts	–	–	–	–	–	26.4	26.4
Termination of lease contracts	–	–	–	–	–	(1.5)	(1.5)
Acquisitions	0.6	–	–	–	0.6	16.1	16.7
Other changes	–	0.4	–	–	0.4	–	0.4
Fair value changes	(0.2)	–	–	(0.2)	(0.4)	–	(0.4)
Transfers	(8.0)	–	–	–	(8.0)	–	(8.0)
Currency effects	0.5	–	–	–	0.5	0.2	0.7
Carrying value December 31, 2022	**547.4**	**1.8**	**–**	**0.5**	**549.7**	**89.7**	**639.4**
Fair value December 31, 2022	**476.9**	**1.8**	**–**	**0.5**	**479.2**	**89.7**	**568.9**
Carrying value January 1, 2023	**547.4**	**1.8**	**–**	**0.5**	**549.7**	**89.7**	**639.4**
Repayments during the period	(120.5)	(4.0)	–	–	(124.5)	(27.9)	(152.4)
New borrowings	60.8	1.6	–	–	62.4	–	62.4
Total change in cash flow from financing activities	**(59.7)**	**(2.4)**	**–**	**–**	**(62.1)**	**(27.9)**	**(90.0)**
New lease contracts	–	–	–	–	–	19.0	19.0
Termination of lease contracts	–	–	–	–	–	(0.9)	(0.9)
Acquisitions[1]	11.8	8.0	74.4	–	94.2	8.1	102.3
Other changes	(1.1)	–	–	–	(1.1)	0.1	(1.0)
Fair value changes	–	–	–	0.2	0.2	–	0.2
Currency effects	1.4	0.5	1.8	–	3.7	(1.4)	2.3
Carrying value December 31, 2023	**499.8**	**7.9**	**76.2**	**0.7**	**584.6**	**86.7**	**671.3**
Fair value December 31, 2023	**482.1**	**7.9**	**76.2**	**0.7**	**566.9**	**86.7**	**653.6**

[1] For additional information refer to Note 24 “Business Combinations”.
[2] For additional information regarding lease obligations refer to Note 9 “Leasing”.

The transfers in 2022 relate to a waiver of receivables of a non-controlling interest.

Lines of credit

Giesecke+Devrient maintains global credit facilities in the amount of EUR 990.8 million (prior year EUR 943.8 million). As of December 31, 2023, G+D used EUR 395.8 million (prior year EUR 328.0 million) of these facilities for bank guarantees, EUR 6.1 million (prior year EUR 4.0 million) for credit orders and EUR 0.0 million (prior year EUR 0.0 million) due to other third parties. Thus, EUR 588.9 million (prior year EUR 611.9 million) in credit lines were not used as of the reporting date.

14. Pensions and Related Liabilities

Description of the plans

Giesecke+Devrient maintains defined benefit pension plans for a considerable number of employees in Germany and Switzerland as well as at a few subsidiaries abroad. These defined benefit or contribution pension plans charge the Group with actuarial risks such as longevity risks, currency exchange risks and interest rate risks.

In addition to the number of years of service, the defined benefit pension plans in Germany are based on the current salary received or are dependent on the final salary. For most of the employees who had employment contracts from January 1, 2002 onwards with a German Group company, the pension plan is based on pension components whose benefits are adjusted each year by 1.0 %. Furthermore, employees in German Group companies are granted the right to use particular salary components for future pension payments. For new employees joining G+D after January 1, 2014, the existing defined contribution plan was terminated. For employees joining the company from January 1, 2014 on, an externally financed pension obligation was introduced.

According to the Swiss Federal Act on Occupational Retirement, Survivors' and Disability Provision (BVG), the second pension pillar – occupational pension provision – is mandatory for the employer. This means that all employees take part in compulsory insurance in a benefit institution (e.g. a pension fund). Contributions are made to the benefit institution, at least half of which must be paid by the employer. The amount of contributions depends on the employee's age, insured salary and the pension regulations. When a benefit event occurs, the corresponding benefit is determined from the retirement savings accumulated in the pension fund.

The measurement date for the calculation of the DBO for the principal pension plans and the other key postretirement benefits is December 31.

Payment obligations exist for defined contribution state pension plans in Germany and abroad.

Total provisions for pensions and related liabilities

Obligations under the Giesecke+Devrient pension plans and other postretirement benefit plans are comprised of the following:

EUR million	Dec 31, 2023	Dec 31, 2022
Pension benefits	414.6	461.2
Other postretirement benefits	4.2	4.0
Other	1.2	1.0
Total accrual for pension and related liabilities	**420.0**	**466.2**

Pensions and other postretirement benefits

Details of the changes in the defined benefit obligation, the current value of plan assets and the other postretirement benefits are summarized in the following tables:

EUR million					Pension benefit plans			Other postretirement benefit plans
	31.12.2023	thereof in Germany	thereof in Switzerland	**31.12.2022**	thereof in Germany	thereof in Switzerland	**31.12.2023**	**31.12.2022**
Change in defined benefit obligation								
Defined benefit obligation at beginning of year	**469.1**	**465.6**	**–**	**690.0**	**686.8**	**–**	**4.0**	**3.5**
Foreign currency exchange rate differences	4.4	–	4.4	0.1	–	–	(0.4)	(0.1)
Service cost	8.9	5.2	2.8	6.9	6.6	–	0.6	0.4
Interest cost	20.3	18.8	1.5	9.6	9.6	–	0.2	0.1
Past service cost	0.1	–	–	0.1	–	–	–	0.7
Curtailments	0.1	–	–	–	–	–	–	–
Plan amendments	–	–	–	0.1	–	–	–	0.7
Plan participants' contributions	0.8	–	0.8	–	–	–	0.1	0.1
Changes in consolidation structure	63.4	–	63.3	(0.1)	–	–	–	–
Actuarial (gains)/losses	50.2	45.3	5.1	(218.0)	(218.0)	–	–	(0.5)
due to financial parameter changes	53.8	48.8	5.1	(223.2)	(223.2)	–	(0.1)	0.1
due to experience adjustments	(3.6)	(3.5)	–	5.2	5.2	–	0.1	(0.6)
Benefits paid	(20.2)	(15.4)	(4.6)	(19.5)	(19.4)	–	(0.3)	(0.2)
Defined benefit obligation at end of year	**597.0**	**519.5**	**73.3**	**469.1**	**465.6**	**–**	**4.2**	**4.0**
Change in plan assets								
Fair value of plan assets at beginning of year	**7.9**	**7.8**	**–**	**9.4**	**9.3**	**–**	**–**	**–**
Foreign currency exchange rate differences	3.0	–	2.8	–	–	–	–	–
Expected interest income	1.5	0.4	1.3	0.2	0.2	–	–	–
Changes in consolidation structure	54.8	–	54.8	–	–	–	–	–
Divestment	–	–	–	(1.0)	(1.0)	–	–	–
Employer contributions	118.9	115.8	3.1	0.6	0.6	–	–	–
Plan participants' contributions	0.8	–	0.8	–	–	–	–	–
Actuarial gains/(losses)	1.1	0.5	0.7	–	–	–	–	–
Other	–	–	–	(1.1)	(1.1)	–	–	–
Benefits paid	(5.6)	(1.0)	(4.6)	(0.2)	(0.2)	–	–	–
Fair value of plan assets at end of year	**182.4**	**123.5**	**58.9**	**7.9**	**7.8**	**–**	**–**	**–**
Net amount recognized at end of year	**414.6**	**396.0**	**14.4**	**461.2**	**457.8**	**–**	**4.2**	**4.0**

Net liability recorded

The development of the net liability recorded in fiscal years ended December 31, 2023 and 2022 is as follows:

EUR million					Pension benefit plans		Other postretirement benefit plans	
	31.12.2023	thereof in Germany	thereof in Switzerland	**31.12.2022**	thereof in Germany	thereof in Switzerland	**31.12.2023**	**31.12.2022**
Net liability at beginning of year	**461.2**	**457.8**	**–**	**680.6**	**677.5**	**–**	**4.0**	**3.5**
Service cost	8.9	5.2	2.8	6.9	6.6	–	0.6	0.4
Past service cost	0.1	–	–	0.1	–	–	–	0.7
Curtailments	0.1	–	–	–	–	–	–	–
Plan amendments	–	–	–	0.1	–	–	–	0.7
Interest expense/(income)	18.8	18.4	0.2	9.4	9.4	–	0.2	0.1
Changes in consolidation structure	8.6	–	8.5	0.9	1.0	–	–	–
Actuarial (gains)/losses	49.1	44.8	4.4	(218.0)	(218.0)	–	–	(0.5)
due to financial parameter changes	52.7	48.3	4.4	(223.2)	(223.2)	–	(0.1)	0.1
due to experience adjustments	(3.6)	(3.5)	–	5.2	5.2	–	0.1	(0.6)
Benefits paid (excluding plan settlements)	(14.6)	(14.4)	–	(19.3)	(19.2)	–	(0.3)	(0.2)
Employer contributions	(118.9)	(115.8)	(3.1)	(0.6)	(0.6)	–	–	–
Plan participants' contributions	–	–	–	–	–	–	0.1	0.1
Other	–	–	–	1.1	1.1	–	–	–
Foreign currency exchange rate differences	1.4	–	1.6	0.1	–	–	(0.4)	(0.1)
Net liability at end of year	**414.6**	**396.0**	**14.4**	**461.2**	**457.8**	**–**	**4.2**	**4.0**

Plan assets

In addition to money directly invested in banks, Giesecke+Devrient manages assets in the amount of EUR 116.2 million (prior year EUR 98.1 million) via a prime bank in a special fund.

On December 15, 2023 these assets were declared as plan assets and introduced into a contractual trust arrangement. In this contractual trust arrangement the company acts as a trustee with additional German subsidiaries as benefactors. The special fund is comprised of two segments that are operated by various portfolio managers.

The plan assets were invested in the following classes of assets:

information as % of plan assets	**31.12.2023**	thereof in Germany	thereof in Switzerland	**31.12.2022**	thereof in Germany	thereof in Switzerland
Cash surrender value of reinsurance	3.7	5.4	–	92.4	92.4	–
Equity securities	0.3	0.5	–	7.3	7.3	–
Shares	35.0	37.1	30.8	–	–	–
Debt instruments	36.6	40.4	28.8	–	–	–
Real estate funds	7.6	–	23.4	–	–	–
Money market funds	–	–	–	0.3	0.3	–
Cash and cash equivalents	4.2	5.8	0.8	–	–	–
Other	12.6	10.8	16.2	–	–	–
	100.0	**100.0**	**100.0**	**100.0**	**100.0**	**–**

The investment strategy includes two segments. The first segment neither reflects a securities index nor does the company base the segment on a fixed benchmark. The portfolio management actively decides on the selection of assets at its own discretion, taking into account analysis and valuations of companies as well as economic and political developments. It aims to achieve positive performance in accordance with the agreed investment objective. The second segment aims to generate added value compared to the benchmark consisting of bonds, shares and alternative investments over a longer period of time.

The additions to plan assets planned for fiscal year 2024 amount to EUR 1.6 million (prior year EUR 0.5 million) in Germany and EUR 3.0 million in Switzerland. There are no minimum funding requirements.

Actuarial assumptions

The discount rates and percentages for salary and pension increases assumed in the determination of the future pension obligations fluctuate in accordance with the economic situation in the countries in which the pension plans exist. The weighted average assumptions for the calculation of the actuarial amounts are as follows:

in %					Pension benefit plans			Other postretirement benefit plans
	Dec 31, 2023	thereof in Germany	thereof in Switzerland	**Dec 31, 2022**	thereof in Germany	thereof in Switzerland	**Dec 31, 2023**	**Dec 31, 2022**
Discount rate/expected return on plan assets	3.5	3.6	1.4	4.2	4.2	–	8.2	8.7
Rate of compensation increase	2.5	2.5	1.6	2.3	2.2	–	8.4	9.0
Rate of pension progression	1.9	2.0	–	2.0	2.0	–	–	–
Mortality tables								
Germany	RT Heubeck 2018 G			RT Heubeck 2018 G				
Switzerland	BVG 2020							

The rate of the expected long-term return on plan assets corresponds with the discount rate. The rate of pension progression as of December 2023 only amounts to 2.0 % for years starting 2026. The effects of 2024 and 2025 are 3.0 %. The rate of pension progression as of December 2022 only amounts to 2.0 % for years starting 2025. The effects of 2023 and 2024 are 3.5 % and 2.25 %. The weighted average term for pension obligations amounts to 15.2 years (prior year 14.7). In Germany it represents 15.5 years (prior year 14.7) and in Switzerland 10.4 years (prior year 0.0) The weighted average term for other benefit obligations amounts to 8.5 years (prior year 8.5).

Sensitivity analysis

The results of the sensitivity analyses for the significant actuarial assumptions for pension obligations as of December 31, 2023 and December 31, 2022 are as follows:

EUR million	**Dec 31, 2023**	thereof in Germany	thereof in Switzerland	**Dec 31, 2022**	thereof in Germany	thereof in Switzerland
Discount rate + 50 basis points	(115.2)	(38.5)	(76.7)	(35.3)	(35.3)	–
Discount rate - 50 basis points	113.3	43.4	69.9	38.2	38.2	–
Rate of pension progression + 25 basis points	10.9	10.9	–	9.9	9.9	–
Rate of pension progression – 25 basis points	(10.3)	(10.3)	–	(9.4)	(9.4)	–
Increase of 2 years in life expectancy	36.5	36.5	–	33.1	33.1	–

The assumptions for all sensitivity calculations were not performed jointly, but rather individually for each calculation assumption examined.

Contributions to pension plans

Contributions to state pension plans in the amount of EUR 38.1 million and EUR 30.5 million were recorded in 2023 and 2022, respectively, thereof EUR 30.2 million and EUR 27.1 million in Germany as well as EUR 3.5 million and EUR 0.0 million in Switzerland. Payments amounting to EUR 3.0 million and EUR 3.3 million were made for the newly established company pension plans in Germany in 2023 and 2022, respectively.

15. Revenue

The following table contains the revenue separated in subgroups, divisions and the time of revenue recognition:

EUR million	2023	2022
Currency Technology	1,075.0	1,004.2
Mobile Security	1,187.1	1,013.3
Veridos	253.8	192.3
secunet	393.7	347.2
Other[1]	63.7	(29.9)
	2,973.3	**2,527.1**

[1] Group consolidation, revenues Corporate Center (a52) und Netcetera (proportionate since acquisition).

In the division Banknote Solutions and in the subgroups Veridos, secunet and Netcetera, revenue is mainly recognized over time, whereas in the subgroup Mobile Security, revenue is largely recognized at a point in time. In the division Currency Management Solutions, revenue is also increasingly recognized over time, but still, predominantly, at a point in time.

16. Income and Expenses Relating to Other Periods

EUR million	2023	2022
Income relating to other periods	24.3	62.9
Expenses relating to other periods	(1.8)	(14.3)
	22.5	**48.6**

Income relating to other periods consists primarily of releases of warranty provisions, provisions for onerous contracts and other provisions included in cost of goods sold, as well as reversals of allowances (see Note 3 “Accounts Receivable Trade and Other Accounts Receivable, net”) recorded in selling expenses. For the most part, expenses relating to other periods comprise tax expenses for prior years recorded in income tax expense.

17. Other Financial Income, net

EUR million	2023	2022
Gains/(losses) from trading securities, net	8.1	(9.4)
Foreign currency exchange gains/(losses), net	(9.7)	(12.9)
Gains/(losses) from derivative financial instruments, net	(1.8)	12.7
Revaluation of former investments in associates	27.9	–
Impairment of investments in associates	(2.7)	–
Other financial income/(expenses), net	2.8	0.2
	24.6	**(9.4)**

The changes in net unrealized gains and losses on trading securities included in earnings during the fiscal years ending December 31, 2023 and 2022 were EUR 6.5 million and EUR – 10.7 million, respectively.

For further information on the revaluation of former investments and impairment on investments in associates, see Notes 6 and 24.

18. Interest Income and Interest Expense

EUR million	2023	2022
Interest income		
Loans and receivables	0.5	0.4
Cash and cash equivalents/short-term investments	3.9	1.4
Trading securities	0.8	0.5
Other	0.6	0.3
	5.8	**2.6**
Interest expense		
Loans and receivables	0.4	0.4
Financial liabilities and lease obligations	18.1	8.8
Other provisions	0.3	0.1
Provisions for pensions	18.8	9.6
Taxes payable	0.7	0.1
Other	0.3	0.2
	38.6	**19.2**

Interest income and expense relating to financial assets and financial liabilities that are not valued at fair value are as follows:

EUR million	2023	2022
Interest income		
Loans and receivables	0.5	0.4
Cash and cash equivalents/short-term investments	3.9	1.4
	4.4	**1.8**
Interest expense		
Loans and receivables	0.4	0.4
Financial liabilities measured at amortized cost	18.1	8.8
	18.5	**9.2**

19. Income Taxes

Income tax expense

Income tax expense for fiscal years 2023 and 2022 is comprised of:

EUR million	2023	2022
Current income tax		
Current year income tax expense	(59.6)	(56.3)
Income tax expense for prior periods	(0.2)	(11.8)
	(59.8)	**(68.1)**
Deferred income tax		
Gross expenditure from origination and reversal of temporary differences and tax loss carryforwards	16.5	2.8
Income tax expense from changes in tax rates and introduction of new taxes	(1.1)	(0.9)
Change in usability of tax loss carryforwards	(7.2)	(4.8)
	8.2	**(2.9)**
Income tax expense, net	**(51.6)**	**(71.0)**

In fiscal year 2023, G+D was subject to German federal corporate tax at a base rate of 15.0 % for the parent company plus a solidarity surcharge of 5.5 % on federal corporate taxes payable. Hence, the statutory rate consisted of a federal corporate tax rate of 15.83 % and trade tax of 15.74 %, resulting in a combined tax rate of 31.6 %.

Reconciliation between the expected and actual income tax expense

Following is a reconciliation of the expected income tax expense to the actual income tax expense which was recorded. The calculation of the expected income tax expense is based on the multiplication of income before income tax at the German corporate combined statutory rate of 31.6 % and 31.5 % in 2023 and 2022, respectively.

EUR million	2023	2022
Expected income tax expense	(45.4)	(47.8)
Foreign taxation differential	6.4	7.5
Non-deductible expenses	(7.2)	(9.6)
Changes in tax rates	(1.1)	(0.9)
Tax-free income	10.7	0.2
Additions due to tax risks and tax payments (refunds) for prior years	(0.6)	(11.3)
Changes in tax loss carryforwards and temporary differences	(7.2)	(4.8)
Withholding taxes	(7.1)	(4.4)
Other	(0.1)	0.1
Actual income tax expense	**(51.6)**	**(71.0)**

Deferred tax assets and liabilities

The gross values of deferred tax assets and liabilities as of December 31, 2023 and 2022 are attributable to the following balance sheet line items:

EUR million

	Assets		Liabilities		Net	
	Dec 31, 2023	Dec 31, 2022	Dec 31, 2023	Dec 31, 2022	Dec 31, 2023	Dec 31, 2022
Financial assets	2.4	2.5	–	(0.2)	2.4	2.3
Accounts receivable trade and other receivables, net	7.3	2.5	(2.5)	(2.2)	4.8	0.3
Contract assets	(0.7)	6.9	(59.9)	(39.2)	(60.6)	(32.3)
Inventories, net	82.5	58.5	(1.7)	–	80.8	58.5
Other assets	1.6	2.2	(3.7)	(0.3)	(2.1)	1.9
Intangible assets	9.8	12.9	(36.7)	(32.9)	(26.9)	(20.0)
Property, plant and equipment (excluding Right-of-Use)	3.5	2.9	(13.6)	(14.0)	(10.1)	(11.1)
Right-of-Use	–	–	(17.3)	(20.2)	(17.2)	(20.2)
Accounts payable trade and other accounts payable	0.1	1.9	(14.1)	–	(14.0)	1.9
Contract liabilities	5.3	1.0	(23.1)	(30.1)	(17.8)	(29.1)
Provisions	14.7	16.1	(3.3)	(1.6)	11.4	14.5
Financial liabilities	21.4	1.1	–	–	21.4	1.1
Lease obligations	17.4	21.6	–	–	17.3	21.6
Deposits received/deferred income	0.4	0.2	–	(0.4)	0.4	(0.2)
Pensions and related liabilities	68.5	46.4	–	–	68.5	46.4
Other liabilities	4.2	3.9	(24.8)	(3.0)	(20.6)	0.9
Tax loss carryforwards	33.9	25.1	–	–	33.9	25.1
Deferred tax assets/(liabilities), gross	272.3	205.7	(200.7)	(144.1)	71.6	61.6
Set-off of tax	(165.8)	(117.1)	165.8	117.1	–	–
Deferred tax assets/(liabilities), net	**106.5**	**88.6**	**(34.9)**	**(27.0)**	**71.6**	**61.6**

[1] Changes in consolidation structure contain changes from intangible assets in the amount of EUR –11.8 million.

As of December 31, 2023 and December 31, 2022 net deferred tax assets relating to leases in the amount of EUR 0.1 million and EUR 1.4 million were recorded, respectively.

The changes in deferred tax assets, net included in net income or other comprehensive income for fiscal years 2023 and 2022 are included in the following summary:

EUR million	2023	2022
Deferred tax assets, net as of January 1	61.6	140.2
Changes affecting net income	8.3	(2.8)
Changes not affecting net income		
Changes in consolidation structure	(11.8)	(8.2)
Changes in net deferred tax assets recognized in other comprehensive income resulting from deferred tax assets on actuarial gains and losses, cash flow hedges and foreign currency translations	13.5	(67.6)
Deferred tax assets, net as of December 31	**71.6**	**61.6**

Net Change	Recorded in Income Statement	Changes in consolidation structure[1]/ Recorded in Other Comprehensive Income	Net Change	Recorded in Income Statement	Changes in consolidation structure[1]/ Recorded in Other Comprehensive Income
2023	**2023**	**2023**	**2022**	**2022**	**2022**
0.1	0.1	–	(1.4)	(0.6)	(0.8)
4.5	4.6	(0.1)	–	–	–
(28.3)	(28.2)	(0.1)	(22.0)	(22.0)	–
22.3	22.5	(0.2)	(5.0)	(5.0)	–
(4.0)	(3.9)	(0.1)	44.1	44.1	–
(6.9)	4.9	(11.8)	(8.4)	(0.1)	(8.3)
1.0	1.2	(0.2)	6.4	6.5	(0.1)
3.0	3.0	–	(10.8)	(10.8)	–
(15.9)	(15.9)	–	(2.5)	(2.5)	–
11.3	11.3	–	(30.7)	(30.7)	–
(3.1)	(3.0)	(0.1)	(5.6)	(5.6)	–
20.3	20.3	–	0.4	0.4	–
(4.3)	(4.3)	–	4.3	4.3	–
0.6	0.6	–	0.1	0.1	–
22.1	7.7	14.4	(69.7)	(2.9)	(66.8)
(21.5)	(21.4)	(0.1)	17.1	16.9	0.2
8.8	8.9	(0.1)	5.1	5.1	–
10.0	8.3	1.7	(78.6)	(2.8)	(75.8)
–	–	–	–	–	–
10.0	**8.3**	**1.7**	**(78.6)**	**(2.8)**	**(75.8)**

Deferred tax assets not recorded in the balance sheet

The amount of deductible timing differences and tax loss carryforwards for which deferred tax assets were not recorded are as follows:

EUR million	2023		2022	
	Gross amount	Tax effect	Gross amount	Tax effect
Deductible temporary differences	9.1	2.7	4.1	1.1
Unused tax losses	186.7	50.6	165.7	46.0
	195.8	**53.3**	**169.8**	**47.1**

The unused tax losses expire as follows:

EUR million	2023		2022	
	Gross amount	Expiration date	Gross amount	Expiration date
Limited	17.1	2024 – 2031	4.5	2027 – 2031
Unlimited	169.6	–	161.2	–

Furthermore, deferred tax assets in the amount of EUR 33.9 million and EUR 25.1 million on tax loss carryforwards in the amount of EUR 119.0 million and EUR 85.8 million were recorded as of December 31, 2023 and 2022, respectively. In the 2023 fiscal year, loss carryforwards were used that had been impaired in the previous year. The use of these loss carryforwards led to a reduction in current tax of EUR 1.9 million.

The determining factor in recognizing deferred tax assets is the probability of the reversal of the temporary differences which resulted in the recognition of the deferred tax assets and future taxable profit against which the unused tax losses can be offset. This is dependent on future taxable profits arising in those periods in which taxable temporary differences reverse and tax losses carryforwards may be utilized. As of December 31, 2023, significant deferred tax assets were recorded on tax loss carryforwards by the following companies: Giesecke+Devrient GmbH, Munich, EUR 26.5 million, GIESECKE + DEVRIENT EPAYMENTS IBERIA S.A., Barcelona, EUR 1.5 million, secunet Security Networks AG, Essen, EUR 1.5 million Giesecke+Devrient Mobile Security Sweden AB, Stockholm, EUR 2.5 million and Netcetera AG, Zurich, EUR 1.8 million. Expected taxable profits based on the forecasts for the next five years are recognized by the respective companies. Based upon the level of historical taxable income and projections of future taxable income, G+D believes that it is not probable that the benefits of deductible timing differences and carryforward tax losses in the amount of EUR 195.8 million and EUR 169.8 million will be realized and therefore has not recognized deferred tax assets for these amounts in 2023 and 2022.

Due to the sustained positive earnings expectations in the G+D tax group in Germany, deferred tax assets in the amount of EUR 26.5 million on tax loss carryforwards and the excess of deferred tax assets in the amount of EUR 44.2 million due temporary differences were recorded in 2023 although a negative tax result was achieved in the fiscal year. As of December 31, 2023, the G+D tax group recorded deferred tax assets in the amount of EUR 70.6 million since the company expects profits as a result of positive business developments.

Income tax on dividends

As of December 31, 2023 and 2022, G+D recorded deferred tax liabilities on cumulative earnings in subsidiaries and investments that are intended for distribution. Furthermore, deferred taxes were recorded on the taxable temporary differences relating to investments in associated companies and joint ventures. As of December 31, 2023 and 2022, the amount of these obligations was EUR 4.0 million and EUR 3.1 million, respectively.

Temporary differences relating to investments in subsidiaries for which deferred tax liabilities were not recorded amounted to EUR 0.0 million and EUR 4.3 million as of December 31, 2023 and 2022, respectively. Deferred tax liabilities were not recognized here, as G+D is able to control the timing of the reversal and the temporary differences will not be reversed in the foreseeable future.

Global minimum tax

The introduction of minimum taxation (MinBestRL-UmsG of December 21, 2023) as of January 1, 2024 has no effect on the tax expense in fiscal year 2023. The company is applying the safe harbor sanction relief for 2024. Nevertheless, the company assumes that in the context of activities abroad and an associated effective tax rate below 15 %, there will be no material minimum taxation for 2024.

20. Equity

As of December 31, 2023 and 2022, the nominal value of the treasury stock amounted to EUR 4.3 million, respectively.

Unappropriated reserves amounted to EUR 698.1 million and EUR 749.9 million as of December 31, 2023 and 2022, respectively.

With respect to capital management, the main objective of Giesecke+Devrient is to secure its continuation as well as generate shareholder value, i.e. in the form of dividend payments. As of December 31, 2023 and 2022, the equity ratio amounted to 28.3 % and 29.2 %, respectively. G+D is not subject to external minimum capital requirements.

21. Financial Instruments

The following tables incorporate the carrying amounts and fair values of G+D's financial instruments. The pure exit price is thereby understood as the fair value of a financial instrument. This is the price at which a transaction to sell an asset or to transfer a liability would take place under current market conditions.

The tables do not contain information relating to fair values of financial assets or liabilities that are not valued at fair value if the carrying amount represents a reasonable approximation of the fair value.

Furthermore, the following tables contain an allocation of the fair value measurement of classes of financial assets and liabilities to levels in accordance with IFRS 13 as of December 31, 2023 and 2022:

EUR million						Dec 31, 2023
	Classification	**Carrying amount**	**Fair value**	Level 1	Level 2	Level 3
Financial assets						
Loans and receivables[1]	Amortized cost	545.8	–	–	–	–
Financial assets held for trading[2]						
Derivative financial assets	FVTPL	5.7	5.7	–	5.7	–
Derivative financial assets (interest swap)	N/A	1.0	1.0	–	1.0	–
Trading securities	FVTPL	–	–	–	–	–
Total		**6.7**				
Investment securities	FVTPL	11.6	11.6	11.6	–	–
Investments in other related parties	FVOCI	18.7	18.7	–	–	18.7
Special classes						
Cash and cash equivalents[3]	Amortized cost	323.5	–	–	–	–
Short-term investments	Amortized cost	4.4	–	–	–	–
Total financial assets		**910.7**				

[1] Amount does not include prepayments in the amount of EUR 77.8 million and EUR 59.4 million as of December 31, 2023 and 2022, respectively, as these are not included in the scope of IFRS 7.
[2] Amount does not include cash surrender value of reinsurance in the amount of EUR 17.1 million and EUR 20.1 million as of December 31, 2023 and 2022, respectively, as this is not included in the scope of IFRS 7.
[3] Cash and cash equivalents include cash in the amount of EUR 0.1 million and EUR 0.1 million, cash in banks in the amount of EUR 302.8 and EUR 595.5 million as well as short-term investments in the amount of EUR 20.6 million and EUR 14.4 million, as of December 31, 2023 and 2022, respectively.

EUR million						Dec 31, 2023
	Classification	**Carrying amount**	**Fair value**	Level 1	Level 2	Level 3
Financial liabilities						
Financial liabilities	Amortized cost	507.7	490.0	–	490.0	–
Accounts payable	Amortized cost	314.9	–	–	–	–
Purchase price consideration	Amortized cost	64.1	–	–	–	–
Liabilities from put-options – current	FVTPL	9.3	9.3	–	–	9.3
Liabilities from put-options – non-current	FVTPL	2.8	2.8	–	–	2.8
Conditional purchase price consideration[1]	FVTPL	8.3	8.3	–	–	8.3
Total		**907.1**				
Derivative financial liabilities	FVTPL	0.7	0.7	–	0.7	–
Special class						
Lease obligations	N/A	86.7	–	–	–	–
Total financial liabilities		**994.5**				

[1] Conditional purchase price consideration as of December 31, 2023 in the amount of EUR 8.3 million is reported under other non-current liabilities (refer to Note 24 “Business Combinations”).

				Dec 31, 2022
Carrying amount	**Fair value**	Level 1	Level 2	Level 3
523.4	–	–	–	–
2.8	2.8	–	2.8	–
2.6	2.6	–	2.6	–
86.8	86.8	86.8	–	–
92.2				
11.0	11.0	11.0	–	–
12.6	12.6	–	–	12.6
610.0	–	–	–	–
0.2	–	–	–	–
1,249.4				

				Dec 31, 2022
Carrying amount	**Fair value**	Level 1	Level 2	Level 3
549.2	478.7	–	478.7	–
383.0	–	–	–	–
15.7	–	–	–	–
–	–	–	–	–
–	–	–	–	–
11.0	11.0	–	–	11.0
958.9				
0.5	0.5	–	0.5	–
89.7	–	–	–	–
1,049.1				

In the tables, financial instruments measured at fair value are allocated to levels in accordance with IFRS 7, Financial Instruments: Disclosures. Thereby, the fair value measurement of a financial instrument is allocated in its entirety to the level for which inputs are material to determine its fair value. At level 1, fair values are mainly determined by using quoted prices from active markets for identical financial assets or liabilities. The fair values at level 2 are determined via market comparison procedures based on observable quoted prices for similar financial assets or liabilities. The discounted cash flow model is used here, in which the future cash flows are discounted with a risk-adjusted interest rate. Fair value measurements at level 3 are mainly based on unobservable market data. The valuation model takes cash flows into account, discounted with a risk-adjusted interest rate. 10 % changes in cash flow and 1 % changes in the discount rate do not result in any significant changes. In 2023, Giesecke+Devrient determined fair values of financial instruments based at level 1, level 2 and level 3. In 2023 and 2022, no material reclassifications between the levels were recorded.

The fair value of foreign currency forward contracts is based on mark-to-market since similar contracts are being traded on active markets. As of December 31, 2023 and 2022, these derivative financial instruments are stated at fair value and recorded on the balance sheet under current financial assets in the amount of EUR 5.7 million and EUR 2.8 million and under current financial liabilities in the amount of EUR 0.7 million and EUR 0.5 million, respectively.

The tables below show the reconciliation of the opening balance to the ending balance for level 3 fair values:

Investments in other related parties

EUR million	2023	2022
As of January 1	12.6	10.8
Acquisitions	3.9	–
Acquisitions/capital increases	7.7	1.8
Sales	(1.4)	–
Devaluations	(3.9)	–
Currency effects	(0.2)	–
As of December 31	**18.7**	**12.6**

Conditional purchase price consideration and liabilities from put-options

EUR million	2023	2022
As of January 1	11.0	–
Acquisitions	12.1	11.0
Income from revaluation	(2.7)	–
As of December 31	**20.4**	**11.0**

The nominal volume of foreign currency forward contracts entered into by Giesecke+Devrient as of December 31, 2023 amounted to:

in million foreign currency units	Purchase contracts	Sales contracts
US dollar	–	201.2
Australian dollar	–	10.0
Brazilian real	–	35.0
British pound	–	7.0
Chinese renminbi	–	100.0
Indonesian rupiah	–	67,000.0
Indian rupee	–	500.0
Canadian dollar	17.0	–
Hong Kong dollar	–	31.0
South African rand	–	190.0
Swiss franc	59.4	20.0

Prior Year 2022

in million foreign currency units	Purchase contracts	Sales contracts
US dollar	–	137.2
Australian dollar	–	9.0
Brazilian real	–	39.0
British pound	–	6.0
Chinese renminbi	–	205.0
Indian rupee	–	1,000.0
Canadian dollar	–	5.0
Hong Kong dollar	–	31.0
South African rand	–	140.0
Swiss franc	25.0	–

Financial instruments not valued at fair value

Cash and cash equivalents, short-term investments, as well as the current portion of accounts receivable, other assets, loans, trade accounts payable and other accounts payable, and other liabilities.
The carrying amounts of these financial instruments are considered to approximate fair value because of the relatively short period of time between origination and their expected realization.

Financial instruments valued at fair value

The fair values of non-derivative financial instruments for the individual classes are as follows:

Marketable securities
Debt and equity securities are carried at fair value, which is based on quoted market prices at the balance sheet date.

Investments
If the fair value cannot be readily determined, acquisition cost are used as the best estimate of the fair value.

Non-current financial assets and financial liabilities
The fair value is determined based on the expected future cash flows discounted using the prevailing market rate as of the balance sheet date for similar maturities and contracts.

As of December 31, 2023 and 2022, there were no significant differences between the fair values and the carrying values of non-current financial assets.

Impairment losses and reversals of impairment losses during fiscal years 2023 and 2022 related solely to financial assets in the class "loans and receivables".

EUR million	2023	2023
Impairment losses	(8.5)	(26.6)
Reversals of impairment losses	3.7	5.0
	(4.8)	**(21.6)**

Net gains and losses from financial assets and liabilities by measurement category amounted to:

EUR million	2023	2023
Financial assets measured at amortized cost	(6.2)	(31.1)
Financial assets and financial liabilities held for trading, at fair value through profit or loss	5.8	(4.0)
Financial assets held for trading, at fair value through other comprehensive income	(3.9)	2.6
Financial liabilities measured at amortized cost	1.0	(0.3)
	(3.3)	**(32.8)**

Net gains and losses on loans and receivables consist of results from impairments, reversals of impairments and foreign currency exchange effects.

Net gains and losses on financial assets and liabilities measured at fair value through profit and loss contain results from changes in fair market values and adjustments on settlement of these financial instruments.

Net gains and losses on financial assets and liabilities measured at fair value through other comprehensive income contain changes in fair market values of these financial instruments.

Net gains and losses from financial liabilities measured at amortized cost comprise foreign currency exchange effects.

22. Financial Risk

Due to the global scope of its operations, the Group is exposed to a variety of financial risks, notably counterparty default risks, liquidity risks as well as market price risks through fluctuations of exchange rates, interest rates and securities. Risks on the procurement side arise from the increase in purchase prices. The financial situation as well as the Group's profitability may be adversely affected by these risks. Giesecke+Devrient manages these risks predominantly within its current business and finance activities as well as based on specific written guidelines. The aim of the financial risk management is the transparency of the risks for the Group and their limitation by suitable countermeasures.

All of the above risks are monitored by risk management procedures. Financial risks are part of the monthly risk reporting for management as well as part of the regular reports to the Supervisory and Advisory Boards.

Counterparty Default Risk

The counterparty default risk is the risk of financial losses in the event that a customer or a contracting partner of a financial instrument fails to meet its contractual obligations. The risk of default generally refers to cash and cash equivalents, trade receivables, contract assets as well as to debt instruments held as financial assets by the Group.

The Group's exposure to counterparty default risk is determined mainly by the individual characteristics of each customer. Giesecke+Devrient handles the risk of default by using an internal evaluation method relating to the solvency of the customer. Factors which are taken into account include those risks of default which may influence the customer base such as country default risk and regions in which the customer operates. Based on the determined internal rating, a classification is undertaken in credit rating categories A to C. Doubtful positions are limited and agreed payment terms are closely monitored. Where customer creditworthiness is an issue, measures to secure payment, such as confirmed and unconfirmed letters of credit, are – wherever possible – requested to minimize the credit risk. To fulfill reporting requirements in accordance with IFRS 7, the maximum exposure to credit risk with regard to financial assets corresponds with the carrying value of these financial assets.

With regard to the exposure of counterparty risks in the financial sector, Giesecke+Devrient ensures that the investment volume is spread widely in order to reduce default risks and excessive dependence on individual financial institutions. The banks with which Giesecke+Devrient conducts financial transactions are selected and regularly reviewed according to various criteria, particularly with regard to creditworthiness.

Liquidity Risk

Liquidity risk is the risk that the Group is potentially unable to fulfill its external financial obligations as contractually agreed in delivering cash or other financial assets. The objective of the Group liquidity management is to ensure, as far as possible, that sufficient liquidity is available to meet its liabilities whenever due under both normal and strained conditions without incurring unacceptable expenses or damaging the Group's reputation.

Securing liquidity therefore has the highest priority and is managed by holding a disposable liquidity reserve appropriate to the size of the company. This means maintaining sufficient cash and unused credit lines with banks. Additionally, appropriate financial instruments such as annual planning for all Group companies and short-term liquidity planning for all major Group companies are used. These planning instruments are complemented by a contractually agreed centralized cash management agreement in which all the main German and foreign Group companies participate.

In addition to sufficient cash deposits, the Group held cash credit lines of EUR 388.9 million (prior year EUR 365.5 million) as of December 31, 2023 to ensure adequate liquidity to cover fluctuations in operating activities. The principal part thereof is a long-term credit line of EUR 280.0 million provided by a consortium to Giesecke+Devrient GmbH running until October 2026. All lenders are banks with an investment grade. As of December 31, 2023, EUR 6.1 million of the total available credit line were drawn (previous year EUR 0.0 million).

In addition, loans incl. accrued interest and excluding charges amounted to EUR 580.1 million (previous year EUR 552.2 million). These consisted of EUR 120.0 million in syndicated loans, EUR 49.1 million in loans from the European Investment Bank EIB and EUR 265.0 million promissory note loans. The remaining amount includes loans from other third parties (including Bundesdruckerei, G+D Stiftung, G+D Stiftung Geldscheinsammlung, KfW development loan in favor of Veridos), local loans for subsidiaries as well as financial lease liabilities and purchase price liabilities which result from acquisitions.

The bank loan of EUR 120 million, reported in the balance sheet as a non-current financial liability, has a contractual term until October 2026. Due to existing loan agreements, repayment could occur significantly earlier: The bank has the right to call in the loan immediately if the leverage ratio (defined as financial debt/EBITDA) exceeds 3.0 at a valuation date (of each financial year-end). As of December 31, 2023, the leverage ratio was 0.8. We assume that there will be no breach of the loan agreements within the term.

The following tables show the carrying amount of all G+D Group liabilities (from financing, trade payables, outstanding purchase price payments and leases) with the contractually agreed (undiscounted) interest and principle payments, as well as derivative financial instruments with negative fair values:

EUR million	Nominal value	Effects from effective interest method	Carrying value	Gross outflows	Up to 1 year		1–2 years		2–3 years		3–4 years		4–5 years		over 5 years	
					Repayment	Interest	Repayment	Interest	Repayment	Interest	Repayment	Interest	Repayment	Interest	Repayment	Interest
Original financial liabilities																
Accounts payable trade and other liabilities			314.9	314.9	314.2		0.7									
Financial liabilities	586.1	2.2	583.9	632.2	98.7	17.9	161.4	15.6	171.3	10.0	77.1	4.4	68.0	1.5	5.6	0.7
Other liabilities	8.3	–	8.3	8.3	8.3											
Lease obligations			86.7	92.2	26.3	1.9	20.2	1.3	13.9	0.9	10.5	0.6	7.0	0.4	8.8	0.4
Derivative financial liablities																
Derivative financial instruments			0.7	(33.7)	(33.7)											
(gross inflow derivates)				33.0	33.0											
Total			**994.5**	**1,046.9**	**446.8**	**19.8**	**182.3**	**16.9**	**185.2**	**10.9**	**87.6**	**5.0**	**75.0**	**1.9**	**14.4**	**1.1**

EUR million	Nominal value	Effects from effective interest method	Carrying value	Gross outflows	Up to 1 year		1–2 years		2–3 years		3–4 years		4–5 years		over 5 years	
					Repayment	Interest	Repayment	Interest	Repayment	Interest	Repayment	Interest	Repayment	Interest	Repayment	Interest
Original financial liabilities																
Accounts payable trade and other liabilities			383.0	383.0	383.0											
Financial liabilities	552.2	3.0	549.2	592.6	111.2	12.9	18.1	10.6	131.7	10.3	154.5	6.7	65.0	3.6	66.9	1.1
Other liabilities	26.7	–	26.7	26.7	15.7		11.0									
Lease obligations			89.7	93.7	22.9	1.2	19.6	1.0	15.5	0.7	11.4	0.5	8.5	0.3	11.8	0.3
Derivative financial liablities																
Derivative financial instruments			0.5	(84.1)	(84.1)											
(gross inflow derivates)				83.6	83.6											
Total			**1,049.1**	**1,095.5**	**532.3**	**14.1**	**48.7**	**11.6**	**147.2**	**11.0**	**165.9**	**7.2**	**73.5**	**3.9**	**78.7**	**1.4**

All financial instruments held as of December 31, 2023 and December 31, 2022 for which payments were already contractually agreed have been included. Plan figures for future liabilities were not included. Amounts in foreign currencies were translated at the respective closing rate. Variable interest payments from financial instruments were determined by applying the last fixed interest rates before December 31, 2023 or December 31, 2022, respectively and can change with market interest rates. Financial liabilities that are repayable at any time are always assigned to the earliest time period.

The available cash liquidity is held in bank accounts. In order to generate positive interest income, term deposits and deposits at notice were utilized with the core banks. The decision on the investment period was based on short-term liquidity planning. There were no deposits at notice as of December 31, 2023 (previous year EUR 0.0 million). In addition, cash investments with a term of more than three months amounting to EUR 4.4 million (previous year EUR 0.3 million) were held by foreign subsidiaries.

Market Risk

Market risk is the risk that market prices such as exchange rates, interest rates or stock prices change, and thereby affect the Group's earnings or the value of the financial instruments held.

A Currency Risk
Due to its international focus, Giesecke+Devrient has both import and export deliveries and cash flows in various currencies. Currency risks arise from future transactions including both purchasing and selling activities. Recognized assets and liabilities as well as highly probable forecasted cashflows in foreign currency are considered. Furthermore, risk exposure occurs due to financing in foreign currency.

Risk exposure can occur at the level of Giesecke+Devrient GmbH as well as on the subsidiary level. For risk control reasons, these currency exposures can be concentrated at group level to Giesecke+Devrient GmbH by using internal foreign currency transactions with corresponding subsidiaries.

The objective of the hedging strategy is to reduce risks from exchange rate fluctuations. Apart from the ongoing endeavor to avoid currency risk or to mitigate the effects naturally, forward exchange contracts, options and swaps with financial institutions with high credit ratings are used to hedge group-wide currency exposures.

Forward exchange contracts as well as options are used exclusively for hedging and not for speculative purposes. In accordance with the risk management standards applicable to international banks, all trading activities are subject to financial controls (back office) that are independent from corporate finance.

Currency risks, which are solely driven by the consolidation effects in the balance sheet and profit and loss statements of foreign subsidiaries (translation risks) are generally not hedged against exchange rate fluctuations.

Giesecke+Devrient uses a multi-level process for its currency hedging:

1. In the hedging of recognized assets and liabilities (economic exposure), the net position for each foreign currency (reporting date) is ideally statically secured to 100 %. This applies to all currencies and includes outstanding trade receivables, outstanding trade payables as well as intercompany loans issued to subsidiaries.

2. In the hedging of forecasted transactions (from major projects) on unrecorded risks, hedging is performed on a case-by-case basis and, if possible, hedge accounting is applied (according to IAS 39).

3. In hedging of forecasted transactions (from other projects) the remaining exposure relating to currencies with significant exposure such as USD and RMB is hedged in phases based on a 12 months net position. The economic exposure exceeding the natural hedge may also be temporarily hedged.

The following table shows intercompany and third party receivables and payables in foreign currency as well as financial derivatives entered into for hedging purposes (net exposure or nominal value of financial derivatives > EUR 5.0 million), which are either accounted for or expected, as of December 31, 2023:

Net Currency Exposure as of December 31, 2023

Foreign currency risk in EUR million	AUD		BRL		CAD		CHF	
	2023	2022	2023	2022	2023	2022	2023	2022
Hedging of booked transactions								
Net exposure	7.2	8.0	6.0	6.0	(16.0)	6.8	23.0	0.8
Firm Comittment	–	–	–	–	–	–	–	–
Financial derivatives	(6.1)	(5.7)	(6.5)	(6.9)	11.6	(3.5)	(21.6)	–
Hedging of expected transactions (Cash Flow Hedge)								
Financial derivatives (FX Forwards)	–	–	–	–	–	–	–	–
Hedging of expected transactions (other projects)								
Financial derivatives (FX Forwards)	–	–	–	–	–	–	63.2	25.5
Financial derivatives (FX Options)	–	–	–	–	–	–	–	–
Net exposure Netting								
Financial derivatives (FX Forwards)	–	–	–	–	–	–	–	–

Net Currency Exposure as of December 31, 2023

Foreign currency risk in EUR million	GBP		HKD		IDR		INR	
	2023	2022	2023	2022	2023	2022	2023	2022
Hedging of booked transactions								
Net exposure	11.5	7.9	–	0.9	2.1	–	7.0	12.2
Firm Comittment	–	–	–	–	–	–	–	–
Financial derivatives	(8.1)	(6.8)	–	–	–	–	(5.4)	(11.3)
Hedging of expected transactions (Cash Flow Hedge)								
Financial derivatives (FX Forwards)	–	–	–	–	–	–	–	–
Hedging of expected transactions (other projects)								
Financial derivatives (FX Forwards)	–	–	(3.6)	(4.0)	(4.0)	–	–	(7.9)
Financial derivatives (FX Options)	–	–	–	–	–	–	–	–
Net exposure Netting								
Financial derivatives (FX Forwards)	–	–	–	–	–	–	–	–

Net Currency Exposure as of December 31, 2023

Foreign currency risk in EUR million	RMB		SEK		USD		ZAR	
	2023	2022	2023	2022	2023	2022	2023	2022
Hedging of booked transactions								
Net exposure	13.7	28.5	(9.2)	(5.0)	133.4	82.1	15.8	11.8
Firm Comittment	–	–	–	–	–	–	–	–
Financial derivatives	(12.7)	(27.9)	–	3.3	(139.4)	(120.0)	(9.3)	(7.7)
Hedging of expected transactions (Cash Flow Hedge)								
Financial derivatives (FX Forwards)	–	–	–	–	–	–	–	–
Hedging of expected transactions (other projects)								
Financial derivatives (FX Forwards)	–	–	–	–	(43.4)	(8.7)	–	–
Financial derivatives (FX Options)	–	–	–	–	–	–	–	–
Net exposure Netting								
Financial derivatives (FX Forwards)	–	–	–	–	–	–	–	–

AUD = Australian Dollar, CAD = Canadian Dollar, CHF = Swiss Franc, BRL = Brasilian Real, CAD = Canadian Dollar, GBP = British Pound, HKD = Hong Kong Dollar, IDR = Indonesian Rupiah, INR = Indian Rupee, RMB = Chinese Renminbi, SEK = Swedish Krona, USD = US-Dollar, ZAR = South African Rand

The valuation effects of the net exposures as of the balance sheet date influence the consolidated income statement.

Sensitivity analyses are used to determine the effects of hypothetical changes of the respective risk variable and their effects on the profits/losses and on the equity of the company as of the balance sheet date. Only major foreign currencies are considered.

The following shows the impacts of primary and derivative financial instruments exceeding EUR 2.0 million on equity as well as the profit and loss statement (without consideration of tax effects), assuming that the EUR had risen or fallen by 10.0 % against the specified foreign currencies as of December 31, 2023 and December 31, 2022, respectively:

Primary Financial Instruments as of December 31, 2023 (Impact > EUR 2 million)

Impact in EUR million

	Equity				Profit/Loss			
	2023		2022		2023		2022	
	–10.0 %	+10.0 %	–10.0 %	+10.0 %	–10.0 %	+10.0 %	–10.0 %	+10.0 %
CHF	–	–	–	–	(2.2)	2.2	–	–
GBP	–	–	–	–	(11.4)	11.4	–	–
RMB	–	–	–	–	(1.4)	1.4	2.8	(2.8)
USD	–	–	–	–	(13.3)	13.3	8.2	(8.2)

Derivative Financial Instruments as of December 31, 2023 (Impact > EUR 2 million)

Impact in EUR million

	Equity				Profit/Loss			
	2023		2022		2023		2022	
	–10.0 %	+10.0 %	–10.0 %	+10.0 %	–10.0 %	+10.0 %	–10.0 %	+10.0 %
CHF	–	–	–	–	4.7	(3.9)	(2.8)	2.3
RMB	–	–	–	–	–	–	(3.1)	2.5
USD	–	–	–	–	(20.2)	16.5	(14.3)	11.7

Global netting agreement and similar agreements

The Group enters into derivative contracts in accordance with the German Master Agreement (GMA). In certain cases – for example, if a payment default occurs - all outstanding transactions are terminated under this agreement, the value determined, and only the net amount is paid.

The German Master Agreement (GMA) does not fulfill the netting criteria for the balance sheet due to the fact that the Group currently has no legal right to net the recorded amounts. Only on future events such as a delay on bank loans or other credit events a legal right to net is enforceable.

The following chart reflects the carrying amount of financial instruments which are included in the described agreements.

December 31, 2023

in EUR million	Gross amounts of financial instruments in the statement of financial position	Related financial instruments that are not offset	Net amount
Financial assets			
Other investments, including derivatives			
Forward exchange contracts used for hedging	4.9	(0.6)	4.3
Other forward exchange contracts	–	–	–
Interest rate derivatives	1.0	–	1.0
Total financial assets	**5.9**	**(0.6)**	**5.3**
Financial liabilities			
Trade and other payables	(0.6)	0.6	–
Forward exchange contracts used for hedging	–	–	–
Other forward exchange contracts	–	–	–
Interest rate derivatives	–	–	–
Total financial liabilties	**(0.6)**	**0.6**	**–**

December 31, 2022

in EUR million	Gross amounts of financial instruments in the statement of financial position	Related financial instruments that are not offset	Net amount
Financial assets			
Other investments, including derivatives			
Forward exchange contracts used for hedging	1.1	(0.4)	0.7
Other forward exchange contracts	–	–	–
Interest rate derivatives	2.6	–	2.6
Total financial assets	**3.7**	**(0.4)**	**3.3**
Financial liabilities			
Trade and other payables	(0.4)	0.4	–
Forward exchange contracts used for hedging	–	–	–
Other forward exchange contracts	–	–	–
Interest rate derivatives	–	–	–
Total financial liabilties	**(0.4)**	**0.4**	**–**

Hedging of the cash flows using Cash Flow Hedge Accounting

As a matter of principle, major projects with a volume of > EUR 8 million are hedged via a forward exchange contract and checked whether they can be accounted for using cash flow hedge accounting (CFH).

As such, fluctuations in valuations of the derivatives can be avoided in the equity due to proportionate accounting of these individual transactions.

The Group designates the spot element of forward foreign exchange contracts to hedge its currency risk and applies a hedge ratio of 1:1. The forward elements of forward exchange contracts are excluded from the designation of the hedging instrument and are separately accounted for as cost of hedging in the financial result in the same way as for standalone derivatives. The Group's policy is to align the critical terms of the forward exchange contracts with the hedged item.

No new hedges were entered into in 2023 using cash flow hedge accounting.

There are strict requirements in order to apply cash flow hedge accounting. These are checked on initiation of the project as well as on an ongoing basis. If the criteria during the term of validity are no longer met, the cash flow hedge accounting is terminated and the effects are subsequently reported in the profit and loss statement. The economic hedge remains in force.

Within the framework of an effectiveness test the Group determines the existence of an economic relationship between the hedging instrument and hedged item based on the currency, amount and timing of their respective cash flows (critical terms match). The Group assesses whether the derivative designated in each hedging relationship has been effective in offsetting changes in cash flows of the hedged item using the hypothetical derivative method.

In these hedging relationships, ineffectiveness can arise from:

- the effect of the counterparties' and the Group's own credit risk on the fair value of the forward foreign exchange contracts which is not reflected in the change in the fair value of the hedged cash flows attributable to the change in exchange rates; and
- to a greater extent changes in the value of the hedged transactions.
- Ineffectiveness can also result from differing interest rate fixings of the hedged item versus the hedging instrument.

As of December 31, 2023 and December 31, 2022 respectively, the Group maintained the following instruments to hedge foreign exchange fluctuations:

2023

	Maturity	
	< 1 year	> 1 year
Foreign currency risk		
Forward exchange contracts		
Net exposure (in EUR million)	–	–
Average EUR/USD forward contract rate	–	–
Interest rate swaps		
Net exposure (in EUR million)	–	60.0
Average fixed rate	–	1.9 %

2022

	Maturity	
	< 1 year	> 1 year
Foreign currency risk		
Forward exchange contracts		
Net exposure (in EUR million)	–	–
Average EUR/USD forward contract rate	–	–
Interest rate swaps		
Net exposure (in EUR million)	–	60.0
Average fixed rate	–	1.9 %

As of December 31, 2023 and December 2022 respectively the amounts related to positions designated to the hedged underlying transactions are following:

December 31, 2023

EUR million	Change in value used for calculating hedge ineffectiveness	Cash flow hedge reserve	Balances remaining in the cashflow hedge reserve from hedging relationships for which hedge accounting is no longer applied
Foreign currency risk			
Sales (receivables)	–	–	–
Purchases (liabilities)	–	–	–
Interest rate risk			
Floating rate instruments	1.0	1.0	–

December 31, 2022

EUR million	Change in value used for calculating hedge ineffectiveness	Cash flow hedge reserve	Balances remaining in the cashflow hedge reserve from hedging relationships for which hedge accounting is no longer applied
Foreign currency risk			
Sales (receivables)	0.3	–	–
Purchases (liabilities)	–	–	–
Interest rate risk			
Floating rate instruments	2.6	2.6	–

Amounts referring to designated hedging instruments and their inefficiencies in relation to the hedging relationship are as follows:

2023

EUR million		Carrying amount					
	Nominal amount	Assets	Liabilities	Balance sheet account in which hedging instrument included	Changes in the value of the hedging instrument recognised in OCI	Amount reclassified from hedging reserve to profit or loss	Profit and loss account which is affected by the reclassification
Foreign currency risk							
Forward exchange contracts – sales (receivables)	–	–	–	Other investments and other payables (liabilities)	–	–	Revenue
Forward exchange contracts – purchases (liabilities)	–	–	–	–	–	–	–
Interest rate risk							
Interest rate swaps	60.0	1.0	–	Other financial assets and other liabilities and other liabilities (liabilities)	(0.8)	0.8	Other financing expenses

2022

EUR million		Carrying amount					
	Nominal amount	Assets	Liabilities	Balance sheet account in which hedging instrument included	Changes in the value of the hedging instrument recognised in OCI	Amount reclassified from hedging reserve to profit or loss	Profit and loss account which is affected by the reclassification
Foreign currency risk							
Forward exchange contracts – sales (receivables)	–	–	–	Other investments and other payables (liabilities)	0.3	(0.4)	Revenue
Forward exchange contracts – purchases (liabilities)	–	–	–	–	–	–	–
Interest rate risk							
Interest rate swaps	60.0	2.6	–	Other financial assets and other liabilities and other liabilities (liabilities)	2.8	0.2	Other financing expenses

The following table contains a reconciliation of the equity component risk categories and the analysis of the positions reported in other comprehensive income after tax, resulting from the reporting of cash flow hedges:

2023

EUR million	Hedging reserve
Balance as of January 1, 2023	**2.6**
Cashflow hedges	
Foreign currency risk – inventory purchases	–
Foreign currency risk – other items	–
Interest rate risk	(0.8)
Amount reclassified to profit or loss:	
Foreign currency risk – other items	–
Interest rate risk	(0.8)
Balance as of December 31, 2023	**1.0**

2022

EUR million	Hedging reserve
Balance as of January 1, 2022	**(0.3)**
Cashflow hedges	
Foreign currency risk – inventory purchases	–
Foreign currency risk – other items	0.3
Interest rate risk	2.8
Amount reclassified to profit or loss:	
Foreign currency risk – other items	(0.3)
Interest rate risk	(0.2)
Balance as of December 31, 2022	**2.6**

B Interest Rate Risk

Giesecke+Devrient's interest rate risks mainly arise from bank loans and the promissory note loans with their respectively agreed interest rates until maturity.

Financial instruments with variable interest rates are subject to cash flow risks which consist of uncertainty regarding the amount of future interest payments. This risk also exists for financial instruments with fixed interest rates when they are due to be reinvested or refinanced.

As of December 31, 2023, the values were as follows:

Interest rate risk financial instruments as of Decmber 31, 2023

EUR million	Average rate of interest		Total amount		Up to 1 year		1–2 years		2–5 years		Over 5 years	
	2023	2022	2023	2022	2023	2022	2023	2022	2023	2022	2023	2022
Fixed-interest financial instruments												
Financial liabilities (current and non-current)	2.1	2.2	459.5	494.7	107.6	107.5	127.9	29.0	223.4	291.3	0.6	66.9
Lease obligations	2.5	1.4	86.7	89.7	26.3	22.8	20.2	19.6	31.4	35.5	8.8	11.8
Variable-interest financial instruments												
Financial liabilities	5.1	1.8	130.9	79.4	(0.6)	19.4	33.5	0.1	93.0	59.9	5.0	–

Interest rate risk financial instruments as of Decmber 31, 2022

EUR million	Average rate of interest		Total amount		Up to 1 year		1–2 years		2–5 years		Over 5 years	
	2022	2021	2022	2021	2022	2021	2022	2021	2022	2021	2022	2021
Fixed-interest financial instruments												
Financial liabilities (current and non-current)	2.2	1.4	494.7	327.1	107.5	18.4	29.0	67.5	291.3	164.2	66.9	77.0
Lease obligations	1.4	1.9	89.7	72.4	22.8	19.0	19.6	17.0	35.5	25.6	11.8	10.8
Variable-interest financial instruments												
Financial liabilities	1.8	0.7	79.4	22.7	19.4	1.8	0.1	19.8	59.9	1.1	–	–

New interest rate derivatives were concluded for the first time again in the course of 2022 for a volume of EUR 60.0 million to hedge part of the variable interest loan provided by a banking syndicate. The interest rate swaps have a floating leg based on the 3-month Euribor. A hedging relationship was designated between the loan and the interest rate swap and this was accounted for in the form of a cash flow hedge. The Group applies a hedge ratio of 1:1.

An interest rate risk sensitivity analysis shows the effect of a 100 basis point (one percentage point) change in market interest rates on the income statement (excluding tax effects) and on equity (excluding tax effects). It is assumed that all other variables remain constant. However, as part of the sensitivity analysis required by IFRS 7, only the impact on net income and equity is considered and not the impact on future cash flows. Therefore, accrued interest payments carried as a liability are recalculated using the hypothetical market interest rate applicable at the reporting date.

The measurement of the interest rate swap designated in cash flow hedge accounting would result in an increase or decrease in equity of EUR 1.6 million (2022: EUR 2.1 million) or EUR –1.6 million as of December 31, 2023 (2022: EUR –2.1 million). The measurement of the cash flows of the remaining variable term loan (amounting to EUR 60.0 million) would result in an increase or decrease in the income statement of EUR 1.7 million (2022: EUR 2.4 million) or EUR –1.7 million (2022: EUR –2.4 million).

The existing cash liquidity is invested in overnight money and time deposits. The decision on the investment period is made on the basis of liquidity planning, partly in short-term investments with terms of < three months (liquid funds) and partly in investments with terms of > three months (current financial assets). With all investments, the security of the counterparty and the lowest possible price risk are paramount. The cash and cash equivalents item is not considered here.

C Special Investment Fund

In addition to money deposited directly at banks, assets as of December 31, 2023 of EUR 116.2 million (previous year EUR 98.1 million) are managed by a well-known German asset management company in a special investment fund. In 2022, a second segment was opened under the umbrella of the special investment fund, which is managed separately by a second asset manager. The investments consists of two portfolios of first-grade bonds (government bonds, corporate bonds and fixed income bonds) investment funds and equities (first-grade companies), as well as commodities and cash.

In 2023, the assets of the special fund were declared as plan assets and transferred to a contractual trust arrangement. Under this contractual trust arrangement, the company acts as trustor with other German subsidiaries as beneficiaries.

The risk of this investment is determined monthly using the Value-at-Risk (VaR) indicator. The Value-at-Risk indicates the loss amount that will not be exceeded in a period of 10 days with a probability of 99.0 %.

The following values arise for the special investment fund (Segment 1 and 2) as of December 31, 2023 (VaR and assets):

Segment 1

EUR million	2023
Market value fund (incl. derivatives)	84.2
99.0 % VaR	1.5
Assets	
Equities incl. equity funds	33.3
Bonds incl. bond funds	38.3
Derivatives	–
Commodity funds	5.9
Cash	6.6
Other assets	0.1

Segment 2

EUR million	2023
Market value fund (incl. derivatives)	32.0
99.0 % VaR	0.9
Assets	
Equities incl. equity funds	12.4
Bonds incl. bond funds	11.6
Derivatives	7.4
Commodity funds	–
Cash	0.6
Other assets	–

The following values arise for the special investment fund as of December 31, 2022 (VaR and assets):

Segment 1

EUR million	2022
Market value fund (incl. derivatives)	78.6
99.0 % VaR	2.5
Assets	
Equities incl. equity funds	34.1
Bonds incl. bond funds	31.2
Derivatives	0.3
Commodity funds	6.3
Cash	6.6
Other assets	0.1

Segment 2

EUR million	2022
Market value fund (incl. derivatives)	19.5
99.0 % VaR	0.7
Assets	
Equities incl. equity funds	5.4
Bonds incl. bond funds	6.0
Derivatives	3.6
Commodity funds	–
Cash	4.5
Other assets	–

In addition to the special investment fund, Giesecke+Devrient holds securities that are classified as available-for-sale securities. The market value of the short-term investments as of December 31, 2023 was EUR 9.8 million (previous year EUR 9.1 million), the market value of the long-term investments as of December 31, 2023 was EUR 1.8 million (previous year 1.9 million euros). This includes, in particular, investments in investment funds that serve to secure pension and semi-retirement obligations against insolvency. Due to minor fluctuations in the value of these shares, no sensitivity analysis is performed. G+D has not identified any concentration of risk as defined in IFRS 7.34.

The information in this section is disclosed in accordance with IFRS 7, Financial Instruments: Disclosures.

23. Contract balances

Descriptions of significant changes in contract assets and contract liabilities:

EUR million	Dec 31, 2023	Dec 31, 2022
Contract assets at the beginning of the period	**201.8**	**269.2**
Currency differences	(3.5)	1.4
Transfers from contract assets to accounts receivable trade at the beginning of the year	(168.0)	(250.4)
Impairment losses on contract assets	(0.5)	(0.6)
Changes in the measure of progress	233.8	182.2
Changes in consolidation structure	4.1	–
Contract assets at the end of the period	**267.7**	**201.8**
Contract liabilities at the beginning of period	**282.4**	**213.0**
Currency differences	(3.1)	0.9
Revenue recognized that was included in the contract liability balance at the beginning of the period	(217.2)	(176.7)
Prepayments received excluding revenue during the period	232.1	245.3
Changes in consolidation structure	7.4	–
Other changes	0.3	(0.1)
Contract liabilities at the end of the period	**301.9**	**282.4**

The Group does not make use of the exemption option of IFRS 15.121. The transaction prices reported in accordance with IFRS 15.120 were not reduced by components that represent consideration from customer contracts.

24. Business Combinations

G+D recognizes the results of operations of the acquired business starting from the date of acquisition for business combinations. The net assets acquired are recorded at fair value at the date of acquisition. The excess of the purchase price over the fair value of tangible and identifiable intangible net assets acquired is recorded as goodwill in the accompanying consolidated balance sheet. In addition, gains and losses on disposals or liquidations are recorded on the exit date of the affiliated company.

Effective May 17, 2022 (time of acquisition) secunet acquired 100 % in the shares in SysEleven GmbH, Berlin, a German provider of hosting and cloud services. The purpose of the acquisition was to provide the customers of secunet with an additional option for a secure cloud infrastructure. The consideration of EUR 61.3 million comprises two components, a cash payment in the amount of EUR 50.3 million as well as a conditional consideration in the amount of EUR 11.0 million. The incidental acquisition costs amounted to EUR 1.5 million and were recorded within general and administrative expenses. As of December 31, 2022, the conditional consideration is recorded as other non-current liabilities. The amount of the conditional consideration depends on qualitative and quantitative objectives for fiscal years 2022 and 2023. The quantitative objectives are based on the cumulated minimum sales and result for fiscal years 2022 and 2023. The qualitative objectives are composed of criteria of operations of SysEleven GmbH. Based on a calculation of probabilities of occurrence as well as a weighting of factors a fair value in the amount of EUR 8.3 million (prior year EUR 11.0 million) was calculated. The subsequent measurement is influenced by the interest situation on the respective valuation date as well as the assessments of target achievement made on the valuation date. The resulting income from the change in valuation of EUR 2.7 million was recorded within cost of goods sold. In addition, the conditional consideration was reclassified to other current liabilities as of December 31, 2023. As G+D holds 75.5 % in the shares of secunet, G+D in turn holds 75.5 % in the shares of SysEleven GmbH. Thus, G+D assumed control over SysEleven GmbH and consolidates the company in full since May 2022.

The difference between the fair value of the consideration in the amount of EUR 61.3 million and the fair value of the identifiable net assets of SysEleven GmbH amounting to EUR 24.0 million at the time of acquisition was recorded as goodwill and amounted to EUR 37.3 million. The goodwill is essentially attributable to the strong market position of SysEleven GmbH in the sector of DSGVO conforming cloud platforms as well as the expansion of the service portfolio and the expected synergies in the public sector that result from the business combination. The goodwill recorded is not deductible for tax purposes.

The identifiable assets acquired and liabilities assumed consist of (in EUR million):

EUR million	
Cash and cash equivalents	1.5
Accounts receivable trade and other receivables, net	1.9
Intangible assets	31.2
Property, plant and equipment	6.5
Accounts payable trade and other accounts payable	2.3
Provisions	0.6
Lease obligations	4.2
Financial liabilities	0.6
Deferred tax liabilities	9.4
Identifiable acquired net assets	**24.0**

In the period between May 17, 2022 and December 31, 2022, SysEleven contributed EUR 8.4 million in net sales and net losses in the amount of EUR 2.1 million to the Group result. If the acquisition had occurred on January 1, 2022, the Group net income in fiscal year 2022 would have amounted to EUR 82.2 million and the Group sales revenues would have amounted to EUR 2,534.8 million.

Effective December 1, 2022, G+D acquired the payment and identity business of VALID USA Inc. and Valid Identity Solutions, LLC in the USA. The consideration of EUR 56.5 million comprises two components, a cash payment in the amount of EUR 40.8 million as well as an additional consideration in the amount of EUR 15.7 million. The additional consideration was paid in almost the same amount. The deviation from the actual payment is immaterial. The incidental acquisition costs in the amount of EUR 2.7 million were recorded within general and administrative expenses. G+D takes over the local production and personalization facilities as well as their employees. Consequently, the transaction was classified as a business combination in accordance with IFRS 3.

Thus, G+D accelerates growth in one of the largest global markets for payment and identity and expands its existing business in regards to drivers' licenses.

The difference between the fair value of the consideration in the amount of EUR 56.5 million and the fair value of the identifiable net assets amounting to EUR 56.5 million at the time of acquisition amounted to EUR 0.0 million.

The identifiable assets acquired and liabilities assumed were determined based on a preliminary purchase price allocation and consist of (in EUR million):

EUR million	
Accounts receivable trade and other receivables, net	26.1
Inventories, net	29.9
Deferred tax assets	3.4
Intangible assets	1.8
Property, plant and equipment	24.3
Accounts payable trade and other accounts payable	12.0
Provisions	1.2
Lease obligations	12.6
Deferred tax liabilities	2.2
Other liabilities	1.0
Identifiable acquired net assets	**56.5**

In the period between December 1, 2022 and December 31, 2022, Valid contributed EUR 13.5 million in net sales and net income in the amount of EUR 1.0 million to the Group result. If the acquisition had occurred on January 1, 2022, the Group net income in fiscal year 2022 would have amounted to EUR 91.2 million and the Group sales revenues would have amounted to EUR 2,675.6 million.

Effective January 16, 2023 Veridos acquired additional 27.0 % of the shares in NetSeT Global Solutions d.o.o., Belgrade, a developer of complex information systems for the administration of citizen data, on the basis of a contract from November 24, 2022 and thus, holds 67.0 % of the shares in this company. As a result of increasing the shares, Veridos expands the global R&D team by about 80 FTE, gains additional control over the R&D activities and IP rights as well as establishes a sales base in the Balkans. Thus, G+D assumes control over NetSeT Global Solutions d.o.o. and consolidates the company in full since January 2023.

The consideration in the amount of EUR 3.3 million is comprised of two components: a cash payment of EUR 2.3 million as well as the fair value of accounts receivable to the company existing on the acquisition date in the amount of EUR 1.0 million.

The business combination was conducted according to the present access method in the consolidated financial statements.

The revaluation of the existing shares resulted in an income in the amount EUR 1.8 million which was recorded in other financial income, net. Therefore, the fair value of the existing shares added up to EUR 2.9 million.

As a result of the acquisition the goodwill was recorded as follows:

EUR million	
Consideration transferred	3.3
Non-controlling interest based on the percentage of the assets and liabilities recorded	0.7
Fair value of the shares previously held	2.9
Fair value of the identifiable net assets	(3.2)
Goodwill	**3.7**

The goodwill is essentially attributable to the skills of the NetSet workforce and the expected synergies. The goodwill recorded is not deductible for tax purposes.

In 2025, Veridos is obligated to acquire the remaining 33.0 % of the shares due to a put option. This option was recorded as non-current financial liabilities in the amount of EUR 2.8 million. The retained earnings were reduced by the corresponding amount.

The identifiable assets acquired and liabilities assumed consist of (in EUR million):

EUR million	
Cash and cash equivalents	1.6
Other assets	0.1
Accounts receivable trade and other receivables, net	1.7
Inventories, net	0.1
Financial assets	0.1
Intangible assets	1.9
Property, plant and equipment	0.4
Other liabilities	1.1
Contract liabilities	0.1
Accounts payable trade and other accounts payable	1.2
Lease obligations	0.2
Deferred tax liabilities	0.1
Identifiable acquired net assets	**3.2**

The assets and liabilities acquired were valued at their fair value as of the acquisition date. The non-controlling interest at the time of acquisition was determined by multiplying the minority interest (in %) by the fair value of the assets acquired (excluding goodwill) less the fair value of the liabilities assumed.

In fiscal year 2023, NetSet contributed EUR 4.3 million in net sales and net income in the amount of EUR 0.5 million to the Group result.

In 2023, G+D acquired a further 65.0 % of the shares in Netcetera Group AG, Zurich in two steps and thus, holds 95.0 % of the shares in this company. In the first step, G+D acquired 31.5 % of the shares on February 10, 2023 at a price of EUR 56.6 million in cash. Through the acquisition, G+D can support and influence the strategic direction of Netcetera Group via a more efficient control structure. Thus, G+D assumes control over Netcetera Group AG and consolidates the company in full since February 2023. In the second step, G+D acquired an additional 33.5 % of the shares on December 1, 2023 as part of the put options exercised by the minority shareholders. The consideration of EUR 62.3 million was recorded as an additional consideration within current financial liabilities. The non-controlling shareholders have further put options to sell the remaining 5.0 % of the shares, which can be exercised at various times (first in May 2024, last in June 2026). The liabilities from this put option amounting to EUR 9.3 million were also reported under current financial liabilities. Retained earnings decreased accordingly by EUR 71.6 million.

The consideration at acquisition date in the amount of EUR 58.6 million is comprised of two components: a cash payment of EUR 56.6 million in fiscal year 2023 as well as loans receivable from the company in the amount of EUR 2.0 million.

The business combination was conducted according to the present access method in the consolidated financial statements.

The revaluation of the existing shares resulted in an income in the amount EUR 26.1 million which was recorded in other financial income, net. Therefore, the fair value of the existing shares added up to EUR 53.7 million.

As a result of the acquisition the goodwill was recorded as follows:

EUR million	
Consideration transferred	58.6
Non-controlling interest based on the percentage of the assets and liabilities recorded	17.9
Fair value of the shares previously held	53.7
Fair value of the identifiable net assets	(48.5)
Goodwill	**81.7**

The goodwill is essentially attributable to the skills of the Netcetera workforce and the expected synergies. The goodwill recorded is not deductible for tax purposes.

The identifiable assets acquired and liabilities assumed consist of:

EUR million	
Cash and cash equivalents	4.8
Contract assets	2.9
Other assets	0.7
Accounts receivable trade and other receivables, net	16.7
Inventories, net	0.2
Deferred tax assets	3.0
Financial assets	0.1
Intangible assets	55.9
Property, plant and equipment	8.6
Investments accounted for under the equity method	8.1
Investments in other related parties	3.9
Contract liabilities	8.4
Other liabilities	5.4
Accounts payable trade and other accounts payable	5.9
Provisions	2.5
Accrual for income taxes and income taxes payable	0.1
Lease obligations	6.8
Deferred tax liabilities	10.8
Financial liabilities	8.0
Pensions and related liabilities	8.5
Identifiable acquired net assets	**48.5**

The assets and liabilities acquired were valued at their fair value as of the acquisition date. The non-controlling interest at the time of acquisition was determined by multiplying the minority interest (in %) by the fair value of the assets acquired (excluding goodwill) less the fair value of the liabilities assumed.

In fiscal year 2023, Netcetera contributed EUR 96.4 million in net sales and net losses in the amount of EUR 16.2 million to the Group result. If the acquisition had occurred on January 1, 2023, the Group net income in fiscal year 2023 would stayed at EUR 92.1 million and the Group sales revenues would have amounted to EUR 2,983.1 million.

This business combination incurred incidental acquisition costs in the amount of EUR 0.1 million. These costs were recorded within general and administrative expenses.

Effective May 17, 2023 (acquisition date), G+D acquired all of the shares in Mecomo Group, Unterschleissheim, a leading German provider of maintenance-free movement tracking of logistics objects. The acquisition expands G+D's solution portfolio for the Internet of Things. The consideration amounts to EUR 23.5 million and was paid in cash. The incidental acquisition costs amounted to EUR 0.1 million and were recorded within general and administrative expenses.

The difference between the fair value of the consideration in the amount of EUR 23.5 million and the fair value of the identifiable net assets of Mecomo amounting to EUR 10.9 million at the time of acquisition was recorded as goodwill and amounted to EUR 12.6 million. The goodwill is essentially attributable to the skills of the Mecomo workforce and the expected synergies. The goodwill recorded is not deductible for tax purposes.

The identifiable assets acquired and liabilities assumed consist of:

EUR million	
Cash and cash equivalents	1.7
Other assets	0.2
Accounts receivable trade and other receivables, net	1.2
Inventories, net	6.9
Financial assets	0.1
Intangible assets	7.5
Property, plant and equipment	0.6
Other liabilities	0.5
Accounts payable trade and other accounts payable	0.7
Provisions	0.1
Accrual for income taxes and income taxes payable	0.5
Lease obligations	0.3
Deferred tax liabilities	2.6
Financial liabilities	2.6
Identifiable acquired net assets	**10.9**

In the period between May 17, 2023 and December 31, 2023, Mecomo contributed EUR 3.8 million in net sales and net losses in the amount of EUR 0.5 million to the Group result. If the acquisition had occurred on January 1, 2023, the Group net income in fiscal year 2023 would have amounted to EUR 92.6 million and the Group sales revenues would have amounted to EUR 2,977.3 million.

Effective August 31, 2023, G+D acquired the remaining 50.0 % of the shares in E-Kart Elektronik Kart Sistemleri Sanayi ve Ticaret Anonim Sirketi, Gebze. The company focuses on the production and personalization of payment cards as well as the government ID and SIM card business. With the acquisition, G+D is strengthening its commitment to the Turkish market with the second largest payment card volume in Europe and accelerating growth there. Thus, G+D assumes control over E-Kart and consolidates the company in full since September 2023.

The consideration amounts to EUR 2.3 million and was fully paid in cash. The fair value of the existing shares was EUR 3.2 million. The sum of both amounts to EUR 5.5 million. The fair value of the identifiable net assets of E-Kart at the time of acquisition is EUR 6.3 million. The expense from the revaluation of the existing shares amounting to EUR 0.8 million is offset by badwill recognized through profit and loss in the same amount, both of which were recorded in other financial income/(expenses), net.

The Identifiable assets acquired and liabilities assumed consist of:

EUR million	
Cash and cash equivalents	2.9
Accounts receivable trade and other receivables, net	4.4
Income taxes receivable	0.9
Inventories, net	13.2
Intangible assets	0.4
Property, plant and equipment	4.8
Accounts payable trade and other accounts payable	8.7
Provisions	0.1
Accrual for income taxes and income taxes payable	0.1
Lease obligations	0.8
Deferred tax liabilities	1.3
Financial liabilities	9.2
Pensions and related liabilities	0.1
Identifiable acquired net assets	**6.3**

In the period between September 1, 2023 and December 31, 2023, E-Kart contributed EUR 23.2 million in net sales and net income in the amount of EUR 0.3 million to the Group result. If the acquisition had occurred on January 1, 2023, the Group net income in fiscal year 2023 would have amounted to EUR 94.1 million and the Group sales revenues would have amounted to EUR 3,001.9 million.

Acquisitions after the balance sheet date

On January 12, 2024 Netcetera acquired an additional 45.1 % of the shares in Braingroup AG, Zurich at a price of EUR 6.6 million in cash and thus, holds 78.1 % of the shares in this company. Thus, G+D assumes control over Braingroup AG in 2024 and consolidates the company in full. By taking over the majority share in Braingroup AG Netcetera complements its solution portfolio for banks and insurance companies and enables Braingroup to market its solution outside of Switzerland. The fair value of the net assets as of December 31, 2023 amounted to EUR 0.6 million. As of December 31, 2023, the fair value of 33.0 % of the shares amounts to EUR 4.8 million. Six months after the conclusion of the annual financial statements of 2026 of Braingroup Netcetera has a call option for the acquisition of the remaining shares. In case this call option is not exercised the non-controlling shareholders have the right to exercise a put option six months after the expiration of the call period and thus, sell their shares to Netcetera. No further information is available at this time.

25. Disclosures on Material Non-controlling Interests

The disclosures on material non-controlling interests (NCI) are as follows:

EUR million	**Giesecke & Devrient Malaysia SDN BHD,** Kuala Lumpur		**Giesecke & Devrient Kabushiki Kaisha,** Tokyo		**Veridos Matsoukis S.A. Security Printing,** Athens	
	Dec 31, 2023	**Dec 31, 2022**	**Dec 31, 2023**	**Dec 31, 2022**	**Dec 31, 2023**	**Dec 31, 2022**
Capital shares NCI	20.0 %	20.0 %	49.0 %	49.0 %	64.0 %	64.0 %
Voting rights NCI	20.0 %	20.0 %	49.0 %	49.0 %	64.0 %	64.0 %
Profit/(loss) attributable to NCI	0.5	1.1	1.0	1.9	1.3	0.9
Dividend paid to NCI	–	–	(1.2)	(1.9)	–	–
Share of equity relating to NCI	9.8	9.3	3.9	4.4	9.7	8.4
Assets[1]	55.7	54.6	8.6	11.0	65.5	58.1
thereof cash and cash equivalents[1]	7.8	7.1	4.4	6.3	3.5	8.0
Liabilities[1]	4.4	5.8	2.7	4.0	49.9	44.6
Revenues[1]	38.1	40.6	22.4	23.7	67.4	43.7
Other comprehensive income[1]	–	–	(0.6)	(0.4)	–	–
Comprehensive income[1]	2.5	5.4	1.4	3.4	2.1	1.4

[1] Before elimination of group transactions; aggregated (not proportional)

EUR million	Veridos GmbH, Berlin		secunet Security Networks AG, Essen including subsidiaries		Netcetera Gruppe, Zurich[2]	
	Dec 31, 2023	Dec 31, 2022	Dec 31, 2023	Dec 31, 2022	Dec 31, 2023	Dec 31, 2022
Capital shares NCI	40.0 %	40.0 %	24.5 %	24.5 %	5.0 %	0,0 %
Voting rights NCI	40.0 %	40.0 %	24.5 %	24.5 %	38.5 %	0,0 %
Profit/(loss) attributable to NCI	0.3	(8.0)	7.1	7.7	(5.5)	–
Dividend paid to NCI	–	–	(4.5)	(8.5)	–	–
Share of equity relating to NCI	9.1	9.1	34.1	31.8	1.8	–
Assets[1]	188.1	179.3	325.4	311.8	100.0	–
thereof cash and cash equivalents[1]	18.3	37.5	41.3	21.5	8.4	–
Liabilities[1]	156.4	150.7	187.5	184.0	76.0	–
Revenues[1]	188.7	140.8	393.7	347.2	96.4	–
Other comprehensive income[1]	(0.9)	6.1	(0.5)	1.5	(5.1)	–
Comprehensive income[1]	3.2	(15.7)	28.5	32.8	(21.3)	–

[1] Before elimination of group transactions; aggregated (not proportional)
[2] Netcetera Group, Zurich accounted for using the equity method in 2022

The cashflows in material non-controlling interests (NCI) as of December 31, 2023 are as follows:

EUR million	Net cash provided by operating activities	Net cash used in investing activities	Net cash used in/provided by financing activities
Giesecke & Devrient Malaysia SDN BHD, Kuala Lumpur	2.5	(1.7)	–
Giesecke & Devrient Kabushiki Kaisha, Tokyo	1.2	(0.1)	(2.5)
Veridos Matsoukis S.A. Security Printing, Athens	(4.1)	(5.7)	5.3
Veridos GmbH, Berlin	0.2	(10.1)	(20.7)
secunet Security Networks AG, Essen incl. subsidiaries	49.4	(8.8)	(20.8)
Netcetera Group, Zurich	(1.0)	(3.4)	12.5

The cashflows in material non-controlling interests (NCI) as of December 31, 2022 are as follows:

EUR million	Net cash provided by operating activities	Net cash used in investing activities	Net cash used in/provided by financing activities
Giesecke & Devrient Malaysia SDN BHD, Kuala Lumpur	9.8	(1.8)	(2.1)
Giesecke & Devrient Kabushiki Kaisha, Tokyo	3.2	(0.2)	(4.0)
Veridos Matsoukis S.A. Security Printing, Athens	–	(2.2)	7.2
Veridos GmbH, Berlin	(3.4)	(5.2)	29.7
secunet Security Networks AG, Essen incl. subsidiaries	(3.9)	(54.5)	(39.6)

26. Related Party Disclosures

Transactions with MC Familiengesellschaft mbH

Since 2012, MC Familiengesellschaft mbH is the Group parent company of Giesecke+Devrient GmbH.

In fiscal year 2022, G+D issued a loan of EUR 5.2 million to MC Familiengesellschaft mbH, which was fully repaid in December 2023. In fiscal year 2023, G+D issued a further loan of EUR 3.0 million to MC Familiengesellschaft mbH with a term until March 31, 2024. As of December 31, 2023 and December 31, 2022, there is a receivable from MC Familiengesellschaft mbH in the amount of EUR 0.1 million and EUR 0.2 million, respectively. As of December 31, 2023 and December 31, 2022, there are no other significant receivables and liabilities or expenses and income from transactions with MC Familiengesellschaft mbH.

Giesecke+Devrient GmbH entered into a service contract with MC Familiengesellschaft mbH. G+D renders accounting/taxes, finance and IT-system services. The allocated fee is immaterial.

Transactions with DER bogen GmbH & Co. KG

As of December 31, 2023 and December 31, 2022, there is a guarantee to DER bogen GmbH & Co. KG in the amount of EUR 30.0 million with a term until June 2024. As of December 31, 2023 and December 31, 2022, no further material transactions involving receivables and payables or income and expenses with DER bogen GmbH & Co. KG existed. The interest expenses amounted to EUR 0.0 million in December 31, 2023 and EUR 0.3 million in December 31, 2022.

Giesecke+Devrient GmbH entered into a service contract with DER bogen GmbH & Co. KG. G+D renders accounting/taxes, finance and IT-system services. The fee charged as of December 31, 2023 and December 31, 2022 was EUR 1.1 million and EUR 1.1 million, respectively.

Transactions with Giesecke+Devrient Foundation and Giesecke+Devrient Foundation Geldscheinsammlung

In fiscal year 2010, G+D established the Giesecke+Devrient Foundation. The company maintained a loan from the Giesecke+Devrient Foundation in the amount of EUR 21.0 million, which was fully repaid in February 2023. In July 2023, the company received a loan from the Giesecke+Devrient Foundation for an amount of EUR 15.0 million. The loan is due at maturity on July 31, 2026. Interest expense amounted to EUR 0.3 million and EUR 0.3 million in 2023 and 2022, respectively (see Note 13 “Financial Liabilities”). The grants amounted to EUR 0.3 million and EUR 0.3 million in fiscal years 2023 and 2022, respectively.

Additionally in fiscal year 2021, the newly established Giesecke+Devrient Foundation Geldscheinsammlung took out a loan in the amount of EUR 0.4 million as of December 31, 2022 which was fully repaid in November 2023. In November 2023, the company received a new loan from the Giesecke+Devrient Foundation Geldscheinsammlung in the amount of EUR 0.6 million. The loan is due at maturity on November 17, 2025. The grants amounted to EUR 0.1 million and EUR 0.1 million in fiscal years 2023 and 2022, respectively.

Transactions between affiliated companies and joint ventures and associated companies

Transactions were carried out between affiliated companies and joint ventures as well as associated companies. The following summary presents these transactions from the viewpoint of the affiliated companies:

EUR million	Services rendered		Services received	
	2023	2022	2023	2022
Joint ventures				
Goods and services	19.3	16.6	1.8	0.5
Other financial transactions	–	–	1.9	–
	19.3	**16.6**	**3.7**	**0.5**
Associated companies				
Goods and services	2.6	3.0	9.3	0.3
Other financial transactions	–	–	1.0	–
	2.6	**3.0**	**10.3**	**0.3**
	21.9	**19.6**	**14.0**	**0.8**

Accounts receivable and accounts payable from joint ventures and associated companies are comprised of the following:

EUR million	Dec 31, 2023	Dec 31, 2022
Joint ventures		
Accounts receivable from joint ventures	1.2	3.5
Associated companies		
Accounts receivable from associated companies	0.1	–
Accounts payable to associated companies	1.1	1.5

None of the balances from joint ventures and associated companies are secured.

Refer to Note 31 “Commitments and Contingent Liabilities” for commitments and contingent liabilities from joint ventures.

Transactions with members of key management personnel

The members of key management personnel include the members of the management board of Giesecke+Devrient GmbH, the chairmen of the management boards of Giesecke+Devrient Currency Technology GmbH, Giesecke+Devrient ePayments GmbH, Giesecke+Devrient Mobile Security Germany GmbH and Veridos GmbH, the chairman of the board of directors of secunet Security Networks AG (equal to Group Executive Committee – GEC) as well as the members of the supervisory board and the advisory board of Giesecke+Devrient GmbH since these bodies are responsible for planning, managing and monitoring the Group activities.

Compensation of key management personnel
The total compensation for active members of key management personnel amounted to EUR 11.1 million and EUR 9.9 million in 2023 and 2022, respectively.

In 2023 and 2022, the short-term benefits amounted to EUR 7.8 million and EUR 7.1 million, respectively. Thereof, EUR 6.9 million (prior year EUR 6.2 million) are attributable to the GEC, EUR 0.4 million (prior year EUR 0.4 million) to the supervisory board, and EUR 0.5 million (prior year EUR 0.5 million) to the advisory board.

The remuneration for services resulting from the termination of the employment relationship amounted to EUR 0.5 million in the financial year (previous year: EUR 0.0 million).

The past service cost for pensions for the GEC (benefits after termination of employment contract) amounted to EUR 0.5 million and EUR 0.6 million in 2023 and 2022, respectively.

Furthermore, long term benefits for active members of the GEC amounted to EUR 2.3 million (prior year EUR 2.2 million).

In the current reporting year, members of the GEC, with exception of the chairman of the board of directors of secunet Security Networks AG, are entitled to receive 40.0 % of their variable salary at the end of two additional years (deferral) in so far as they already held their positions and obtained consent in the prior year. The payment is based on the achievement of target average ROCE (return on capital employed) for fiscal years 2023 and 2022 and each of the two following years. The right to deferral only exists if employment continues or is terminated because of specific predetermined reasons. The related expense is included in other long-term payments.

The consolidated financial statements include provisions for pensions for the GEC amounting to EUR 4.4 million and EUR 2.8 million as of December 31, 2023 and 2022, respectively, as well as provisions or payables relating to compensation for members of key management personnel in the amount of EUR 10.2 million and EUR 8.6 million, respectively.

Total remuneration of the supervisory board and the advisory board in accordance with commercial law equals the stated short-term benefits. Total remuneration of the active members of the management body of the parent company in accordance with commercial law are not disclosed according to Section 315e (1) in conjunction with Sections 314 (3) no. 2, 286 (4) no. 2 German Commercial Code (HGB).

Business transactions with members of key management personnel or other related parties
In the course of ordinary business activities, Giesecke+Devrient receives advisory and consultancy services from companies and personnel with connections to the members of the supervisory board and advisory board or to the shareholder as well as to the members of the supervisory board and the advisory board itself.

No prepayments or loans to members of key management personnel were granted in fiscal years 2023 and 2022.

Former key management personnel of Giesecke+Devrient GmbH
Compensation to former members of the management board of the parent company and their survivors amounted to EUR 1.7 million and EUR 1.4 million in 2023 and 2022, respectively.

Pension obligations to former members of the management board of the parent company and their survivors amounted to EUR 16.9 million and EUR 17.2 million as of December 31, 2023 and 2022, respectively.

27. Number of Employees

The average number of full-time equivalent employees (excluding trainees and employees on maternity leave):

	2023	2022
Production	8,895	7,744
Sales	1,440	1,407
Research and development	1,637	1,248
Administration	1,967	1,661
	13,939	**12,060**

28. Personnel Expenses

EUR million	2023	2022
Wages and salaries	859.3	697.3
Social security contributions	147.1	114.7
Pension entitlements	10.6	10.1
	1,017.0	**822.1**

29. Disclosure in accordance with Section 161 German Stock Corporations Act (AktG)

The consolidated financial statements include secunet Security Networks AG, a publicly traded company. In accordance with Section 161 AktG (German Stock Corporation Act), the management of secunet AG has filed the required declaration and made it permanently available to the public on their website (http://www.secunet.com).

30. Exemption from the disclosure of the annual financial statements and management report in accordance with Section 264 /Section 264b HGB

The following companies will exercise their right not to prepare annual financial statements as well as not to prepare management reports in accordance with the regulations for corporate entities and certain registered partnerships as corporate entities (Section 264 (3) HGB) or partnerships that do not have an individual person either directly or indirectly as a general partner ("Kapitalgesellschaft und Co.") (Section 264b HGB). They also exercise their right not to have them audited or to disclose them:

- Giesecke+Devrient Currency Technology GmbH, Munich
- Papierfabrik Louisenthal GmbH, Gmund am Tegernsee
- Giesecke+Devrient ePayments GmbH, Munich
- Giesecke+Devrient Mobile Security Germany GmbH, Munich
- Giesecke+Devrient Secure Data Management GmbH, Neustadt b. Coburg
- EPC Electronic Payment Cards GmbH & Co. KG, Munich
- Giesecke+Devrient advance52 GmbH, Munich
- Giesecke+Devrient Ventures GmbH, Munich
- MC Holding GmbH & Co. KG, Gruenwald
- Giesecke+Devrient Grundstücksgesellschaft mbH & Co. KG, Gruenwald
- Giesecke+Devrient Immobilien Management GmbH, Munich
- Giesecke+Devrient Group Services GmbH & Co. KG, Munich
- Giesecke+Devrient TrustTech Fund I GmbH & Co. KG, Munich

31. Commitments and Contingent Liabilities

Legal proceedings/contingent liabilities
Giesecke+Devrient is involved in pending claims and legal proceedings arising in the ordinary course of business. Provisions have been made for estimated liabilities for certain items. G+D believes the resolution of all such matters will not have a material impact on G+D's net assets, results of operations and financial position.

Contingent liabilities in the amount of EUR 25.7 million as of December 31, 2023 (as of December 31, 2022 EUR 22.0 million) relating to tax risks outside Germany exist. As of December 31, 2023, additional contingent liabilities relating to legal disputes amounting to EUR 5.7 million (December 31, 2022: EUR 0.2 million) as well as other contingent liabilities in the amount of EUR 0.8 million (December 31, 2022: EUR 0.2 million) exist. G+D believes claims relating to these tax risks, legal disputes and other contingent liabilities are improbable.

With regard to financial guarantees, the maximum credit risk is the maximum amount that the Group would have to pay.

Guarantees
Giesecke+Devrient does not hold material amounts of financial assets which serve as collateral for liabilities or contingent liabilities. Moreover, G+D does not hold collateral which it would be permitted to sell or repledge in the event of default by the owner of the collateral.

G+D has issued guarantees for deposits received in the amount of EUR 164.1 million as of December 31, 2023 and EUR 98.8 million as of December 31, 2022.

Giesecke+Devrient guarantees indebtedness of a joint venture concerning contractual performance to third parties. These arrangements cover credit lines of the joint venture in the amount of up to EUR 4.0 million in 2022. Amounts relating to interest charges were also guaranteed. In the event of default of the joint venture, G+D was required to repay the borrowings covered by these guarantees.

Furthermore, Giesecke+Devrient guarantees indebtedness for DER bogen GmbH & Co. KG in the amount of EUR 30.0 million in 2023 and EUR 30.0 million in 2022 as well as for a third party concerning contractual performance to third parties in the amount of up to EUR 0.0 million in 2023 and EUR 1.0 million in 2022. The guarantees secure credit lines used for DER bogen GmbH & Co. KG. Amounts relating to interest charges are also guaranteed. In the event of default, G+D is required to repay the borrowings covered by these guarantees. The maximum exposure relating to these guarantees amounted to EUR 30.0 million as of December 31, 2023 and EUR 31.0 million as of December 31, 2022, respectively.

Commitments
As of December 31, 2023, Giesecke+Devrient has material purchase commitments which mainly consist of short-term agreements that were entered into during the 2023 fiscal year for the purchase of supplies, inventories, property, plant and equipment, land and services.

The aggregate amount of required payments for commitments as of December 31, 2023 is allocated to the respective years as follows:

EUR million	
2024	381.4
2025	77.7
2026	31.9
2027	13.0
2028	1.0
thereafter	2.8
	507.8

32. Grants

In fiscal years 2023 and 2022, G+D received other miscellaneous grants for operational investments in the amount of EUR 1.7 million and EUR 0.9 million. In addition, G+D received grants for expenses of EUR 8.5 million and EUR 3.6 million in 2023 and 2022, respectively. At present, there is reasonable assurance that the attached conditions will be fulfilled.

33. Risks

Refer to section 4 of the Group management report, "Opportunities and Risk Report", for the related disclosures.

34. Audit fees in accordance with Section 314 (1) no. 9 HGB

The audit fees for KPMG AG for the fiscal year ended 2023 amounted to EUR 2.6 million. The breakdown into categories is as follows: fees for audit services EUR 1.7 million, fees for audit-related services EUR 0.3 million, fees for tax-related services EUR 0.2 million and fees for all other services EUR 0.4 million.

35. Group to which the Company belongs

MC Familiengesellschaft mbH is the parent company of the Giesecke+Devrient Group (see Note 26 "Related Party Disclosures"). As of December 31, 2023, consolidated financial statements and a group management report will be prepared in accordance with International Financial Reporting Standards (IFRS) as adopted by the EU. The consolidated financial statements of MC Familiengesellschaft mbH will be published electronically in the German Federal Gazette.

36. Events after the Balance Sheet Date

With regard to the acquisition of Braingroup AG, see note 24 "Business combinations".

There have been no further significant events after the balance sheet date which are expected to have a material impact on the net assets, financial position, and results of operations of the Group.

37. Shareholdings

Direct and indirect investments held by Giesecke+Devrient GmbH in affiliated companies

	Shareholding in %
Giesecke+Devrient ePayments GmbH, Munich	100.0
Giesecke+Devrient Currency Technology GmbH, Munich	100.0
MC Holding GmbH & Co. KG, Gruenwald	100.0
Giesecke & Devrient Grundstücksgesellschaft mbH & Co. KG, Gruenwald	100.0[1]
Giesecke & Devrient Immobilien Verwaltungsgesellschaft mbH, Gruenwald	100.0
Papierfabrik Louisenthal GmbH, Gmund am Tegernsee	100.0
Giesecke+Devrient Mobile Security Germany GmbH, Munich	100.0
Giesecke+Devrient Secure Data Management GmbH, Neustadt near Coburg	100.0
Giesecke+Devrient advance52 GmbH, Munich	100.0
Giesecke+Devrient Ventures GmbH, Munich	100.0
Giesecke+Devrient Immobilien Management GmbH, Munich	100.0
EPC Electronic Payment Cards GmbH & Co. KG, Munich	100.0
Giesecke+Devrient Group Services GmbH & Co. KG, Munich	100.0
Giesecke+Devrient Ventures Management GmbH i. G., Munich	100.0
Giesecke+Devrient Ventures GP GmbH, Munich	100.0
Giesecke+Devrient TrustTech Fund I GmbH & Co. KG, Munich	100.0
Giesecke+Devrient Ventures TrustTech Carry Pool GmbH & Co. KG, Munich	100.0
MECOMO AG, Unterschleissheim	100.0
mecSOLAR smart devices GmbH, Unterschleissheim	100.0
GIESECKE + DEVRIENT EPAYMENTS IBERIA S.A., Barcelona	100.0
Giesecke + Devrient Mobile Security TCD Iberia, S.L., Barcelona	100.0
Giesecke + Devrient Currency Technology Iberia S.L., Madrid	100.0
Data Connectivity Podsystem SL, Seville	100.0
GIESECKE+DEVRIENT EPAYMENTS UK LTD, Wembley/Middlesex	100.0
GIESECKE+DEVRIENT MOBILE SECURITY TCD UK LIMITED, Wembley/Middlesex	100.0
Giesecke+Devrient Currency Technology GB Ltd, Milton Keynes	100.0
Podsystem Limited, Buckingham	100.0
Luner.io Limited, Buckingham	100.0
Giesecke+Devrient Mobile Security Slovakia, s.r.o., Nitra	100.0
Giesecke+Devrient ePayments Italia S.R.L., Milan	100.0
Giesecke+Devrient Currency Technology Italia S.R.L., Rome	100.0
GIESECKE+DEVRIENT MOBILE SECURITY TCD ITALIA S.R.L., Milan	100.0
Giesecke+Devrient ePayments France S.A.S., Craponne	100.0
Giesecke+Devrient Mobile Security TCD France SASU, Puteaux	100.0
Giesecke+Devrient Mobile Security Sweden AB, Stockholm	100.0
Giesecke+Devrient ePayments Finland Oy, Helsinki	100.0
Giesecke+Devrient Currency Technology Netherlands B.V., Amsterdam	100.0
Giesecke+Devrient Currency Technology Istanbul Ticaret ve Servis Limited Sirketi, Istanbul	100.0

[1] The general partner is Giesecke & Devrient Immobilien Verwaltungsgesellschaft mbH, Gruenwald

Direct and indirect investments held by Giesecke+Devrient GmbH in affiliated companies

	Shareholding in %
E-Kart Elektronik Kart Sistemleri Sanayi ve Ticaret Anonim Sirketi, Gebze	100.0
Giesecke+Devrient Mobile Security Russia, OOO, Moscow	100.0
Giesecke+Devrient Currency Technology FZE, Dubai	100.0
Giesecke+Devrient Holding FZE, Dubai	100.0
Giesecke & Devrient Egypt Ltd., Cairo	100.0
G+D epayments Egypt LLC, New Cairo	100.0
Giesecke & Devrient Saudi Arabia for Currency Technology, Riyadh	100.0
Giesecke and Devrient Currency Technology South Africa (Pty) Ltd, Johannesburg	100.0
GIESECKE AND DEVRIENT MOBILE SECURITY TCD SOUTH AFRICA (PTY) LTD, Johannesburg	100.0
Giesecke+Devrient Currency Technology Africa Ltd., Lagos	100.0
Giesecke+Devrient Currency Technology America, Inc., Dulles/Virginia	100.0
Giesecke+Devrient Mobile Security America, Inc., Dulles/Virginia	100.0
Giesecke+Devrient ePayments America, Inc., Dulles/Virginia	100.0
Podsystem Corporation, San Francisco/California	100.0
Giesecke+Devrient ePayments Canada, Inc., Toronto/Ontario	100.0
Giesecke y Devrient de México S.A. de C.V., Mexico City	100.0
GIESECKE+DEVRIENT MOBILE SECURITY TCD MÉXICO S.A. de C.V, Mexico City	100.0
Giesecke y Devrient Currency Technology de México, S.A. de C.V., Mexico City	100.0
Giesecke+Devrient ePayments Brasil Ltda, São Paulo	100.0
GIESECKE+DEVRIENT MOBILE SECURITY TCD BRASIL LTDA, São Paulo	100.0
Giesecke+Devrient Currency Technology Brasil Serviços e Comércio de Soluções Tecnológicas Ltda., São Paulo	100.0
MECOMO do Brasil S.A., Porto Alegre	100.0[1]
GyD Latinoamericana S.A., Buenos Aires	100.0
Giesecke and Devrient Mobile Security Australia Pty Ltd, Knoxfield/Victoria	100.0
GIESECKE+DEVRIENT MOBILE SECURITY TCD AUSTRALIA PTY LTD, Knoxfield/Victoria	100.0
Giesecke+Devrient Mobile Security Asia Pte. Ltd., Singapore	100.0
Giesecke+Devrient ePayments Asia Pte Ltd, Singapore	100.0
Giesecke & Devrient Asia Pacific Banking Systems (Shanghai) Co., Ltd., Shanghai	100.0
Giesecke+Devrient (China) Technologies Co., Ltd., Nanchang/Jiangxi	100.0
Giesecke+Devrient (Jiangxi) Technology Co., Ltd., Nanchang/Jiangxi	100.0
Giesecke & Devrient Asia Pacific Ltd., Hongkong	100.0
Giesecke and Devrient Mobile Security Hong Kong Limited, Wan Chai	100.0
Giesecke & Devrient India Private Limited, New Delhi	100.0
Giesecke & Devrient MS India Private Limited, New Delhi	100.0
Giesecke and Devrient Currency Technology Korea Co., Ltd., Seoul	100.0
PT Giesecke & Devrient Indonesia, Jakarta	100.0

[1] Not consolidated due to immateriality

Direct and indirect investments held by Giesecke+Devrient GmbH in affiliated companies

	Shareholding in %
PT Giesecke and Devrient Mobile Security Indonesia i.L., Jakarta	100.0
Giesecke & Devrient Egypt Services LLC, Cairo	99.0
Netcetera Group AG, Zurich	95.0
Netcetera AG, Zurich	95.0
Netcetera GmbH, Munich	95.0
Netcetera GmbH, Linz	95.0
Netcetera SAS, Paris	95.0
Netcetera AG, Vaduz	95.0
Netcetera DOOEL, Skopje	95.0
Netcetera Pte. Ltd., Singapore	95.0
Netcetera d.o.o., Ljubljana	95.0
Route Rank SA, Lausanne	95.0
Netcetera FZ-LLC, Dubai	95.0
Keynode AG i.L., Zurich	95.0
Giesecke & Devrient LOMO, ZAO, St. Petersburg	84.7
GIESECKE AND DEVRIENT EPAYMENTS SOUTH AFRICA (PTY) LTD, Johannesburg	84.0
Giesecke & Devrient Malaysia SDN BHD, Kuala Lumpur	80.0
secunet Security Networks AG, Essen	75.5
Secunet Inc., Austin (Shell company)	75.5[1]
secunet International GmbH & Co.KG, Essen	75.5
secunet International Management GmbH, Essen	75.5
finally safe GmbH, Essen	75.5
stashcat GmbH, Hanover	75.5
SysEleven GmbH, Berlin	75.5
Veridos GmbH, Berlin	60.0
Veridos Canada Ltd., Toronto/Ontario	60.0
Veridos America Inc., Wilmington/Delaware	60.0
Veridos FZE, Dubai	60.0
VERIDOS IDENTITY SOLUTIONS UGANDA SMC LIMITED, Kampala	60.0
Firdaus Al Aman for general Trading, Baghdad	60.0
Veridos Brasil Comércio de Smart Cards e Soluções para Identificação Segura e Autenticação Ltda. i.L., São Paulo	60.0
Veridos México S.A. de C.V., Mexico City	60.0
E-SEEK Inc., San Diego/California	60.0
Giesecke & Devrient Kabushiki Kaisha, Tokyo	51.0
Netset Global Solutions D.O.O., Belgrade	40.2
secustack GmbH, Dresden	38.5
Veridos Matsoukis. Security Printing S.A, Athens	36.0
VERIDOS TRADING AND SERVICES L.L.C., Abu Dhabi	29.4

[1] Not consolidated due to immateriality

Investments held by Giesecke+Devrient GmbH in associated companies and joint ventures

	Shareholding in %
Podsystem MX SAPI de CV, León, Gto	51.0
Shenzhen Giesecke & Devrient Currency Automation Systems Co. Ltd., Shenzhen	50.0
NETLYNC MOBILE COMMUNICATION TECHNOLOGIES LIMITED, Dublin	36.0
Blokverse d. o. o., Skopje	32.3
Braingroup AG, Zurich	31.3
UGANDA SECURITY PRINTING COMPANY (USPCL) LIMITED, Entebbe	29.4
Unitek Engineering AG, Zurich	28.5
D ONE Value Creation AG, Zurich	27.5
Securities Grid Ltd., London	23.7
Build38 GmbH, Munich	20.9
Relief Validation Ltd, Dhaka	15.0

Investments held by Giesecke+Devrient GmbH in other related parties

	Shareholding in %
Brighter AI Technologies GmbH, Berlin	7.8
FINANCIAL NETWORK ANALYTICS LTD, London	6.5
Nano Corp., Paris	5.7
Yields NV, Ghent	5.6
Tenzir GmbH, Hamburg	5.3
IDnow GmbH, Munich	5.0
SALV TECHNOLOGIES OÜ, Tallinn	4.3
Blindflug Studios AG, Zurich	4.1
Cognism Ltd., London	1.2
Verimi GmbH, Frankfurt am Main	0.5

The affiliated companies Ayanta AG and Keynode AG are deconsolidated in the 2023 financial year.

Munich, March 22, 2024

Giesecke+Devrient GmbH
The Management Board

Dr. R. Wintergerst
Group CEO

Dr. P. Zattler
Group CFO

Corporate Bodies

Supervisory Board

Prof. Klaus Josef Lutz
(Chairman) Munich

Eva Schäflein-Bohsewe[1]
(Deputy Chairman) Neubiberg

Achim Berg
Munich

Prof. Dr. Gabi Dreo Rodosek
Haar

Ralf Gerlach[1]
Gilching

Raimund Litters[1]
Rosenheim

Astrid Meier-Sikorski[1]
Munich

Horst Müller[1]
Ernsgaden

Claudia Scheck[1]
Königsmoos

Dr. Walter Schlebusch
Munich

Verena von Mitschke-Collande
Tutzing

Stefan Winners
Munich

Advisory Board

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(Chairman) Munich

Verena von Mitschke-Collande
(Deputy Chairman) Tutzing

Achim Berg
Munich

Prof. Dr. Gabi Dreo Rodosek
Haar

Dr. Walter Schlebusch
Munich

Stefan Winners
Munich

Management Board

Dr. Ralf Wintergerst
(Group CEO, Giesecke+Devrient)

Dr. Peter Zattler
(Group CFO, Giesecke+Devrient)

[1] Employee representatives

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